Calibration Uncertainty in Adaptive Large Language Model Evaluation: Theory and Empirical Evidence (Extended Version)

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Abstract

Adaptive testing with item response theory needs item parameters calibrated on other systems, and new, private or costly benchmarks for language models often have only a handful. We study this calibration uncertainty with EOST-M, which uses each item's expected response function under an empirical-Bayes prior and selects items by expected information gain. This model is exact given the calibrators' abilities, and a difficulty with Gaussian uncertainty s caps an item's Fisher information at $2/(\pi s^2)$ for any link. Two frozen-plan studies on eight benchmarks give three messages. First, plug-in Fisher-information testing fails with small panels: with eight calibrators and one target per family cluster, EOST-M predicted held-out verdicts better (median 5.9%, significant on 5 of 8 benchmarks). Second, accounting for the uncertainty fixes much of this: EOST-M beat a regularized Rasch-type test (2.4%), Bayesian tests that also integrate the calibrators' abilities were not significantly different from it, and its score error was lower than with random items under the same model, ATLAS-style testing, tinyBenchmarks and MINCE, all run on the same eight calibrators, on every benchmark, meeting the frozen criterion only against MINCE. Third, the information criterion then matters little. EOST-M's 90% intervals still under-cover (0.74; 0.86 after recalibration).

Keywords: adaptive testing, item response theory, evaluation of large language models, calibration uncertainty, expected information gain

1 Introduction

Large language models (LLMs) are compared by running them on static benchmarks and reporting an average. This is expensive—agentic and code benchmarks execute hidden tests or whole repositories for every item—and statistically wasteful, because items differ enormously in how much they reveal about a system (Lalor et al., 2016; Rodriguez et al., 2021). Computerized adaptive testing (CAT), the psychometric answer to the same problem for human examinees (Lord, 1980; Wainer, 2000), has therefore been transferred to LLMs:

item response theory (IRT) models fitted on the item-level verdicts of previously evaluated systems are used to pick anchor items (Vivek et al., 2024; Maia Polo et al., 2024), to distill benchmarks (Kipnis et al., 2025), or to administer items adaptively by maximum Fisher information (Hofmann et al., 2025; Li et al., 2026; Zheng et al., 2026; Zhou et al., 2026).

These methods calibrate on hundreds to thousands of systems and then treat the item parameters as known. Many practical settings have only a handful of calibration systems: a new or private benchmark, a freshly generated item pool, a domain where only a few open models can be run, or an agentic benchmark where every evaluation is costly. With four to eight calibrators, item parameters—especially discriminations—are poorly identified, and adaptive selection then *capitalizes on calibration error*: it preferentially picks the items whose informativeness is most overestimated (van der Linden and Glas, 2000; Patton et al., 2013). Psychometrics knows remedies for this: expected response functions that integrate the item-parameter posterior (Mislevy et al., 1994) and fully Bayesian adaptive testing (van der Linden and Ren, 2020; Fink et al., 2025). What is not known is how much these remedies matter in the small-panel regime of LLM evaluation, against which competitors, and with what consequences for the uncertainty that an adaptive test reports.

This paper studies that question with a calibration-marginal adaptive test, EOST-M: items enter the test through their expected response function under an empirical-Bayes item prior, are selected by expected information gain (EIG), and are graded by an independent oracle. Two studies with analysis plans frozen before execution—the second designed after an external review, with one target per family cluster and the strongest competitors we could identify—lead to three messages.

1. **Small-panel calibration uncertainty is the real problem.** With 4–8 calibrators, plug-in Fisher-information CAT treats item parameters that a small panel cannot determine—poorly located difficulties and inflated discriminations—as known, and both its predictions and its intervals suffer.
2. **Accounting for that uncertainty fixes a substantial part of it.** Integrating the item-parameter posterior (EOST-M) and constraining the discriminations (a regularized Rasch-type test) are two remedies for the same failure. In study 2, with one target per family cluster, EOST-M also predicted held-out verdicts better than the regularized Rasch-type test (median 2.4%, 5 of 8 benchmarks significant), and Bayesian CATs that also integrate the calibrators' abilities were not significantly different from it.
3. **Once uncertainty is handled, the choice of information criterion matters much less than first hypothesized.** EIG, maximum and posterior-weighted Fisher information and the score-targeted criterion performed almost identically under the calibration-marginal model (a descriptive comparison, not an equivalence test), and the original hypothesis that information beats predictive entropy on held-out prediction is consistent in sign but small and not supported by the frozen rule; for the benchmark score, entropy differs more, as the one-step score-targeted analysis predicts.

Contributions.

1. **Theory** (Section 4). The calibration-marginal model is the exact Bayesian model given the calibrators' abilities and the hyperprior (Lemma 1), which locates it with

respect to fully Bayesian CAT; an item whose difficulty has a Gaussian posterior with standard deviation s can never carry more than $2/(\pi s^2)$ of Fisher information, for the logistic link and any other, with a general bound for other posteriors (Proposition 5); entropy, Fisher information and EIG are related exactly and agree under a common discrimination; and a closed form for one-step score-targeted design.

2. **Evidence** (Sections 6–9). Study 1: eight real benchmarks from five domains (629 systems), panels of 4–64 calibrators, re-analysed with family clusters as units. Study 2: independent targets (one per strict family cluster), competitors including a regularized Rasch-type test, a posterior-draw marginal Bayesian CAT, ATLAS-style 3PL CAT, tinyBenchmarks, MINCE and Random:mg, a leave-one-calibrator-out recalibration of score intervals, and a temporal validation on later SWE-bench submissions. A cost analysis gives break-even points.

3. **Reproducibility.** A Docker-only pipeline over pinned public sources; every number in the paper is a macro generated from hashed artifacts and the qualitative statements about results are re-checked on the artifacts by an automatic test; both analysis plans were hashed and committed before execution (Section 5).

Scope. All evidence is a replay of recorded verdicts produced by deterministic oracles or by recorded aggregate verdicts (Section 5); a protocol for live deployment is released (Section 10) but could not be run in our execution environment. The framework also allows models to generate their own items (self-testing); this was explored only in a historical pilot program (Section 5.4) and is not part of the evidence.

Main findings. In study 1 (629 systems, leave-family-out replay), EOST-M with eight calibrators predicted held-out verdicts better than plug-in Fisher CAT on all 8 benchmarks (median 5.2%, Holm-significant on 7), estimated the benchmark score with a median 54.4% smaller error than uniform subsampling and a median 25.5% smaller than random items scored by the same model, and gave less overconfident intervals; the gains shrink as the panel grows. With strict family clusters as units the prediction gain keeps its size (5.4%) but is significant on only 4 benchmarks, below the frozen threshold. Study 2, with one target per cluster, supported the replications of all three study-1 results and the superiority of EOST-M over the regularized Rasch-type test; a posterior-draw marginal Bayesian CAT was not significantly better or worse than EOST-M on any benchmark but met the frozen equivalence margin on too few of them, before and after a post hoc correction of its burn-in, and in follow-ups with long chains and better-mixing samplers the median difference stayed below half a per cent; and in the eight-calibrator regime EOST-M was ahead of ATLAS-style CAT, tinyBenchmarks and MINCE on every benchmark, significantly on four or five of eight. Of its 13 frozen decisions, 6 were supported, and four of the unsupported superiority hypotheses missed the threshold by one benchmark; in a secondary random-effects meta-analysis that treats the benchmarks as independent their pooled effects exclude zero, but only that of C2 does so when the five benchmarks that share their systems count as one source. The two prediction results (against plug-in Fisher CAT and the Rasch-type test) did not depend on how the family clusters were built. Recalibration raised the coverage of EOST-M's 90% intervals from 0.74 to 0.86, short of the frozen criterion and without a better interval

score. A temporal validation on later SWE-bench submissions favoured EOST-M in 10 of 11 contrasts.

2 Related work and what is new here

Calibration error in adaptive testing. Treating estimated item parameters as true inflates measurement precision and biases ability estimates, and adaptive selection amplifies the problem because it seeks the items that look most informative, the *capitalization on calibration error* (van der Linden and Glas, 2000; Olea et al., 2012; Patton et al., 2013). Remedies propagate parameter uncertainty into scale scores (Zhang et al., 2011; Yang et al., 2012), replace the item response function by its expectation over the parameter posterior—the expected response function (Mislevy et al., 1994), which we use—or treat all parameters in a fully Bayesian way. van der Linden and Ren (2020) give a fast Markov chain Monte Carlo algorithm for Bayesian adaptive testing that samples the joint posterior of ability and item parameters during the test, using stored posterior draws of the item parameters from a Bayesian calibration. Fink et al. (2025) compare corrections for calibration error in simulated 2PL tests with 100–500 calibration responses per item and find the fully Bayesian approach, with item selection by Bayesian maximum information averaged over ability and item draws, the most accurate. We therefore do *not* claim to be the first to integrate calibration uncertainty into adaptive testing. Lemma 1 shows that our calibration-marginal model is exactly the fully Bayesian model with the calibrators' abilities and the hyperprior held at point estimates; study 2 compares the two directly.

Item selection. Classical CAT selects the item with maximum Fisher information at the current ability estimate (Lord, 1980); Bayesian variants weight information by the ability posterior (Owen, 1975; van der Linden, 1998), use Kullback–Leibler information (Chang and Ying, 1996) or other criteria (Veerkamp and Berger, 1997), and differ mainly early in a test (Chen et al., 2000); randomesque selection among the most informative items limits exposure (Kingsbury and Zara, 1989). Mutual-information selection was proposed for classification testing (Weissman, 2007) and multidimensional CAT (Wang and Chang, 2011); in experimental design it is Lindley's information (Lindley, 1956; Chaloner and Verdinelli, 1995; Rainforth et al., 2024) and in active learning BALD (Houlsby et al., 2011), whose motivation—entropy conflates epistemic and aleatoric uncertainty—is Proposition 2. Uncertainty sampling (Lewis and Gale, 1994) and expected error reduction (Roy and McCallum, 2001) are the active-learning counterparts of our entropy and score-targeted criteria.

IRT and reduced evaluation for LLMs. IRT has been used to build evaluation scales (Lalor et al., 2016), to weight leaderboard examples (Rodriguez et al., 2021), to select anchor points and compressed benchmarks (Vivek et al., 2024; Maia Polo et al., 2024; Kipnis et al., 2025) and to amortize calibration with content-based difficulty predictors (Truong et al., 2025). tinyBenchmarks (Maia Polo et al., 2024) choose about 100 anchor items by clustering item embeddings from hundreds of training models and combine the anchors with IRT predictions (gp-IRT). ATLAS (Li et al., 2026) runs Fisher-information CAT with a 3PL calibrated on thousands of leaderboard models and reports up to 90% fewer items; Fluid Benchmarking (Hofmann et al., 2025) does the same with about one hundred calibration

models. MINCE (Das et al., 2026) uses a few calibration models only to size a random subset by Monte Carlo, without an IRT model. Other work extends adaptive testing to continuous scores (Balkır et al., 2026), accounts for prompt sensitivity with parameters treated as known (Jo and Wilson, 2025), links new datasets by fixed-parameter calibration (Habba et al., 2026), uses neural or medical item banks (Zhou et al., 2026; Zheng et al., 2026), or applies a Laplace approximation to neural IRT difficulties with selection by posterior-expected Fisher information (Mandujano Reyes, 2026); a position paper argues for psychometric testing of AI systems (Zhuang et al., 2025), and a simulation study finds item-level IRT inferences unreliable when few calibration models are available (Jiang et al., 2026)—the regime of this paper. Error bars for LLM evaluation are studied by Madaan et al. (2024) and Miller (2024).

What is new. The ingredients—expected response functions, Bayesian adaptive testing, information-based selection—are classical. What this paper adds is (i) their systematic study in the regime that LLM evaluation actually faces, with calibration panels of 4 to 64 systems instead of hundreds to thousands of examinees, on eight real benchmarks from five domains and with independent targets; (ii) theory for that regime: exactness of the calibration-marginal model given the calibrators (Lemma 1), an information cap that holds exactly for any link under a Gaussian difficulty posterior (Proposition 5), the relation between entropy, Fisher information and EIG under calibration uncertainty, and a one-step score-targeted criterion; (iii) head-to-head comparisons, under analysis plans frozen before execution, with the competitors that matter in this regime—a regularized Rasch-type test, Bayesian adaptive tests that also integrate the calibrators' abilities, ATLAS-style 3PL CAT, tinyBenchmarks, MINCE, and uniform subsampling with and without the same calibration; and (iv) an honest account of uncertainty quantification, including a leave-one-calibrator-out recalibration of score intervals and a cost analysis.

Self-evaluation and circularity. Using models to judge models raises circularity concerns (Zheng et al., 2023). The framework allows models to generate items but requires an independent oracle for verdicts; this aspect was explored only in a historical pilot (Section 5.4) and is not part of the evidence of this paper.

3 Calibration-marginal adaptive testing

3.1 Items, oracles and recorded verdicts

A *test item* i is a task together with an *oracle* O_i that maps a system output to a verdict $y_i \in \{0, 1\}$: exact numeric match for math word problems, execution of hidden unit tests for code and repository-level software engineering, or a log-likelihood comparison of fixed answer options for multiple-choice tasks. The framework (called EOST in our project, for entropy-oracled information-optimal self-testing, where optimality is meant in the one-step sense of Section 4) imposes one structural rule: *the system under evaluation may propose or select items, but it is never the oracle of its own verdicts.* In this paper all items come from fixed banks, which isolates the measurement question; self-generated items were explored only in a historical pilot (Section 5.4).

Two kinds of verdicts occur in our data and are kept distinct. A *deterministic verdict* is the output of the oracle on one recorded system output (GSM8K, ARC-C, HellaSwag, WinoGrande, MMLU: a single greedy or log-likelihood evaluation; SWE-bench Verified: the

official harness on the submitted patch). A *recorded aggregate verdict* summarizes several sampled outputs: for LiveCodeBench and CRUXEval-O the public data give a pass rate over samples, which we binarize at one half. Replaying either kind is exact for the recorded data, but only the first corresponds to one administration of the item to one deterministic system; Section 7 reports the main contrasts separately for the two groups.

3.2 Measurement model and calibration panel

Let s be the target system with latent competence θ_s. Responses follow the two-parameter logistic (2PL) model

$$\Pr(y_{si} = 1 \mid \theta_s, a_i, b_i) = \sigma\big(a_i(\theta_s - b_i)\big), \qquad a_i > 0. \tag{1}$$

Item parameters $\psi_i = (\log a_i, b_i)$ are unknown and are calibrated on a *panel* $\mathcal{C}$ of N other systems whose verdicts on the bank are available. The panel excludes the target and every related system: in study 1 the target's heuristic *family* (same model line, reasoning-effort or version variant, or, for agentic systems, any scaffold sharing an underlying LLM); in study 2 its *strict cluster*, which also joins near-duplicates detected from the verdicts themselves (Section 5).

Calibration proceeds in four steps, shared by every policy: (i) joint maximum a posteriori (MAP) estimation of the calibrators' abilities $\hat{\theta}_{\mathcal{C}}$ and item parameters with $\theta_j \sim \mathcal{N}(0,1)$, $\log a_i \sim \mathcal{N}(0,1)$, $b_i \sim \mathcal{N}(0, 2^2)$; (ii) an empirical-Bayes item hyperprior $\log a_i \sim \mathcal{N}(\mu_a, \sigma_a^2)$, $b_i \sim \mathcal{N}(\mu_b, \sigma_b^2)$ obtained by maximizing the marginal likelihood of *all* items given $\hat{\theta}_{\mathcal{C}}$ (EM on a parameter grid); the hyperparameters pool hundreds of items and are therefore well determined even when $N = 4$; (iii) plug-in item parameters $(\hat{a}_i, \hat{b}_i)$, the conditional MAP given $\hat{\theta}_{\mathcal{C}}$ under that hyperprior; (iv) the exact grid posterior $\pi_i(\psi) \propto p(\psi) \prod_{j \in \mathcal{C}} p(y_{ji} \mid \hat{\theta}_j, \psi)$ on a 26×129 grid over $(\log a, b) \in [-2.5, 2.5] \times [-8, 8]$.

Two measurement models follow. The **plug-in** model uses $P_i(\theta) = \sigma(\hat{a}_i(\theta - \hat{b}_i))$, as in classical computerized adaptive testing and in recent LLM work (Hofmann et al., 2025; Li et al., 2026). The **marginal** model uses the expected response function

$$\bar{P}_i(\theta) = \int \sigma\big(a(\theta - b)\big)\, \pi_i(d\psi) \tag{2}$$

(Mislevy et al., 1994). By Lemma 1, $\bar{P}_i$ is the exact response probability of the target given $\hat{\theta}_{\mathcal{C}}$ and the hyperprior, so no further approximation is involved in the ability posterior, the predictive distribution or the information criteria below. Both models share $\hat{\theta}_{\mathcal{C}}$ and the hyperprior: they differ only in plugging in a mode versus integrating the item posterior.

Numerically, the joint MAP estimate of step (i) uses L-BFGS-B with analytic gradients. The hyperprior of step (ii) is obtained by EM on a 14×65 grid (at most 60 iterations, relative tolerance 10^{-7}), with $\sigma_b \in [0.3, 4]$ and $\sigma_a \in [0.1, 1.5]$. The plug-in parameters of step (iii) are computed by vectorized Fisher scoring with step halving. For the grid posteriors of step (iv), with $\Lambda_1 = \log \sigma(z)$ and $\Lambda_0 = \log \sigma(-z)$ evaluated on the $N \times G$ matrix $z_{jg} = a_g(\hat{\theta}_j - b_g)$, the item log-likelihoods are $Y^\top \Lambda_1 + (1 - Y)^\top \Lambda_0$ (two GEMMs), and the expected response functions on the ability grid are one further GEMM.

3.3 Adaptive testing and selection criteria

The target's ability posterior π_t is kept on a 201-point grid over $[-5, 5]$ with prior $\mathcal{N}(0, 1)$ and is updated after each verdict. At step t an item is chosen from the remaining candidate pool by one of:

- Random: uniform choice (the model is used only for scoring);
- Entropy: maximize $U_{\mathrm{H}}(i) = h(\bar{p}_i)$, the predictive entropy (uncertainty sampling);
- MFI: maximize Fisher information at the posterior mean (classical CAT);
- PWFI: maximize posterior-weighted Fisher information (van der Linden, 1998);
- EIG: maximize $U_{\mathrm{EIG}}(i) = I(\theta; y_i \mid \mathcal{D}_t)$, the expected information gain about competence (the EOST criterion);
- EVR: maximize the expected reduction of the posterior variance of the unobserved bank total (score-targeted EOST, Proposition 7).

Each criterion is run under the plug-in (`:pi`) and the marginal (`:mg`) model, except PWFI, which is run under the marginal model only. The proposed method, **EOST-M**, is EIG:mg (Algorithm 1, Figure 1). The classical state of practice is MFI:pi. A model-free baseline, Naive, samples items uniformly and reports the running sample mean with a Wilson interval and finite-population correction.

The ability posterior is kept in log space. Entropy, EIG and PWFI are matrix–vector products with precomputed item×grid matrices; MFI interpolates Fisher information at the posterior mean; EVR uses the closed form of Proposition 7 evaluated exactly on the grid (unit-tested against a brute-force enumeration). Ties are broken uniformly at random with a dedicated random stream.

3.4 Recalibrating the score interval

The model-based interval ignores what the model gets wrong (misspecification, dependence between systems) and, as we will see, is too narrow on real data. A simple correction uses only the calibration panel. For every calibrator c, the item posteriors are recomputed without c's cluster (abilities and hyperprior held at the panel fit), the same adaptive test is replayed on c, whose full-bank score is known, and its absolute score errors $|\hat{A}_{c,t} - A_c|$ are recorded. The recalibrated half-width at step t is the 90% quantile of these errors pooled over calibrators and over the steps within a factor 1.5 of t. This leave-one-calibrator-out rule ("+R") is a cross-validated correction, not a conformal guarantee: with $N \leq 8$ pseudo-targets no finite-sample coverage guarantee at 90% is available (Vovk et al., 2005). The rule was chosen among three variants on development banks before study 2 (Section 5.4).

3.5 Estimands and endpoints

Latent competence is a model construct whose scale is fixed only by the prior, so endpoints are defined on observable quantities.

1. **Held-out predictive log loss.** A fixed holdout H (20% of the bank, drawn once per target and replicate and shared by *all* policies and panel sizes, a design adopted

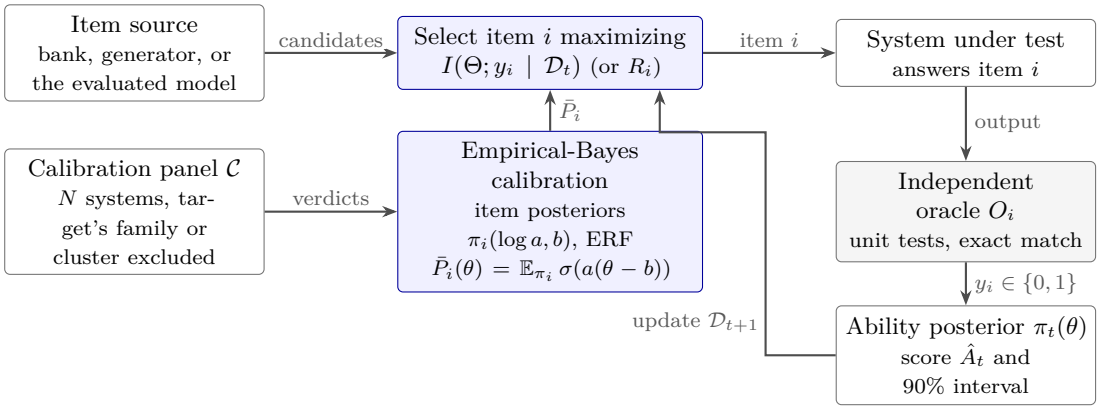


Figure 1: EOST-M, the calibration-marginal adaptive test studied in this paper. Item parameters are not plugged in: each item enters the test through its expected response function (ERF) under its empirical-Bayes posterior given a small calibration panel, items are selected one at a time by expected information gain about competence (or by the score-targeted criterion R_i when the estimand is the benchmark score), and verdicts come from an independent oracle. The framework allows items from any source, including the evaluated model, but all evidence in this paper uses fixed banks.

after an audit of the pilot program, Section 5.4) is never administered. After each step, every policy predicts the target's verdicts on H with $\mathbb{E}_{\pi_t} P_i(\theta)$; the integrated log loss averages the holdout log loss over steps $t = 1, \ldots, T$. A strictly proper score, it rewards both accuracy and calibration of what the test has learned about the target.

2. **Score error.** The full-bank accuracy $A = \frac{1}{I} \sum_i y_{si}$ is estimated by $\hat{A}_t = \frac{1}{I}\big(\sum_{\text{asked}} y_{si} + \sum_{\text{not asked}} \mathbb{E}_{\pi_t} P_i(\theta)\big)$; the integrated absolute error (IAE) averages $|\hat{A}_t - A|$ over $t \le T$.
3. **Interval coverage and interval score.** The proportion of steps at which the 90% predictive interval for A contains A, and the integrated interval score $(u - l) + \frac{2}{\alpha}(l - A)_+ + \frac{2}{\alpha}(A - u)_+$ with $\alpha = 0.1$, a proper score that penalizes both width and misses (Gneiting and Raftery, 2007). The predictive distribution of the unobserved total is a mixture over the ability grid of sums of independent Bernoulli variables; the interval uses its exact mean and variance and a normal approximation.
4. Secondary: holdout Brier score, final-step errors, sustained items-to-tolerance (the first t after which $|\hat{A}_s - A| \le \varepsilon$ for all $s \ge t$), and, for synthetic banks, error and coverage of the true θ.

Algorithm 1 EOST-M: calibration-marginal adaptive test with one-step (myopic) expected-information-gain selection

Require: bank $\mathcal{I}$ with oracles O_i; panel verdicts $Y_{\mathcal{C}} \in \{0,1\}^{N\times I}$; budget T; ability grid $\{\theta_k\}_{k=1}^{K}$

$(\hat{\theta}_{\mathcal{C}}, \cdot) \leftarrow$ joint MAP of the 2PL on $Y_{\mathcal{C}}$ ▷ calibrator abilities

$\eta \leftarrow \arg\max_\eta \sum_i \log \sum_g p_\eta(\psi_g)\, p(Y_{\mathcal{C},i} \mid \hat{\theta}_{\mathcal{C}}, \psi_g)$ ▷ empirical-Bayes item hyperprior (EM)

$W_{ig} \propto p_\eta(\psi_g)\, p(Y_{\mathcal{C},i} \mid \hat{\theta}_{\mathcal{C}}, \psi_g)$ ▷ item posteriors on the $(\log a, b)$ grid

$\bar{P}_{ik} \leftarrow \sum_g W_{ig}\, \sigma\big(a_g(\theta_k - b_g)\big)$, $H_{ik} \leftarrow h(\bar{P}_{ik})$ ▷ expected response functions

$\log \pi_k \leftarrow \log \phi(\theta_k)$; $\mathcal{U} \leftarrow \mathcal{I}$ (candidate pool)

for $t = 1, \ldots, T$ **do**

  $\pi \leftarrow \mathrm{softmax}(\log \pi)$; $\bar{p}_i \leftarrow \sum_k \bar{P}_{ik}\pi_k$ for $i \in \mathcal{U}$

  $i_t \leftarrow \arg\max_{i\in\mathcal{U}}\ h(\bar{p}_i) - \sum_k H_{ik}\pi_k$ ▷ EIG; or R_i of Eq. (7)

  $y \leftarrow O_{i_t}$(system's answer to i_t) ▷ independent oracle

  $\log \pi_k \leftarrow \log \pi_k + y \log \bar{P}_{i_t k} + (1-y)\log(1-\bar{P}_{i_t k})$; $\mathcal{U} \leftarrow \mathcal{U} \setminus \{i_t\}$

end for

$\pi \leftarrow \mathrm{softmax}(\log \pi)$; $\bar{p}_i \leftarrow \sum_k \bar{P}_{ik}\pi_k$ for all i

return posterior π, score estimate $\hat{A}_T = \frac{1}{I}\big(\sum_{\text{asked}} y + \sum_{i\notin\text{asked}} \bar{p}_i\big)$ and its interval

4 Theory: what calibration uncertainty does to an adaptive test

Throughout, π_t denotes the target's ability posterior after t administered items, μ_t and σ_t^2 its mean and variance, $h(p) = -p\log p - (1-p)\log(1-p)$ the binary entropy (nats), and P_i the response function used for item i (plug-in $P_i(\theta) = \sigma(\hat{a}_i(\theta - \hat{b}_i))$ or the expected response function $\bar{P}_i$ of Eq. (2)). Each result is followed by its proof. All optimality statements below are *myopic*: they concern the best next item given the current posterior, not the best sequential policy over a whole test.

The calibration-marginal model is an exact Bayesian model given the calibrators.

Lemma 1 (Exactness given the panel's abilities) *Let the calibrators' abilities θ_C and the item hyperparameters η be fixed. Then the item parameters $\psi_1, \ldots, \psi_I$ are independent a posteriori given the panel's responses, and for any set S of items answered by a new target with ability θ,*

$$p(y_S \mid \theta, \mathcal{D}_{\text{panel}}, \theta_C, \eta) = \prod_{i\in S} \bar{P}_i(\theta)^{y_i}\{1-\bar{P}_i(\theta)\}^{1-y_i}, \qquad \bar{P}_i(\theta) = \mathbb{E}\big[P_{\psi_i}(\theta) \mid \mathcal{D}_{\text{panel}}, \theta_C, \eta\big]. \tag{3}$$

Integrating θ_C and η as well (a fully Bayesian treatment) replaces the product by

$$\mathbb{E}_{\theta_C,\eta\mid\mathcal{D}}\Big[\prod_{i\in S} \bar{P}_i^{(\theta_C,\eta)}(\theta)^{y_i}\{1-\bar{P}_i^{(\theta_C,\eta)}(\theta)\}^{1-y_i}\Big],$$

which no longer factorizes, because shared abilities and hyperparameters make the items dependent.

Proof Given θ_C and η, the joint posterior of the item parameters is proportional to $\prod_i p(\psi_i \mid \eta)\prod_n p(y_{ni} \mid \theta_n, \psi_i)$, a product of factors that each involve one item only; the items are therefore independent a posteriori. The target's responses are conditionally independent given (θ, ψ) and each involves one item, so $p(y_S \mid \theta, \cdot) = \int \prod_{i\in S} p(y_i \mid \theta, \psi_i)\prod_i p(\psi_i \mid \cdot)\, d\psi =$

$\prod_{i\in S}\int p(y_i \mid \theta,\psi_i)p(\psi_i \mid \cdot)\,d\psi_i$, and the inner integral is $\bar{P}_i(\theta)^{y_i}\{1-\bar{P}_i(\theta)\}^{1-y_i}$. Integrating over (θ_C,η) afterwards averages products, not factors, which does not factorize unless the posterior of (θ_C,η) is a point mass. ■

The ability posterior, the predictive distribution and the information criteria of EOST-M are therefore exact Bayesian quantities *conditional on* point estimates of the calibrators' abilities and of the hyperprior; nothing is approximated about the item parameters except the grid. This locates EOST-M precisely with respect to fully Bayesian adaptive testing (van der Linden and Ren, 2020; Fink et al., 2025): those methods also integrate θ_C and η, by Markov chain Monte Carlo. When their sampler draws stored item parameters independently for each administered item at every iteration, as the algorithm of van der Linden and Ren (2020) does, the ability posterior it targets uses the per-item averages $\mathbb{E}_{\theta_C,\eta}[\bar{P}_i^{(\theta_C,\eta)}]$, i.e. Eq. (3) with item marginals that also average over (θ_C,η); we call this the *posterior-draw marginal* Bayesian CAT, and the version that keeps the product inside the expectation the *joint fully Bayesian* CAT. The difference between the two is thus an empirical question about how much the uncertainty in (θ_C,η) matters; study 2 answers it (Section 7).

Proposition 2 (Entropy–information decomposition) *For every item i,*

$$U_{\mathrm{EIG}}(i) = I(\theta; y_i \mid \mathcal{D}_t) = \underbrace{h(\bar{p}_i)}_{U_{\mathrm{H}}(i)} - \mathbb{E}_{\pi_t}\big[h(P_i(\theta))\big], \qquad \bar{p}_i = \mathbb{E}_{\pi_t}[P_i(\theta)], \tag{4}$$

so that $0 \le U_{\mathrm{EIG}}(i) \le U_{\mathrm{H}}(i)$, with equality on the right if and only if the item is deterministic given θ, π_t-almost surely.

Proof $I(\theta; y_i \mid \mathcal{D}_t) = H(y_i \mid \mathcal{D}_t) - H(y_i \mid \theta, \mathcal{D}_t)$. Given $\mathcal{D}_t$, y_i is Bernoulli with mean $\bar{p}_i = \mathbb{E}_{\pi_t}P_i(\theta)$, so $H(y_i \mid \mathcal{D}_t) = h(\bar{p}_i)$; given θ it is Bernoulli$(P_i(\theta))$, so $H(y_i \mid \theta, \mathcal{D}_t) = \mathbb{E}_{\pi_t}h(P_i(\theta)) \ge 0$, with equality iff $P_i(\theta) \in \{0,1\}$ π_t-a.s. Non-negativity of mutual information gives the lower bound. For the plug-in model the item parameters are fixed; for the marginal model the response of the target to item i given θ is Bernoulli$(\bar{P}_i(\theta))$ because $\psi_i = (\log a_i, b_i)$ appears only in item i's likelihood and its posterior given the panel does not depend on the target's other responses, so the same argument applies with $P_i = \bar{P}_i$. ■

Predictive entropy over-values items whose outcome is uncertain *even for a known competence*: the subtracted term is the item's irreducible noise. EIG is the myopically optimal choice for information about θ; it is not claimed to be optimal over a whole test, for which greedy guarantees require additional structure (Golovin and Krause, 2011).

Proposition 3 (Local equivalence with Fisher information) *Let P_i be three times continuously differentiable with bounded derivatives near μ_t, with $P_i(\mu_t) \in (0,1)$, and let π_t concentrate, $\sigma_t \to 0$ with $\mathbb{E}_{\pi_t}|\theta-\mu_t|^3 = o(\sigma_t^2)$. Then*

$$U_{\mathrm{EIG}}(i) = \tfrac{1}{2}\,\sigma_t^2\,\mathcal{I}_i(\mu_t) + o(\sigma_t^2), \qquad \mathcal{I}_i(\theta) = \frac{P_i'(\theta)^2}{P_i(\theta)\{1-P_i(\theta)\}}, \tag{5}$$

while $U_{\mathrm{H}}(i) = h\big(P_i(\mu_t)\big) + O(\sigma_t^2)$.

Proof Assume $P_i \in C^3$ with bounded derivatives near $\mu = \mu_t$ and $P_i(\mu) \in (0,1)$. Write $\delta = \theta - \mu$, $P = P_i(\mu)$, $P' = P_i'(\mu)$, $P'' = P_i''(\mu)$. Taylor's theorem gives $P_i(\theta) = P + P'\delta + \frac{1}{2}P''\delta^2 + O(|\delta|^3)$, hence $\bar{p}_i = P + \frac{1}{2}P''\sigma^2 + o(\sigma^2)$ because $\mathbb{E}\delta = 0$, $\mathbb{E}\delta^2 = \sigma^2$ and $\mathbb{E}|\delta|^3 = o(\sigma^2)$. Since h is smooth on $(0,1)$, $h(\bar{p}_i) = h(P) + \frac{1}{2}h'(P)P''\sigma^2 + o(\sigma^2)$ and $\mathbb{E}\, h(P_i(\theta)) = h(P) + \frac{1}{2}h'(P)P''\sigma^2 + \frac{1}{2}h''(P)P'^2\sigma^2 + o(\sigma^2)$. (The expansions hold on $|\delta| \le \varepsilon$ for a fixed small ε; since h and P_i are bounded, the contribution of $|\delta| > \varepsilon$ is at most a constant times $\Pr(|\delta| > \varepsilon) \le \mathbb{E}|\delta|^3/\varepsilon^3 = o(\sigma^2)$.) Subtracting and using $h''(p) = -1/\{p(1-p)\}$, $U_{\mathrm{EIG}}(i) = \frac{1}{2}\sigma^2 P'^2/\{P(1-P)\} + o(\sigma^2) = \frac{1}{2}\sigma^2\mathcal{I}_i(\mu) + o(\sigma^2)$. The statement for U_{H} is the first expansion. ■

Corollary 4 (Entropy is blind to discrimination) *Under the 2PL model,*

$$\mathcal{I}_i(\mu) = a_i^2 P_i(\mu)\{1 - P_i(\mu)\}.$$

Two items with the same success probability $P_i(\mu_t) = p$ are tied by predictive entropy while their information ratio is $(a_1/a_2)^2$, so the two criteria rank them differently whenever $a_1 \neq a_2$. When all items share one discrimination, both $h(P_i(\mu_t))$ and $\mathcal{I}_i(\mu_t)$ are strictly decreasing in $|P_i(\mu_t) - \frac{1}{2}|$, hence entropy sampling, maximum Fisher information and EIG induce the same ranking to leading order: under a common discrimination, entropy sampling is optimal by accident.

Proof For the 2PL, $P_i' = a_i P_i(1 - P_i)$, so $\mathcal{I}_i = a_i^2 P_i(1 - P_i)$. With a common a, $\mathcal{I}_i$ and $h(P_i)$ are both strictly decreasing functions of $|P_i - \frac{1}{2}|$, hence order-equivalent. ■

The corollary gives a sufficient condition for agreement (a common discrimination) and a constructive source of disagreement (ties in p with unequal a); it does not claim that equal discriminations are necessary for two rankings to coincide on a particular bank. Proposition 3 also implies that EIG, posterior-weighted Fisher information and maximum Fisher information agree to first order; differences between them are early-test phenomena (Chen et al., 2000). The decisive difference comes from calibration uncertainty.

Proposition 5 (Information cap under calibration uncertainty) *Let G be a continuous distribution function (logistic for the 2PL, Φ for the normal ogive), let the item response function be $P_\psi(\theta) = G(a(\theta - b))$, and let the difficulty given the panel be $b \mid a \sim \mathcal{N}(\bar{b}, s^2)$. Then the Fisher information about θ carried by a response under the expected response function satisfies, for every θ and every a,*

$$\bar{\mathcal{I}}(\theta) = \frac{\bar{P}'(\theta)^2}{\bar{P}(\theta)\{1 - \bar{P}(\theta)\}} \le \frac{2}{\pi s^2}, \tag{6}$$

and the bound is attained in the limit $a \to \infty$. For the normal ogive the expected response function is exactly $\Phi\big(a^{\mathrm{eff}}(\theta - \bar{b})\big)$ with $a^{\mathrm{eff}} = a/\sqrt{1 + a^2 s^2} < 1/s$.

Proof Fix $a > 0$ and let L have distribution function G, independent of $b \sim \mathcal{N}(\bar{b}, s^2)$ (conditionally on a). Since $G(a(\theta - b)) = \Pr(L/a + b \le \theta \mid b)$, the expected response function

is $\bar{P}(\theta) = \Pr(L/a + b \le \theta)$, so a response can be represented as $y = \mathbf{1}\{L/a + b \le \theta\}$. Consider the augmented observation (y, L). The distribution of L does not depend on θ, so its Fisher information is $\mathcal{I}_{(y,L)}(\theta) = \mathbb{E}_L[\mathcal{I}_{y|L}(\theta)]$. Given $L = \ell$, y is Bernoulli with success probability $\Pr(b \le \theta - \ell/a) = \Phi(u)$, $u = (\theta - \ell/a - \bar{b})/s$, whose information is $s^{-2}\phi(u)^2/\{\Phi(u)(1 - \Phi(u))\} \le s^{-2} \cdot \frac{2}{\pi}$, the maximum being attained at $u = 0$ where $\phi(0)^2 = (2\pi)^{-1}$ and $\Phi(0)(1 - \Phi(0)) = \frac{1}{4}$. Hence $\mathcal{I}_{(y,L)}(\theta) \le 2/(\pi s^2)$. (For a difficulty posterior with any distribution function F and density f, the success probability given $L = \ell$ is $F(\theta - \ell/a)$, with information $f^2/\{F(1 - F)\}$ at that point, which gives the general bound stated after this proof.) The response y is a statistic of (y, L), and the Fisher information of a statistic never exceeds that of the data it is computed from: the score of y is the conditional expectation of the score of (y, L) given y, whose variance is at most the variance of the latter (Zamir, 1998). Therefore $\bar{\mathcal{I}}(\theta) = \mathcal{I}_y(\theta) \le \mathcal{I}_{(y,L)}(\theta) \le 2/(\pi s^2)$. If a is itself uncertain, the same argument conditions on (a, L). As $a \to \infty$, $L/a \to 0$ and $\bar{P}(\theta) \to \Phi((\theta - \bar{b})/s)$, whose information at $\theta = \bar{b}$ is $2/(\pi s^2)$, so the bound is sharp. For the normal ogive, $L \sim \mathcal{N}(0, 1)$ and $L/a + b \sim \mathcal{N}(\bar{b}, a^{-2} + s^2)$, which gives $\bar{P}(\theta) = \Phi\big(a^{\text{eff}}(\theta - \bar{b})\big)$ with $a^{\text{eff}} = a/\sqrt{1 + a^2 s^2}$ and maximal information $\frac{2}{\pi}(a^{\text{eff}})^2 < 2/(\pi s^2)$. ■

The bound holds exactly for the logistic 2PL used in this paper; no probit approximation is involved. In words, the proof represents a response as $y = \mathbf{1}\{L/a + b \le \theta\}$ with L distributed as G independently of b: given L, the response is a probit item with discrimination $1/s$, whose information never exceeds $2/(\pi s^2)$, and discarding L cannot increase Fisher information (Zamir, 1998). A unit test checks the bound numerically for the logistic link over a grid of (a, s, θ). The Gaussian form of the difficulty posterior enters only through the conditional item given L: the same argument gives $\bar{\mathcal{I}}(\theta) \le \sup_u f(u)^2/\{F(u)(1 - F(u))\}$ for any difficulty posterior with distribution function F and density f (for a logistic posterior with standard deviation s this is $\pi^2/(12s^2) \approx 0.82/s^2$, against $2/(\pi s^2) \approx 0.64/s^2$ for a Gaussian one), whereas a posterior concentrated on a few points gives no useful bound. Our grid posteriors are discrete approximations of smooth posteriors, so for them the Gaussian cap indicates the order of magnitude rather than a guarantee.

Corollary 6 (Averaging information is not integrating uncertainty) *For any item-parameter posterior, the information of the expected response function never exceeds the posterior-expected Fisher information,* $\bar{\mathcal{I}}_i(\theta) \le \mathbb{E}_{\pi_i}[\mathcal{I}_{i,\psi}(\theta)]$.

Proof For a binary response, $\mathcal{I} = P'^2/\{P(1 - P)\} = P'^2/P + P'^2/(1 - P)$. The map $(x, y) \mapsto x^2/y$ is jointly convex on $\mathbb{R} \times (0, \infty)$, so $\mathcal{I}$ is a jointly convex function of (P, P'). The expected response function and its derivative are the posterior averages $(\bar{P}_i, \bar{P}'_i) = \mathbb{E}_{\pi_i}[(P_{i,\psi}, P'_{i,\psi})]$, and Jensen's inequality gives $\bar{\mathcal{I}}_i(\theta) \le \mathbb{E}_{\pi_i}[\mathcal{I}_{i,\psi}(\theta)]$. ■

Criteria that average Fisher information over parameter uncertainty—the Bayesian maximum-information criterion of fully Bayesian adaptive testing (van der Linden and Ren, 2020; Fink et al., 2025) and Laplace-approximated variants (Mandujano Reyes, 2026)—can therefore be optimistic about poorly calibrated items and do not inherit the cap of Proposition 5. (Empirically this optimism did not change which items were worth asking; Sections 6.8 and 7.) Under the plug-in model the information of an item is $\hat{a}_i^2/4$ at its

difficulty, unbounded in the point estimate $\hat{a}_i$. Items whose responses are perfectly separated in a small panel (solved by every strong calibrator and by no weak one, or by none) are exactly those with inflated $\hat{a}_i$ and large s_i; plug-in selection therefore concentrates on them, the classical *capitalization on calibration error* (van der Linden and Glas, 2000; Patton et al., 2013). The expected response function removes the incentive automatically: its information is capped—by $2/(\pi s_i^2)$ for a Gaussian difficulty posterior, by the general bound above otherwise—whatever $\hat{a}_i$ is. A shrinkage prior on a_i limits the inflation of $\hat{a}_i$, but not the uncertainty s_i about where the item sits.

Myopic score-targeted design. Practitioners often care about the full-bank accuracy $A = \frac{1}{I}\sum_i y_i$ rather than about θ. Let U be the set of unobserved items and $S_U = \sum_{k\in U} y_k$. The score-targeted criterion (EVR) asks the item that minimizes the expected posterior variance of S_U after one more response.

Proposition 7 (One-step score-targeted design) *For $i \in U$ let*

$$v_i = \bar{p}_i(1-\bar{p}_i), \qquad M_{-i}(\theta) = \sum_{k\in U\setminus\{i\}} P_k(\theta), \qquad c_i = \mathrm{Cov}_{\pi_t}\big(P_i(\theta), M_{-i}(\theta)\big).$$

The expected one-step variance reduction obtained by asking i is exactly

$$R_i = \mathrm{Var}(S_U \mid \mathcal{D}_t) - \mathbb{E}_{y_i}\big[\mathrm{Var}(S_{U\setminus\{i\}} \mid \mathcal{D}_t, y_i)\big] = \frac{(v_i + c_i)^2}{v_i}. \tag{7}$$

Under the conditions of Proposition 3, and if $P_i'(\mu_t)M_{-i}'(\mu_t) \neq 0$, then

$$c_i = \sigma_t^2 P_i'(\mu_t)M_{-i}'(\mu_t)\{1+o(1)\}, \qquad R_i = v_i + 2c_i + \sigma_t^4 M_{-i}'(\mu_t)^2\,\mathcal{I}_i(\mu_t)\{1+o(1)\}.$$

Proof Write $S_U = y_i + S_{-i}$. By the law of total variance, $\mathbb{E}_{y_i}[\mathrm{Var}(S_{-i} \mid y_i)] = \mathrm{Var}(S_{-i}) - \mathrm{Var}(\mathbb{E}[S_{-i} \mid y_i])$, so $R_i = \mathrm{Var}(y_i) + 2\,\mathrm{Cov}(y_i, S_{-i}) + \mathrm{Var}(\mathbb{E}[S_{-i} \mid y_i])$ (all moments conditional on $\mathcal{D}_t$). Here $\mathrm{Var}(y_i) = v_i$. Responses are conditionally independent given θ, so $\mathrm{Cov}(y_i, S_{-i}) = \mathrm{Cov}_{\pi_t}(P_i(\theta), M_{-i}(\theta)) = c_i$. Because y_i is binary, $\mathbb{E}[S_{-i} \mid y_i]$ is affine in y_i with slope c_i/v_i, so $\mathrm{Var}(\mathbb{E}[S_{-i} \mid y_i]) = c_i^2/v_i$ and $R_i = (v_i + c_i)^2/v_i$. Under the assumptions of Proposition 3, write $P_i(\theta) = P + P'\delta + r_1(\delta)$ and $M_{-i}(\theta) = M + M'\delta + r_2(\delta)$ with $|r_k(\delta)| \le C\delta^2$ (derivatives at μ, as in the proof of Proposition 3). Since $\mathbb{E}\delta = 0$, $c_i = P'M'\sigma^2 + P'\,\mathbb{E}[\delta r_2] + M'\,\mathbb{E}[\delta r_1] + \mathrm{Cov}(r_1, r_2)$; the last three terms are $O(\mathbb{E}|\delta|^3 + \mathbb{E}\delta^4) = o(\sigma^2)$ because the ability grid is bounded, so that $\mathbb{E}\delta^4 \le C'\,\mathbb{E}|\delta|^3$. Hence $c_i = \sigma^2 P'M'\{1+o(1)\}$ when $P'M' \neq 0$, and with $v_i = P(1-P)\{1+O(\sigma^2)\}$, $c_i^2/v_i = \sigma^4 M'^2 P'^2/\{P(1-P)\}\{1+o(1)\} = \sigma^4 M'^2 \mathcal{I}_i(\mu)\{1+o(1)\}$. ■

The first term of R_i is a monotone transform of predictive entropy (own outcome uncertainty); the last is Fisher information weighted by the squared sensitivity of the remaining bank to θ. One-step score-targeted selection therefore behaves like uncertainty sampling once competence is pinned down ($\sigma_t^2 M' \to 0$) and like information sampling while it is not; because M' grows with the number of unobserved items, the information regime lasts longer on larger banks (Section 6.3 finds this tendency). "Information-optimal" is thus relative to an estimand and to a one-step horizon: EIG maximizes the next response's information about competence, and R_i the next response's reduction of score variance.

Table 1: Real response matrices. Verdict: one deterministic evaluation per system and item (single) or a recorded pass rate over several samples binarized at one half (aggregate; TruthfulQA: continuous MC2 score binarized). J systems, F heuristic families, I items; accuracy range across systems; items solved by no system. Role: confirmatory in studies 1 and 2, or development bank used only to design the protocols (Sections 5.2 and 5.3). SE: software engineering.

Bank	Role	Domain	Oracle	Verdict	J	F	I	Accuracy	Unsolved
GSM8K	1, 2	math	numeric match	single	395	227	1319	0.00–0.77	13
ARC-C	1, 2	multiple choice	log-lik. choice	single	395	227	1172	0.26–0.73	51
HellaSwag	1, 2	multiple choice	log-lik. choice	single	395	227	2000	0.31–0.90	43
WinoGrande	1, 2	multiple choice	log-lik. choice	single	395	227	1267	0.59–0.87	1
MMLU	1, 2	multiple choice	log-lik. choice	single	395	227	2000	0.26–0.80	0
SWE-bench V.	1, 2	agentic SE	repository tests	single	175	123	500	0.00–0.79	26
LiveCodeBench	1, 2	code generation	hidden tests	aggregate	28	15	1055	0.23–0.87	46
CRUXEval-O	1, 2	code reasoning	execution	aggregate	31	12	800	0.22–0.78	98
SWE-bench Lite	dev.	agentic SE	repository tests	single	84	49	300	0.00–0.60	35
CRUXEval-I	dev.	code reasoning	execution	aggregate	31	12	800	0.14–0.78	59
TruthfulQA	dev.	multiple choice	log-lik. choice	aggregate	395	227	817	0.30–0.73	47

5 Experimental design

5.1 Data

We use public item-level verdicts of real systems (Table 1). Open LLM Leaderboard (Beeching et al., 2023) verdicts for 395 models on GSM8K (Cobbe et al., 2021), ARC-Challenge (Clark et al., 2018), HellaSwag (Zellers et al., 2019), WinoGrande (Sakaguchi et al., 2020), MMLU (Hendrycks et al., 2021) and TruthfulQA (Lin et al., 2022) are taken from the release of Maia Polo et al. (2024); HellaSwag and MMLU are uniformly subsampled to 2,000 items with a fixed seed, and TruthfulQA's continuous MC2 score is binarized at 0.5. SWE-bench Verified and Lite (Jimenez et al., 2024) verdicts (instance resolved by the official harness) come from the public submissions repository (175 and 84 systems). LiveCodeBench (Jain et al., 2025) pass rates for 28 models on 1,055 problems and CRUXEval (Gu et al., 2024) pass rates of temperature-0.2 samples for 31 models are binarized at one half; these two banks therefore carry recorded aggregate verdicts (Section 3), the other six deterministic ones. Every source is pinned to a git commit and every processed matrix carries a SHA-256 hash.

Synthetic banks. Three simulated banks with known abilities complement the real data (300 systems, 600 items each, $\theta \sim \mathcal{N}(0,1)$, $\log a \sim \mathcal{N}(0, 0.5^2)$, $b \sim \mathcal{N}(0, 1.5^2)$): a well-specified 2PL bank; a bank in which 60% of the items have a guessing floor of 0.25; and a bank in which half of the items depend on a second ability dimension correlated 0.6 with the first. The fitted model is always the unidimensional 2PL. Synthetic results never enter a confirmatory decision.

Families and strict clusters. Systems that share an underlying model are not independent. *Heuristic families* are defined by organization namespace (Open LLM Leaderboard), by model line including reasoning-effort and version variants (LiveCodeBench, CRUXEval), and, for SWE-bench, by the set of underlying LLMs declared in a submission's metadata

(two agentic systems are related if they share any LLM). Heuristic families miss renamed copies, merges and fine-tunes across organizations, so study 2 uses *strict clusters*: connected components of the graph that joins two systems if they share a heuristic family or if the correlation of their standardized 2PL residuals is at least 0.5—they share at least half of their idiosyncratic error pattern beyond what ability explains. Residuals are pooled over banks that share systems (the six Open LLM Leaderboard banks; CRUXEval-O and -I). Agreement measures such as Cohen's κ were not usable for this purpose: on HellaSwag, $\kappa \geq 0.8$ chains 353 of 395 systems into one component because all strong models agree on most items. The strict clusters number 176 on the Open LLM Leaderboard (largest 39 systems), 92 on SWE-bench Verified, 15 on LiveCodeBench and 12 on CRUXEval.

5.2 Study 1: frozen plan, leave-family-out replay

For each bank, up to 60 targets were drawn stratified by accuracy decile (all systems for LiveCodeBench and CRUXEval). For each target and replicate $r = 1, \ldots, 5$, a holdout of 20% of the items was drawn and shared by all policies and panel sizes; for each panel size $N \in \{4, 8, 16, 32, 64\}$ a panel was drawn outside the target's heuristic family. The twelve policies of Section 3 (five criteria under both models, PWFI under the marginal model only, plus Naive) were replayed on the target's recorded verdicts with a budget of $T = 100$ items. All random streams derive from one master seed by hashing (bank, purpose, target, replicate, N, policy).

The protocol was designed on development banks only (SWE-bench Lite, CRUXEval-I, TruthfulQA, synthetic 2PL; Section 5.4), where one problem appeared: with fixed weakly-informative priors, items solved by no calibrator were shrunk towards an arbitrary difficulty, so empirical-Bayes hyperpriors were adopted for every policy. The plan (banks, protocol, endpoints, four hypotheses, tests, multiplicity control, decision rule), the analysis code and the data manifests were then hashed and committed before any confirmatory run, and the run wrapper refuses to start if a frozen file changes. No public registry or external time stamp was used, so we call this a *frozen analysis plan* rather than a preregistration (Nosek et al., 2018); the git history shows the freeze commit before every confirmatory artifact.

Hypotheses (primary panel size $N = 8$): **H1** EIG:mg has lower integrated holdout log loss than Entropy:mg; **H2** EIG:mg has lower integrated holdout log loss than MFI:pi; **H3** EIG:mg has lower integrated score error than Naive subsampling; **H4** EIG:mg's 90% score intervals are better calibrated than MFI:pi's ($|\text{coverage} - 0.9|$). Each contrast is a two-sided Wilcoxon signed-rank test (Wilcoxon, 1945) on paired per-target values (replicates averaged within target), with Holm correction (Holm, 1979) over the whole family (8 banks $\times$ panel sizes $\times$ 4 hypotheses); a hypothesis is *supported* if, at $N = 8$, the adjusted test favours EOST-M in at least five of the eight banks and the baseline in none. Because targets of one family are not independent, every study-1 contrast is also re-analysed post hoc with heuristic families and with strict clusters as units (cluster means, the same tests and decision rule).

5.3 Study 2: independent targets and strong competitors

Study 2 was designed after an external review of study 1 and frozen before execution (plan, hypotheses, cluster file and code hashed and committed; the plan and its hash were also

sent to the first author, outside the execution environment, before the first run). It differs from study 1 in four ways. *Targets* are one randomly chosen representative per strict cluster (systems that were not study-1 targets preferred), at most 40 per bank, stratified by accuracy, so targets are independent units; *panels* ($N \in \{4, 8, 16\}$, three replicates, new seeds) exclude the target's whole strict cluster. *Competitors* are the strongest we could identify:

- MFI:1PR, a regularized Rasch-type plug-in CAT (one common discrimination with $\log a \sim \mathcal{N}(0, 0.05^2)$, empirical-Bayes prior on b), and EIG:1PRMG, the same model integrated (both found post hoc in study 1);
- FB:BMI and FB:EIG, a *posterior-draw marginal Bayesian CAT* (PDM Bayesian CAT; called the fully Bayesian CAT in the frozen plan): a Gibbs sampler for the hierarchical 2PL of the panel (abilities, item parameters on the grid, hyperparameters, with a translation move for the overall location), eight retained draws, and the target's posterior computed with each item's response function averaged over the draws separately (the target of van der Linden and Ren, 2020; Lemma 1 shows that this integrates the calibrators' abilities and the hyperprior item by item, not jointly); selection by Bayesian maximum information (posterior-weighted expected Fisher information, the strategy recommended by Fink et al., 2025) or by EIG ($N \in \{4, 8\}$);
- RANDOM:MG, random items scored with the EOST-M model, which isolates the value of adaptivity from that of the calibration;
- ATLAS, an ATLAS-style CAT (Li et al., 2026): 3PL with the lower asymptote fixed at the chance rate of the item format (ARC-C, HellaSwag, MMLU 0.25, WinoGrande 0.5, others 0), first item closest in difficulty to 0, then a uniform draw among the five most informative items at the expected a posteriori (EAP) ability (Kingsbury and Zara, 1989);
- TINYBENCHMARKS (Maia Polo et al., 2024): anchor items by K-Means on the panel's item embeddings and the gp-IRT estimator, at budgets of 25, 50 and 100 items; MINCE (Das et al., 2026): Monte Carlo sizing on the panel and the plain mean of one frozen random subset, at the same budgets and at its own size $n^\star$;
- MFI:PI and NAIVE as in study 1, two empirical-Bayes variants of EOST-M (cross-fitted hyperprior; Student-t hyperprior family) and an exploratory mixed policy (every fourth item uniform at random).

Intervals of EOST-M and of the Rasch-type test are also reported after the leave-one-calibrator-out recalibration of Section 3. *Hypotheses* (all at $N = 8$, Holm over 11 superiority hypotheses $\times$ 8 banks; supported with ≥ 5 banks for EOST-M and none against): replications of H2–H4 with independent targets (R2–R4); EOST-M versus the Rasch-type plug-in (C1) and marginal (C2) tests on log loss; versus Random:mg on score error (C4); recalibrated versus raw EOST-M intervals (C5b) and versus Naive's Wilson intervals (C6) on the interval score; versus ATLAS-style CAT on score error (C7); versus tinyBenchmarks (C8) and MINCE (C9) on the absolute score error after 50 items. In addition, the PDM Bayesian CAT and EOST-M are declared practically equivalent (C3) if the 90% bootstrap interval of their relative log-loss difference lies within $\pm 1\%$ on at least six banks, and the recalibrated EOST-M interval is declared calibrated (C5a) if its mean coverage lies in $[0.85, 0.95]$ on

at least six banks. A *temporal component* on SWE-bench Verified uses the latest quarter of submissions by date as targets and only earlier submissions as calibrators ($N = 8$); the synthetic banks are a secondary component. During the run, before any result was inspected, an independent review of the code found two problems—tinyBenchmarks can use fewer anchors than its budget, and the burn-in of the Bayesian sampler was too short on some banks—whose post hoc corrections were specified in the research log before the results were looked at; the frozen run was completed unchanged and every frozen decision is reported as computed (Section 5.5, deviation xiii).

Implementation of the competitors. The *Bayesian CATs* run a Gibbs sampler on the panel's hierarchical 2PL: (1) every item's grid point is drawn from its exact conditional grid posterior; (2) every calibrator's ability is drawn by griddy Gibbs on 65 points spanning ± 0.8 around its current value; (2b) all abilities and difficulties and the difficulty mean are shifted by a common multiple of the difficulty grid step (at most 16 steps either way) drawn from its conditional, which only involves the ability prior (the likelihood and the difficulty hyperprior are invariant); shifts that would move any sampled difficulty off the grid are excluded, so the move is exact only within these bounds and rarely acts when many difficulties sit near the edge of the grid; (3) the hyperparameters are drawn from their conjugate conditionals (flat means, Jeffreys standard deviations). The chain starts at EOST-M's estimates; in the frozen plan, 8 draws are kept every 4 sweeps after a burn-in of 10 sweeps, and each contributes its exact conditional item posterior (Rao–Blackwellization). As noted above, this burn-in proved too short on some banks; the post hoc correction (60 sweeps) is described in Section 7.2. In 300-sweep chains on the development banks (`scripts/fb_mixing.py`), the translation move removed the slow mixing of the overall location: the lag-3 autocorrelation of the mean calibrator ability fell from 0.86–0.94 without the move to at most 0.16 with it. That location is genuinely uncertain—its posterior standard deviation was 0.44, 0.32 and 0.24 for $N = 4$, 8 and 16, close to $1/\sqrt{N}$ as implied by the $\mathcal{N}(0, 1)$ ability prior—but it shifts abilities and difficulties together. Scale hyperparameters mix more slowly (lag-3 autocorrelation of σ_a between 0.65 and 0.89), so the eight retained draws represent their uncertainty only roughly. Follow-up RF2 (Section 8.2) measured this with four long chains, and exploratory follow-ups added the missing moves.

The *ATLAS-style* test fits a 3PL with fixed lower asymptote by grid MAP given the panel's abilities and the empirical-Bayes prior. *tinyBenchmarks* uses K-Means (best of five k-means++ restarts) on $(a_i, a_i b_i)$ of the panel's plug-in 2PL fit, the item nearest to each centroid as anchor, cluster fractions as weights, the bounded maximum-likelihood ability for p-IRT, and $\lambda = \hat{b}^2/(\hat{v}/(4n) + \hat{b}^2)$ with $\hat{b}$ the leave-one-calibrator-out error of the IRT prediction and $\hat{v}$ the mean Bernoulli variance of the calibrators. *MINCE* draws 2,000 random subsets per candidate size (steps of 25) and stops when the 95th percentile of the worst-case drift over calibrators improves by less than one point.

5.4 From the pilot program to the frozen plans

Pilot program (historical). Before study 1, the framework (EOST in our project) was developed through a staged pilot program on math word problems with `Qwen3-8B` as target, greedy decoding and exact numeric oracles, including self-generated items with an independent oracle (the self-testing aspect of the framework). Every stage had an explicit

pass/fail gate fixed in advance. We report the stages *as recorded in the frozen project log*, including failed gates and the subsequent corrections, because they motivated the design of Sections 3–5. Stages S0–S4 were executed before the project's Docker-only rule; the S4B run of one model was executed in the project's earlier container runtime, which differs from the runtime of this paper (Section 5.6). Their raw artifacts are archived with SHA-256 hashes but they were *not* re-executed for this paper, and none of the paper's claims rests on them. Table 2 summarizes them.

Table 2: Pilot program (historical values from the frozen project log; not re-executed here). AUIC: area under the information curve; IG: expected-information-gain policy (EOST).

Stage	Setting	Recorded outcome and gate
S0	Synthetic latent-trait simulator, 30 seeds; IG vs. entropy, random, difficulty-matched	IG > entropy on entropy reduction (6.77 vs. 6.51; Holm $p = 2.8 \times 10^{-9}$) and AUIC (5.29 vs. 5.12; rank-biserial 1.0). **Pass**.
S1-R2	GSM-Symbolic, 45 items; 4 calibration conditions (two Qwen3 checkpoints × thinking on/off)	Conditions were not independent systems. **Gate fail**.
S1-R3	Same bank, regularized 2PL	IG AUIC 0.950 vs. entropy 0.923 (29/30 seeds); identical final holdout. Exploratory.
S2	LLM-generated items, procedural oracle (75 items, 5 families)	After fixing a numeric-tolerance bug: IG AUIC 1.224 vs. 1.139 (27/30 seeds, $p = 1.2 \times 10^{-4}$); gains early in the test, endpoint saturated, log-loss tail risk. Publication gate **fail**.
S3	Circularity: self-, external- and procedurally-generated items (45 matched triplets)	Self minus external accuracy 0.089 (95% CI $[0, 0.2]$, McNemar $p = 0.22$): no *detectable* circularity gap at this power (not equivalence).
S4	GSM8K, GSM-Hard, SVAMP, ASDiv (240 items); 4 calibration conditions	Macro AUIC IG 0.967 vs. entropy 0.858 (30/30 seeds). Original gate **fail** (predictive non-inferiority). An audit found the original Brier holdout was policy-dependent; with a common fixed holdout, integrated Brier 0.0936 vs. 0.0969 (24/30 seeds, $p = 3.1 \times 10^{-5}$), final Brier tie (16/30). GSM-Hard went the other way (Brier 0.195 vs. 0.183). 2PL weakly identified: 7 distinct b, 50.8% of a at the upper bound.
S4B	Expanding the calibration panel to independent systems	Six calibrators passed a post-hoc parser audit (parse failures within the project's internal 5% threshold); a seventh, community checkpoint produced 235/240 truncated generations—a generation-protocol failure, not a capability measurement—and is classified as invalid for calibration. Incomplete; not analysed.

Three lessons from the pilot shaped EOST v2. First, the information advantage of IG over predictive entropy was consistent across synthetic, self-generated and multi-benchmark settings, but it was concentrated early in the test, as Proposition 3 predicts. Second, predictive evaluation must use a holdout shared by all policies; the policy-dependent holdout of the original S4 analysis produced a spurious non-inferiority failure, which we keep reported as such. Third, and most important, a calibration panel of four conditions drawn from two checkpoints cannot identify a 2PL: half of the discriminations saturated at the optimizer bound, the textbook setting for capitalization on chance (Proposition 5). The confirmatory

Table 3: Development phase (exploratory; 30 targets, 3 replicates). The switch from fixed to empirical-Bayes (EB) item hyperpriors was decided here, before freezing the confirmatory plan.

			holdout log loss		score IAE		coverage	
Bank	prior	N	EIG:mg	MFI:pi	EIG:mg	MFI:pi	EIG:mg	MFI:pi
Synth. 2PL	fixed	4	0.5762	0.6410	0.0433	0.0543	0.78	0.55
SWE-bench Lite	fixed	4	0.4133	0.4110	0.0430	0.0406	0.74	0.69
CRUXEval-I	fixed	4	0.4833	0.5054	0.0342	0.0441	0.88	0.63
TruthfulQA	fixed	4	0.4215	0.4179	0.0335	0.0395	0.87	0.66
Synth. 2PL	fixed	8	0.5371	0.5696	0.0382	0.0427	0.77	0.59
SWE-bench Lite	fixed	8	0.3743	0.3773	0.0349	0.0292	0.74	0.75
CRUXEval-I	fixed	8	0.4460	0.4659	0.0315	0.0400	0.80	0.55
TruthfulQA	fixed	8	0.3726	0.3755	0.0264	0.0305	0.83	0.64
Synth. 2PL	fixed	64	0.4914	0.4931	0.0290	0.0275	0.77	0.80
SWE-bench Lite	fixed	64	0.3254	0.3267	0.0264	0.0250	0.80	0.81
TruthfulQA	fixed	64	0.3204	0.3214	0.0223	0.0237	0.70	0.64
Synth. 2PL	EB	4	0.5711	0.5884	0.0424	0.0475	0.87	0.78
SWE-bench Lite	EB	4	0.4036	0.4207	0.0393	0.0371	0.77	0.69
CRUXEval-I	EB	4	0.4833	0.5113	0.0365	0.0452	0.83	0.64
TruthfulQA	EB	4	0.4110	0.4323	0.0367	0.0392	0.77	0.64
Synth. 2PL	EB	8	0.5332	0.5403	0.0359	0.0407	0.92	0.82
SWE-bench Lite	EB	8	0.3649	0.3730	0.0296	0.0293	0.83	0.79
CRUXEval-I	EB	8	0.4450	0.4702	0.0326	0.0416	0.79	0.53
TruthfulQA	EB	8	0.3679	0.3851	0.0284	0.0308	0.78	0.65
Synth. 2PL	EB	64	0.4902	0.4915	0.0261	0.0273	0.89	0.87
SWE-bench Lite	EB	64	0.3228	0.3243	0.0237	0.0240	0.86	0.84
TruthfulQA	EB	64	0.3202	0.3223	0.0236	0.0248	0.67	0.63

study therefore targets exactly this small-panel regime, with independent systems, leave-family-out panels and calibration-marginal selection.

Development runs. The development runs on the development banks that shaped both protocols are summarized in Table 3; the switch to empirical-Bayes hyperpriors described in Section 5.2 was decided there, before the plan of study 1 was frozen.

Choosing the recalibration rule. Three leave-one-calibrator-out rules were implemented: multiplying the model half-width by the 90% quantile of the pseudo-targets' absolute standardized errors divided by $z_{0.95}$ (Rq) or by their root mean square (Rr), and replacing it by the 90% quantile of their absolute errors (Ra). On the development banks only, and by a selection rule written in the research log before that run, the rule whose mean coverage at $N \in \{4, 8\}$ was closest to 0.90 was chosen: Ra (0.83, against 0.80 for Rq and 0.80 for Rr; Table 4). The same run showed that no rule helps with four calibrators, which is why the recalibration hypotheses concern $N = 8$.

5.5 Deviations from the frozen plans

No deviation affected a confirmatory computation. For completeness: (i) the confirmatory and sensitivity runs were executed as Docker containers running exactly the frozen-plan gate and the commands of `./run confirm` and `./run sensitivity`; (ii) infrastructure restarts interrupted the confirmatory run twice while HellaSwag was in progress; completed banks were kept unchanged, and the remaining banks and the sensitivity analyses were run with a

Table 4: Choice of the interval recalibration on the development banks (before study 2; 20 targets or fewer, two replicates). Mean coverage of EOST-M's 90% score interval without recalibration and with the three leave-one-calibrator-out rules: quantile of standardized errors (Rq), their root mean square (Rr), quantile of absolute errors (Ra). The rule written in advance selects the rule with mean coverage closest to 0.90: Ra.

Bank	N	raw	Rq	Rr	Ra
CRUXEval-I	4	0.85	0.81	0.82	0.84
CRUXEval-I	8	0.84	0.88	0.89	0.92
TruthfulQA	4	0.77	0.75	0.77	0.77
TruthfulQA	8	0.70	0.75	0.75	0.82
SWE-bench Lite	4	0.80	0.74	0.74	0.77
SWE-bench Lite	8	0.76	0.82	0.82	0.87
Synth. 2PL	4	0.85	0.77	0.80	0.79
Synth. 2PL	8	0.86	0.85	0.85	0.85
mean		0.80	0.80	0.80	0.83

checkpointing wrapper (`scripts/run_benchmark_resumable.py`) that calls the same frozen functions with the same seeds but saves each finished target, so that a restart only loses the targets in flight; on a test bank this wrapper reproduced the frozen runner's output exactly (maximum absolute difference 0); (iii) in the one-parameter sensitivity analysis on HellaSwag, one target produced expected response probabilities exceeding 1 by floating-point rounding, which made the frozen score-targeted (EVR) computation overflow and stop the run; the sensitivity runs from that bank on were executed with a guard in the wrapper that clips these probabilities to $[0, 1]$ inside the EVR criterion, without modifying any frozen file; the guard is an exact no-op whenever the probabilities are valid, and it reproduced stored confirmatory and sensitivity outputs exactly (maximum absolute difference 0); the confirmatory runs had completed without it; (iv) unit tests for Propositions 5 and 7 and Corollary 6, the figure and table generator, the exploratory ranking analysis (Section 6.4) and the automatic check of the manuscript's qualitative claims (`scripts/check_claims.py`) were written after the freeze; (v) the planned one-parameter sensitivity analysis turned out to be uninformative because its joint posterior mode is degenerate (Section 6.7); it is reported but not interpreted, and a variant with an informative prior on the common discrimination was added post hoc and is labelled as such; (vi) the late-test comparisons of Section 6.3 (plug-in levelling off, entropy crossover) and the mechanism analysis of Figure 4 were not in the frozen plan and are labelled exploratory; (vii) two post hoc experiments were added after all planned runs—strong baselines (Section 6.8) and a regularization path of the discrimination prior (Section 6.8)—with decision rules recorded in the project's research log before running; they use the confirmatory targets, holdouts and panels, reproduce the recomputed confirmatory rows exactly, and do not enter the confirmatory decision rule; an exploratory decomposition of interval coverage into width and score error (Table 16) analyses the stored strong-baseline runs without new runs, with its interpretation rule also recorded before computing; (viii) the hash check of every frozen file was repeated after all runs (`PREREG_GATE=PASS`); (ix) after an external review, every study-1 contrast was re-analysed with families as units (post hoc;

Table 7), and a cost analysis and checks of the empirical-Bayes hyperprior were added (post hoc); (x) study 2 is a new study with its own frozen plan (`configs/study2_plan.md`), not a modification of study 1; its interval-recalibration rule was chosen on development banks by a rule written before that development run (Section 5.4); (xi) because no external time stamp was available, the paper uses "frozen analysis plan" instead of "preregistration" for both studies; (xii) the prospective live protocol could not be executed (no access to model weights in the execution environment) and was replaced by a temporal split of SWE-bench Verified submissions, specified in the study-2 plan. (xiii) While study 2 was running, and before any of its results was inspected, an independent review of its code found two problems and three points of wording, and the handling below was recorded in the research log before the results were looked at; the frozen run was completed unchanged and every frozen decision is reported as computed. First, with eight calibrators the items have at most 2^8 distinct response patterns, so tinyBenchmarks' K-Means can return fewer anchors than its budget, and the frozen code did not record how many it used; the number was recomputed post hoc with the same seeds (the recomputed errors reproduce the frozen ones exactly), and C8 is also reported on full-budget cells and matched on the number of items actually used (Section 7.3). Second, the burn-in of the fully Bayesian sampler (10 sweeps) was too short on some banks: the translation move is restricted to shifts that keep every sampled difficulty on the grid, so it rarely acts when many difficulties sit near the edge of the grid, and the overall location then drifted during the retained sweeps; both fully Bayesian arms were re-run post hoc with a burn-in of 60 sweeps at $N = 8$ and $N = 4$ (same targets, panels, holdouts and chains; with the frozen burn-in the re-run reproduces the frozen rows exactly), C3 is reported for both versions, and a joint fully Bayesian model was added as an exploratory arm. Third, the recalibration window is $[t/1.5, 1.5t]$ (the plan says "±50%"), the translation move is exact only within its bounds, and the direction of a significant contrast is taken from the sign of the mean difference although the test is rank-based (as in study 1); 0 of the 815 Holm-significant contrasts of the two studies have a mean difference and a rank-biserial correlation of opposite signs. (xiv) After a second external review, three robustness follow-ups were planned in the research log before they were run (Section 8): panels drawn under cluster rules that do not use the target's responses and under other residual thresholds (RF1; the recomputed frozen clusters reproduce every frozen row exactly), longer multi-chain runs of the Bayesian CATs with convergence diagnostics (RF2; on a subset of 48 panels), and a random-effects meta-analysis across benchmarks (RF3); none changes a frozen decision, and the frozen arm called the fully Bayesian CAT in the plan is called the posterior-draw marginal Bayesian CAT in the paper.

5.6 Computation, reproducibility and environment

All computations of this paper are executed by `./run` inside one pinned CPU-only container image; scientific containers have no network access, and data were fetched beforehand with sparse git checkouts of pinned commits. Study 1 (172,620 adaptive tests of 100 items) required about 3.8 h on a two-core machine and study 2 about 3 h, plus 2.6 h for its post hoc corrections.

- *Runtime image* `eost-runtime:cpu`, image ID `sha256:5b21ccfeb1fd...` (full ID in `configs/ENVIRONMENT.json`), built on 2026-10-29 from `ubuntu:24.04` pinned by

digest `sha256:b97134452ef5...`; Python 3.12.3, NumPy 2.5.4, SciPy 1.18.1, pandas 3.1.0; complete package list in `configs/ENVIRONMENT.json` and `requirements.lock`. Study 1, its post hoc analyses, study 2 and the post hoc corrections of study 2 all ran in this image; the post hoc recomputation of study 1 one week after the frozen runs reproduced the stored rows bit for bit, and the correction runs of study 2 reproduce the frozen Bayesian-CAT rows exactly when given the frozen burn-in.

- *Paper image*: TeX Live, image ID `sha256:b8c7f599855b...`.
- *Pilot image* (historical, Section 5.4): an earlier runtime of the project that shares the tag `eost-runtime:cpu` but is a different image (`python:3.12-slim-bookworm`, Python 3.12.15, PyTorch 2.7.0+cpu, image ID `sha256:8423de84e45c...`, on a different workstation); no result of this paper was computed in it.
- *Sources* pinned to git commits: SWE-bench experiments `40f164d`, tinyBenchmarks `e9a8b10`, LiveCodeBench site `f186ab5`, CRUXEval `190faf1`, and, for the live protocol only, GSM8K `3101c7d`. Processed matrices, both frozen plans and every artifact are SHA-256 hashed.
- *Tests and checks*: unit tests cover the analytic gradients, parameter recovery, the empirical-Bayes estimator, the information cap for the logistic link (Proposition 5), the small-variance limit of EIG, the Jensen inequality of Corollary 6, the EVR closed form and its numerical guard, the equivalence of the study-2 replay and live engines with the frozen replay, the Gibbs sampler, its post hoc copy and the reduction baselines. Every number in the text is a macro generated from the artifacts, and every qualitative statement about results in Sections 6–11 is encoded as a predicate that `scripts/check_claims.py` re-evaluates on the artifacts (`CLAIMS_GATE=PASS`; `artifacts/CLAIMS.json`).

6 Study 1: calibration-marginal versus plug-in adaptive testing

6.1 Decisions under the frozen plan, and with dependent targets

Table 5 reports the four frozen contrasts at $N = 8$ and Table 6 all panel sizes. **H2 (calibration-marginal EIG vs. plug-in Fisher CAT)** is **supported**: at $N = 8$ EOST-M lowered the integrated holdout log loss in 8 of 8 benchmarks (median improvement 5.2%, range 2.4% to 9.6%), Holm-significantly in 7 and never significantly in favour of the baseline. At $N = 4$ the median improvement was 7.1% (8/8 significant), at $N = 64$ only 1.1%. **H3 (EOST-M vs. naive subsampling)** is **supported**: the integrated score error fell by a median of 54.4% (7 of 8 benchmarks significant). **H4 (interval calibration)** is **supported**: EOST-M reduced the miscalibration $|\text{coverage} - 0.9|$ of plug-in Fisher CAT by a median of 44.2% (7 of 8 significant). **H1 (information vs. entropy)** is **not supported**: under the same calibration-marginal model, EIG lowered held-out log loss in 8 of 8 benchmarks, but by a median of only 1.9% (range 0.4% to 5.5%), Holm-significantly in 4, short of the frozen threshold. At the other panel sizes the same contrast was significant more often (7/8 benchmarks at $N = 4$, 5/6 at $N = 64$; Table 6), never in favour of entropy, but always with median gains of a few per cent. The sign predicted by the EOST thesis is therefore observed throughout, but its size on held-out prediction is small once calibration uncertainty

Table 5: Study 1 at the primary panel size $N = 8$: relative improvement (%) of EOST-M over the baseline of each hypothesis (log loss for H1–H2, score error for H3, |coverage − 0.9| for H4); * Holm-significant in favour of EOST-M (none in favour of a baseline). Last row: benchmarks significant for EOST-M / for the baseline, and the frozen decision. Full statistics in Table 8.

Benchmark	targets	H1 vs Entropy:mg	H2 vs MFI:pi	H3 vs Naive	H4 vs MFI:pi
GSM8K	60	+2.7*	+2.4	+16.8	+11.6
ARC-C	60	+0.4	+4.7*	+61.0*	+41.6*
HellaSwag	60	+5.5*	+9.6*	+68.4*	+46.8*
WinoGrande	60	+3.1*	+4.9*	+53.4*	+57.3*
MMLU	60	+2.9*	+5.0*	+55.3*	+59.0*
SWE-bench V.	60	+0.7	+6.1*	+46.9*	+31.7*
LiveCodeBench	28	+1.0	+6.4*	+51.3*	+48.1*
CRUXEval-O	31	+0.5	+5.4*	+57.7*	+33.4*
decision		4/0 not supported	7/0 supported	7/0 supported	7/0 supported

Table 6: Confirmatory tests at every panel size: benchmarks with a Holm-significant difference favouring EOST-M / favouring the baseline / tested, and (in parentheses) the median relative improvement across benchmarks.

	$N = 4$	$N = 8$	$N = 16$	$N = 32$	$N = 64$
H1	7/0/8 (+2.5%)	4/0/8 (+1.9%)	7/0/8 (+1.5%)	5/0/6 (+2.1%)	5/0/6 (+1.8%)
H2	8/0/8 (+7.1%)	7/0/8 (+5.2%)	3/0/8 (+3.6%)	1/0/6 (+1.9%)	0/0/6 (+1.1%)
H3	7/0/8 (+45.9%)	7/0/8 (+54.4%)	8/0/8 (+60.9%)	6/0/6 (+62.5%)	6/0/6 (+62.1%)
H4	7/0/8 (+56.4%)	7/0/8 (+44.2%)	4/0/8 (+29.4%)	3/0/6 (+21.5%)	1/0/6 (+10.3%)

is integrated (Corollary 4); on score estimation the gap is larger (Table 9). GSM8K is the one benchmark on which the H2–H4 contrasts, while pointing the same way, did not reach significance. It is also the confirmatory bank on which an eight-system panel leaves the fewest items unlocated (21% of items with identical panel verdicts, against a median of 39% elsewhere), because its leaderboard population spans the whole accuracy range, from format failures (15% of the models score below 5%) to strong models; by the mechanism of Figure 4, there is less for marginalization to correct. Math word problems were also the heterogeneous case of the pilot program (GSM-Hard, Section 5.4).

Targets of one family are not independent, and on the Open LLM Leaderboard they are numerous: hundreds of systems are fine-tunes or merges of a few base models. Table 7 repeats every contrast with families as units. With heuristic families the frozen decisions are unchanged. With strict clusters—which also merge near-duplicates found from the verdicts and leave between 12 and 50 units per benchmark—H3 and H4 remain supported, but H2 is significant on 4 of 8 benchmarks only, below the frozen threshold of five, although its median effect is unchanged (5.4%, 95% cluster-bootstrap interval 4.3–6.4%). The effect of marginalization on prediction is therefore not in doubt, but the per-benchmark significance

Table 7: Study 1 with dependent targets taken into account (post hoc). Benchmarks with a Holm-significant difference favouring EOST-M / favouring the baseline at $N = 8$, with the frozen decision in parentheses, when the unit of analysis is the target (frozen plan), the heuristic family or the strict cluster (targets averaged within unit; same 144-test Holm family). Last column: median relative improvement over benchmarks with strict clusters as units, and its 95% cluster-bootstrap interval.

	Contrast	target	heuristic family	strict cluster	median % [95% CI]
H1	EIG:mg vs Entropy:mg (log loss)	4/0 (no)	4/0 (no)	4/0 (no)	1.8 [1.3, 2.5]
H2	EIG:mg vs MFI:pi (log loss)	7/0 (yes)	5/0 (yes)	4/0 (no)	5.4 [4.3, 6.4]
H3	EIG:mg vs Naive (score IAE)	7/0 (yes)	7/0 (yes)	7/0 (yes)	55.4 [50.8, 57.6]
H4	EIG:mg vs MFI:pi ($\lvert\text{cov} - 0.9\rvert$)	7/0 (yes)	5/0 (yes)	5/0 (yes)	44.6 [37.0, 50.5]

claimed with targets as units overstates the evidence; study 2 was designed with one target per cluster to settle the question.

6.2 Where the gain comes from

Because EOST-M differs from plug-in Fisher CAT in two respects—the criterion and the measurement model—Table 9 crosses them. At $N = 8$, replacing plug-in parameters by expected response functions improves held-out log loss by a median of +4.4% even with entropy selection, whereas switching the criterion under plug-in parameters changes little (entropy −1.8%, EIG +0.0%). Combining both gives +5.2%. EIG beat entropy on log loss in 8 of 8 benchmarks under the marginal model and in 7 under plug-in parameters. For the score, the criterion matters more: entropy selection is the worst adaptive rule under both models (score IAE −9.2% relative to MFI:pi under the marginal model), while EIG:mg improves it by +7.5%. Under the marginal model, the four information-based criteria (MFI, PWFI, EIG and EVR) are practically indistinguishable: their mean held-out log losses differ by at most 0.1% and their score errors by at most 3% (Table 10), as the first-order equivalence of Proposition 3 predicts. Most of the gain over current practice thus comes from integrating calibration uncertainty; selecting by information rather than by entropy is what turns that model into an efficient test, and EIG is the principled member of an equivalence class rather than a uniquely better rule. The gain of EOST-M over plug-in Fisher CAT also decreases monotonically with the panel size on every benchmark—largest with four calibrators, nearly gone with 64 (Figure 2)—as Proposition 5 predicts, since the uncertainty of item difficulties shrinks as calibrators are added. With four calibrators, 56% of the items of a confirmatory bank were answered identically by every calibrator, against 15% with 64 (Figure 3a); the difficulty of these items is located only by the prior. Under the empirical-Bayes prior, plug-in discriminations above 4 are practically absent with small panels (0% of items at $N = 4$) and appear only with large ones (10% at $N = 64$; panel c), so the overconfidence of plug-in testing at small N comes mainly from treating poorly located difficulties as known rather than from extreme discriminations: the expected response function shrinks the median effective discrimination from 1.99 to 0.98 at $N = 4$ (panel b), as Proposition 5 prescribes. Across benchmarks and panel sizes, the log-loss gain of EOST-M

Table 8: Study 1 in full at the primary panel size $N = 8$. Per-target means over 5 replicates; n targets; improvement is relative to the baseline (positive favours EOST-M); W/L wins/losses over targets; r_{rb} matched-pairs rank-biserial; p two-sided Wilcoxon, Holm-adjusted over the whole family; ↑/↓ significant in favour of EOST-M/baseline.

Benchmark	n	Baseline	EOST-M	Improv.	W/L	r_{rb}	p_{Holm}
H1: EIG:mg vs Entropy:mg (log loss)							
GSM8K	60	0.3885	0.3779	+2.7%	52/8	+0.88	$< 10^{-4}$ ↑
ARC-C	60	0.3478	0.3464	+0.4%	39/21	+0.37	0.369
HellaSwag	60	0.1880	0.1777	+5.5%	55/5	+0.90	$< 10^{-4}$ ↑
WinoGrande	60	0.3792	0.3674	+3.1%	52/8	+0.78	$< 10^{-4}$ ↑
MMLU	60	0.4817	0.4677	+2.9%	53/7	+0.89	$< 10^{-4}$ ↑
SWE-bench V.	60	0.3838	0.3811	+0.7%	41/19	+0.35	0.560
LiveCodeBench	28	0.2813	0.2784	+1.0%	16/12	+0.32	1.000
CRUXEval-O	31	0.3838	0.3817	+0.5%	22/9	+0.31	1.000
decision rule: 4/8 favour EOST-M, 0 favour baseline ⇒ **not supported**							
H2: EIG:mg vs MFI:pi (log loss)							
GSM8K	60	0.3874	0.3779	+2.4%	33/27	+0.28	1.000
ARC-C	60	0.3634	0.3464	+4.7%	45/15	+0.70	0.0002 ↑
HellaSwag	60	0.1967	0.1777	+9.6%	41/19	+0.64	0.0010 ↑
WinoGrande	60	0.3865	0.3674	+4.9%	38/22	+0.52	0.020 ↑
MMLU	60	0.4922	0.4677	+5.0%	44/16	+0.63	0.0012 ↑
SWE-bench V.	60	0.4060	0.3811	+6.1%	48/12	+0.78	$< 10^{-4}$ ↑
LiveCodeBench	28	0.2973	0.2784	+6.4%	19/9	+0.70	0.032 ↑
CRUXEval-O	31	0.4036	0.3817	+5.4%	26/5	+0.81	0.0010 ↑
decision rule: 7/8 favour EOST-M, 0 favour baseline ⇒ **supported**							
H3: EIG:mg vs Naive (score IAE)							
GSM8K	60	0.0526	0.0438	+16.8%	41/19	+0.46	0.084
ARC-C	60	0.0732	0.0285	+61.0%	60/0	+1.00	$< 10^{-4}$ ↑
HellaSwag	60	0.0572	0.0181	+68.4%	58/2	+0.99	$< 10^{-4}$ ↑
WinoGrande	60	0.0630	0.0293	+53.4%	58/2	+1.00	$< 10^{-4}$ ↑
MMLU	60	0.0727	0.0325	+55.3%	58/2	+0.99	$< 10^{-4}$ ↑
SWE-bench V.	60	0.0666	0.0353	+46.9%	55/5	+0.94	$< 10^{-4}$ ↑
LiveCodeBench	28	0.0591	0.0288	+51.3%	28/0	+1.00	$< 10^{-4}$ ↑
CRUXEval-O	31	0.0734	0.0311	+57.7%	31/0	+1.00	$< 10^{-4}$ ↑
decision rule: 7/8 favour EOST-M, 0 favour baseline ⇒ **supported**							
H4: EIG:mg vs MFI:pi ($\lvert\text{cov} - 0.9\rvert$*)*							
GSM8K	60	0.2451	0.2166	+11.6%	38/20	+0.22	1.000
ARC-C	60	0.2935	0.1715	+41.6%	49/10	+0.85	$< 10^{-4}$ ↑
HellaSwag	60	0.3178	0.1690	+46.8%	48/12	+0.76	$< 10^{-4}$ ↑
WinoGrande	60	0.3990	0.1704	+57.3%	54/6	+0.96	$< 10^{-4}$ ↑
MMLU	60	0.3880	0.1593	+59.0%	55/5	+0.92	$< 10^{-4}$ ↑
SWE-bench V.	60	0.2502	0.1710	+31.7%	41/19	+0.52	0.020 ↑
LiveCodeBench	28	0.3091	0.1604	+48.1%	21/7	+0.77	0.0072 ↑
CRUXEval-O	31	0.2680	0.1786	+33.4%	22/9	+0.67	0.048 ↑
decision rule: 7/8 favour EOST-M, 0 favour baseline ⇒ **supported**							

Table 9: Decomposing the gain over plug-in Fisher CAT (MFI:pi): selection criterion (entropy vs. EIG) × measurement model (plug-in vs. calibration-marginal). Median across confirmatory benchmarks of the relative improvement over MFI:pi, in %; B = number of benchmarks.

		holdout log loss				score IAE			
N	B	Ent:pi	EIG:pi	Ent:mg	EIG:mg	Ent:pi	EIG:pi	Ent:mg	EIG:mg
4	8	−0.8	−0.1	+5.0	+7.1	−12.3	−0.8	−15.2	+11.7
8	8	−1.8	0.0	+4.4	+5.2	−17.5	+0.7	−9.2	+7.5
16	8	−1.4	0.0	+2.0	+3.6	−18.7	−0.7	−10.9	+5.9
32	6	−1.8	0.0	−0.5	+1.9	−23.4	+0.2	−22.8	+7.4
64	6	−2.0	+0.1	−1.3	+1.1	−17.5	+0.6	−16.7	+6.6

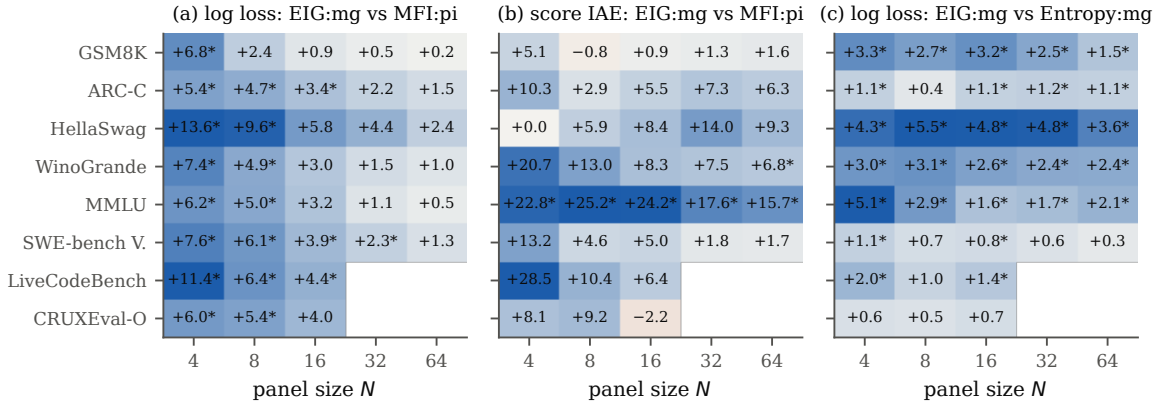


Figure 2: Relative improvement (%) of EIG:mg over the baseline for every benchmark and calibration-panel size; * marks Holm-adjusted $p < 0.05$ (panels a and c: confirmatory family; panel b: secondary family). Blue favours EIG:mg.

over plug-in Fisher CAT increases with the fraction of unlocated items (Spearman $\rho = 0.85$; Figure 4).

6.3 Learning curves

Figure 5 shows the score error as a function of the number of administered items. Every model-based policy starts below the naive estimator, because the calibrated model predicts the bank before any item is asked: after one item, the model-based error is 17–60% of the naive error, except on GSM8K (81%). Two regimes follow. In the first items, plug-in policies, which take their item parameters at face value, reduce the error fastest: at $t = 5$ a plug-in policy is ahead of EOST-M on 7 of 8 benchmarks. Their error then levels off. Between $t = 25$ and $t = 100$ the error of plug-in Fisher CAT falls by a median of 27%, that of EOST-M

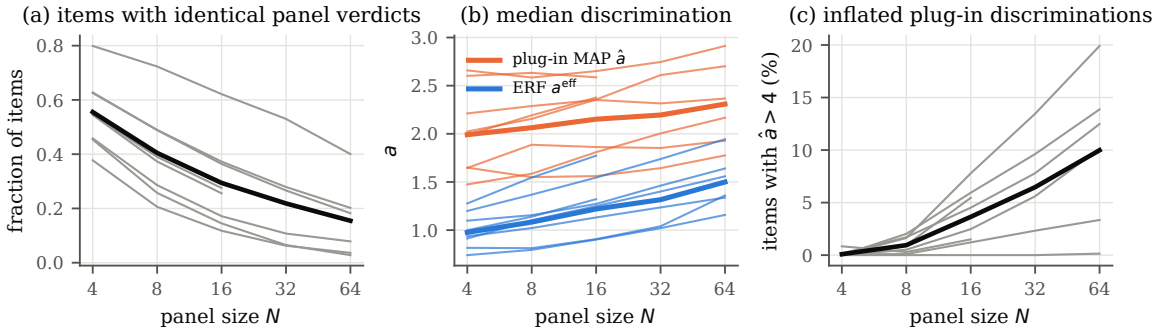


Figure 3: Calibration diagnostics on the study-1 benchmarks (thin lines: individual benchmarks; thick: mean). (a) Fraction of items that every calibrator answered identically, whose difficulty is identified only through the prior. (b) Median plug-in discrimination versus the effective discrimination of the expected response function. (c) Fraction of items whose plug-in discrimination exceeds 4 under the empirical-Bayes prior.

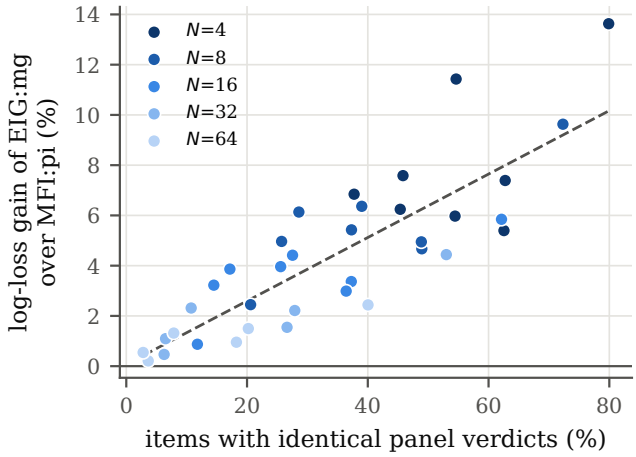


Figure 4: Exploratory: the log-loss gain of EIG:mg over plug-in Fisher CAT grows with the fraction of items that the calibration panel cannot locate (identical verdicts across calibrators). One point per benchmark and panel size; dashed: least-squares line.

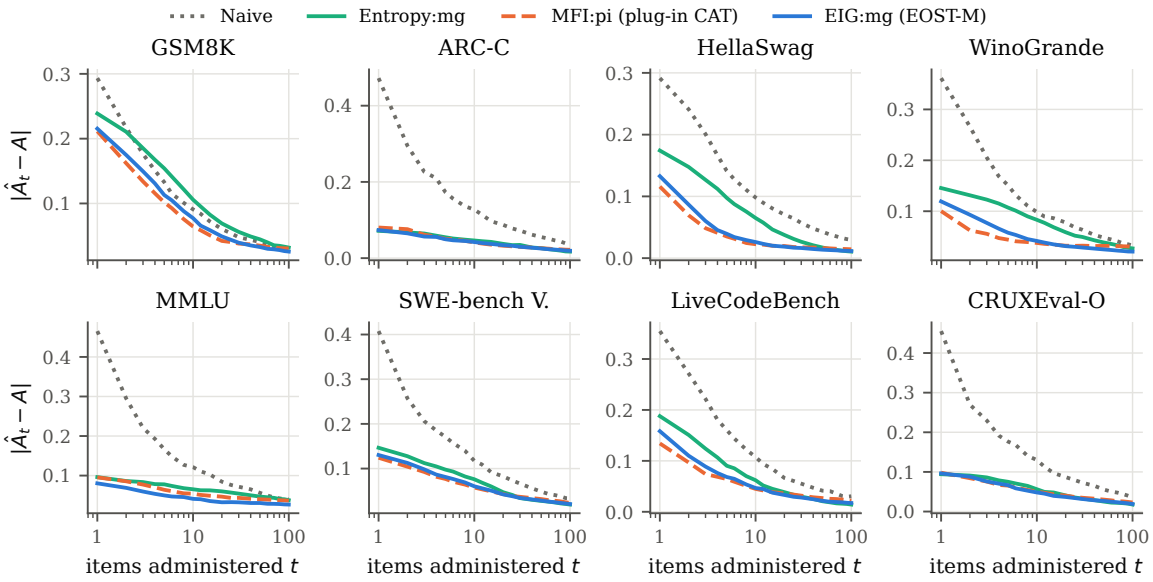


Figure 5: Absolute error of the full-bank score estimate as items are administered ($N = 8$; mean over targets of per-target means over 5 replicates; logarithmic t axis).

by 41%, and after 100 items EOST-M is below plug-in Fisher CAT on 8 of 8 benchmarks. This is the signature of a bias floor: with eight calibrators, the predicted outcomes of the unobserved items inherit the errors of the plug-in parameters, and administering further items does not remove them. Table 11 converts the curves into test lengths: at $N = 8$, EOST-M matched the score error of 100 uniformly sampled items after a median of 22.5 items (between 8 and 70 across benchmarks), and with the largest panels after a median of 8 items. Table 10 compares all twelve policies: averaged over benchmarks at $N = 8$, EOST-M has a score IAE of 0.0309 versus 0.0341 for plug-in Fisher CAT and 0.0647 for naive subsampling (a 52% reduction), and reaches a sustained error below 0.03 after 51 items on average versus 86 for naive subsampling. Adaptivity itself contributes: under the same calibration-marginal model, random item selection—a model-assisted estimator in the spirit of IRT-based benchmark compression (Maia Polo et al., 2024)—has a score IAE of 0.0432, and EOST-M's error is significantly lower on 7 of 8 benchmarks (secondary contrast). Table 12 gives the held-out log loss, the score error and the coverage of six of the policies on every bank, real or synthetic, and every panel size, and Figure 6 the holdout Brier score as items are administered.

Late in the test, entropy selection under the marginal model catches up: after 100 items it has the lowest error of all policies on 5 of 8 benchmarks (SWE-bench V., CRUXEval-O, LiveCodeBench, ARC-C and HellaSwag). Figure 7 locates the crossover, the item after which entropy selection stays below EIG:mg. At $N = 8$ it occurs on SWE-bench V. (500 items) after 30; CRUXEval-O (800 items) after 50; LiveCodeBench (1,055 items) after 50; ARC-C (1,172 items) after 70; HellaSwag (2,000 items) after 90, and not within 100 items

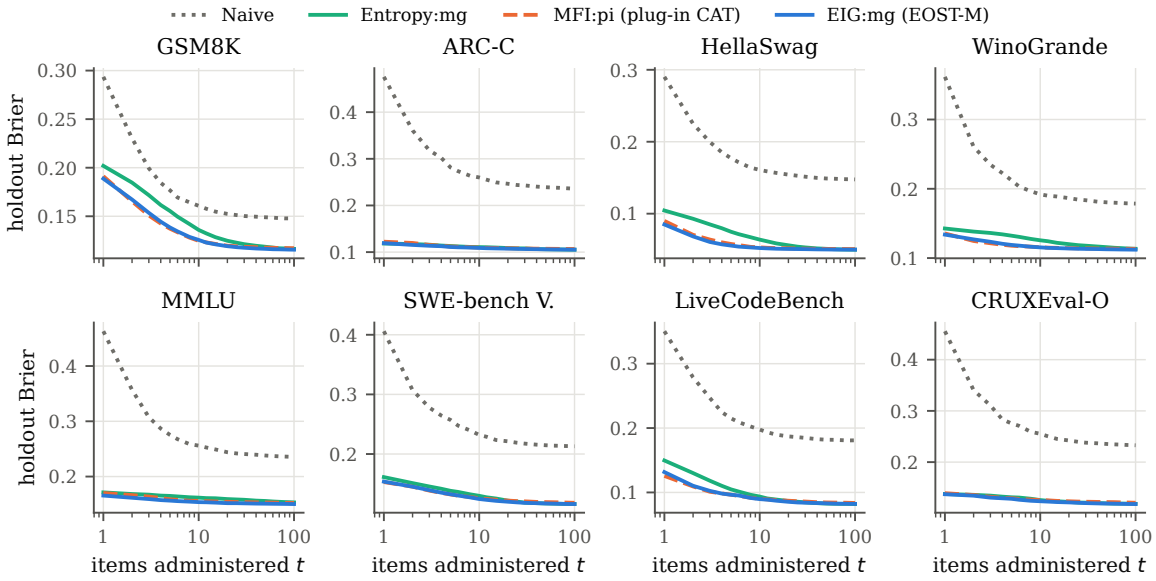


Figure 6: Holdout Brier score as items are administered ($N = 8$).

Table 10: All twelve policies at $N = 8$, averaged over the eight confirmatory benchmarks (mean of per-benchmark means; in parentheses the mean rank across benchmarks, 1 = best). IAE: integrated absolute error of the full-bank score; $|\text{cov} - 0.9|$: miscalibration of the 90% interval; $t_{0.03}$: sustained items-to-tolerance at $\varepsilon = 0.03$ (101 = never).

Policy	holdout log loss	score IAE	$\lvert\text{cov} - 0.9\rvert$	$t_{0.03}$
Naive	–	0.0647 (11.5)	0.0644 (1.1)	86.3 (11.6)
Random:pi	0.3720 (10.4)	0.0424 (9.8)	0.0972 (2.8)	71.2 (10.0)
Random:mg	0.3564 (6.1)	0.0432 (10.0)	0.0897 (2.2)	71.5 (10.4)
Entropy:pi	0.3724 (10.0)	0.0398 (8.9)	0.1598 (6.6)	61.5 (8.8)
Entropy:mg	0.3543 (5.6)	0.0394 (8.4)	0.1190 (4.9)	64.0 (8.9)
MFI:pi (plug-in CAT)	0.3666 (8.1)	0.0341 (6.2)	0.3088 (10.8)	53.4 (5.1)
MFI:mg	0.3470 (2.5)	0.0301 (2.5)	0.1677 (6.6)	50.3 (4.0)
PWFI:mg	0.3469 (2.1)	0.0301 (2.1)	0.1686 (7.1)	49.9 (3.1)
EIG:pi	0.3665 (7.8)	0.0339 (6.0)	0.3142 (11.4)	53.6 (4.8)
EIG:mg **(EOST-M)**	0.3473 (3.0)	0.0309 (3.8)	0.1746 (7.6)	50.5 (3.8)
EVR:pi	0.3665 (8.0)	0.0340 (6.0)	0.3051 (10.5)	53.0 (4.2)
EVR:mg	0.3471 (2.4)	0.0303 (2.9)	0.1664 (6.4)	50.0 (3.4)

Table 11: Items needed to match the full-bank score error of 100 uniformly sampled items (Naive): first checkpoint t at which the mean $|\hat{A}_t - A|$ over targets falls below the Naive error at $t = 100$, at the primary panel size $N = 8$ and at the largest panel available for the benchmark.

Benchmark	N	Naive error @100	Entropy:mg	MFI:pi	EIG:mg	EVR:mg
GSM8K	8	0.0296	>100	>100	70	80
	64	0.0258	>100	90	80	80
ARC-C	8	0.0364	25	15	20	12
	64	0.0408	12	5	3	3
HellaSwag	8	0.0295	30	8	8	10
	64	0.0306	10	3	3	4
WinoGrande	8	0.0326	80	20	20	15
	64	0.0311	30	8	6	5
MMLU	8	0.0395	100	60	15	12
	64	0.0396	25	6	5	6
SWE-bench V.	8	0.0334	40	50	40	40
	64	0.0314	30	40	40	30
LiveCodeBench	8	0.0305	30	40	25	30
	16	0.0298	25	20	15	20
CRUXEval-O	8	0.0359	25	30	25	20
	16	0.0379	25	10	10	10

on WinoGrande, GSM8K and MMLU. Across all panel sizes the crossover tends to come later the larger the bank (Spearman $\rho = 0.73$ over 36 benchmark–panel combinations, no crossover counted as 101): it occurred within the budget in 94% of the combinations on the four smallest banks (500–1,172 items) and in 15% on the four larger ones. This is the regime change described by Proposition 7: once competence is pinned down relative to the number of unobserved items, an item's own outcome uncertainty dominates the expected reduction of score variance, and the larger the bank, the later this happens. The score-targeted criterion, which implements this trade-off under the fitted model, ended below EIG:mg on 4 of the 5 benchmarks on which entropy finished best and on 0 of the others: it moved in the predicted direction, but only part of the way. These late-test comparisons were not part of the frozen plan and are exploratory. Bank size is confounded with source and domain, and WinoGrande (1,267 items), GSM8K (1,319 items) and MMLU (2,000 items) show no crossover at any panel size, so size is at most one determinant of when the regime changes.

6.4 Ranking systems

Leaderboards need the order of systems rather than one score. In an exploratory analysis with eight calibrators, Kendall's τ between the score estimated after 25 items and the true full-bank accuracy, across the targets of a benchmark, had a median of 0.77 for EOST-M, 0.74 for plug-in Fisher CAT and 0.54 for 25 uniformly sampled items (Table 13). Short adaptive tests therefore rank systems far better than random subsets of the same size, and the two adaptive tests are close to each other.

Table 12: Full results: holdout log loss / score IAE / coverage of the 90% interval (per-target means, then averaged over targets; leading zeros omitted).

Bank	N	Naive	Entropy:mg	MFI:pi (plug-in CAT)	EIG:pi	EIG:mg	EVR:mg
GSM8K	4	–/.051/.92	.422/.069/.62	.438/.054/.60	.438/.055/.58	.408/.052/.67	.411/.055/.65
GSM8K	8	–/.053/.89	.388/.058/.67	.387/.043/.67	.387/.042/.68	.378/.044/.70	.379/.044/.69
GSM8K	16	–/.054/.90	.367/.053/.70	.358/.038/.70	.358/.038/.70	.355/.037/.73	.356/.039/.72
GSM8K	32	–/.053/.91	.348/.045/.73	.341/.035/.70	.341/.035/.70	.340/.034/.73	.341/.035/.73
GSM8K	64	–/.053/.89	.334/.041/.75	.330/.035/.68	.330/.035/.68	.329/.035/.70	.329/.035/.70
ARC-C	4	–/.072/.90	.383/.038/.87	.400/.037/.56	.401/.037/.58	.378/.033/.78	.377/.033/.77
ARC-C	8	–/.073/.90	.348/.029/.89	.363/.029/.61	.364/.030/.61	.346/.029/.74	.346/.027/.76
ARC-C	16	–/.075/.89	.325/.027/.87	.332/.026/.58	.332/.025/.59	.321/.024/.70	.321/.023/.71
ARC-C	32	–/.076/.88	.310/.026/.86	.314/.024/.57	.314/.024/.57	.307/.022/.65	.307/.023/.65
ARC-C	64	–/.074/.88	.303/.026/.82	.304/.024/.55	.303/.024/.55	.299/.023/.60	.299/.023/.60
HellaSwag	4	–/.057/.90	.213/.033/.76	.236/.023/.60	.237/.023/.58	.204/.023/.78	.205/.024/.80
HellaSwag	8	–/.057/.90	.188/.029/.80	.197/.019/.59	.195/.018/.58	.178/.018/.76	.178/.019/.77
HellaSwag	16	–/.056/.91	.167/.022/.84	.168/.016/.60	.168/.016/.61	.158/.015/.73	.159/.015/.74
HellaSwag	32	–/.060/.88	.156/.019/.83	.156/.014/.64	.157/.015/.63	.149/.012/.72	.149/.013/.72
HellaSwag	64	–/.058/.91	.148/.017/.82	.146/.014/.64	.145/.013/.64	.143/.012/.68	.143/.013/.68
WinoGrande	4	–/.062/.91	.408/.055/.76	.427/.043/.47	.427/.045/.45	.396/.034/.79	.397/.036/.79
WinoGrande	8	–/.063/.89	.379/.047/.81	.387/.034/.50	.386/.033/.50	.367/.029/.75	.366/.029/.76
WinoGrande	16	–/.061/.91	.356/.041/.82	.357/.029/.55	.358/.029/.54	.347/.026/.68	.347/.026/.68
WinoGrande	32	–/.063/.90	.342/.036/.79	.339/.026/.54	.339/.026/.54	.334/.024/.62	.334/.023/.63
WinoGrande	64	–/.062/.90	.332/.034/.76	.327/.027/.49	.327/.027/.49	.324/.025/.54	.324/.026/.53
MMLU	4	–/.072/.89	.530/.079/.70	.536/.052/.59	.542/.055/.58	.503/.040/.80	.503/.039/.81
MMLU	8	–/.073/.91	.482/.051/.87	.492/.043/.51	.493/.045/.48	.468/.032/.77	.468/.032/.77
MMLU	16	–/.073/.90	.445/.036/.91	.452/.038/.47	.451/.037/.47	.438/.029/.69	.437/.029/.70
MMLU	32	–/.071/.91	.420/.037/.86	.418/.032/.47	.418/.032/.48	.413/.026/.64	.413/.026/.64
MMLU	64	–/.072/.89	.412/.036/.84	.405/.030/.47	.406/.031/.48	.403/.026/.60	.403/.025/.60
SWE-bench V.	4	–/.067/.90	.416/.046/.79	.445/.044/.62	.445/.043/.66	.411/.038/.79	.410/.038/.80
SWE-bench V.	8	–/.067/.89	.384/.038/.83	.406/.037/.66	.408/.038/.65	.381/.035/.76	.380/.034/.79
SWE-bench V.	16	–/.063/.92	.356/.033/.84	.367/.033/.68	.368/.035/.67	.353/.032/.75	.353/.031/.75
SWE-bench V.	32	–/.066/.90	.339/.031/.83	.345/.033/.65	.345/.032/.66	.337/.032/.69	.337/.031/.70
SWE-bench V.	64	–/.065/.91	.329/.030/.83	.332/.032/.67	.332/.031/.68	.328/.031/.70	.326/.030/.72
LiveCodeBench	4	–/.059/.94	.324/.038/.81	.359/.043/.53	.356/.042/.53	.318/.031/.80	.321/.034/.78
LiveCodeBench	8	–/.059/.94	.281/.031/.85	.297/.032/.60	.298/.033/.59	.278/.029/.76	.278/.028/.76
LiveCodeBench	16	–/.064/.91	.262/.026/.87	.270/.025/.68	.271/.025/.67	.258/.023/.77	.260/.024/.76
CRUXEval-O	4	–/.068/.92	.419/.038/.89	.443/.038/.69	.438/.036/.70	.416/.035/.83	.415/.034/.84
CRUXEval-O	8	–/.073/.89	.384/.032/.89	.404/.034/.64	.401/.032/.66	.382/.031/.77	.380/.030/.79
CRUXEval-O	16	–/.070/.89	.361/.030/.87	.374/.026/.69	.375/.028/.66	.359/.027/.75	.359/.027/.75
Synth. 2PL	4	–/.072/.89	.566/.051/.89	.577/.048/.77	.576/.048/.77	.561/.042/.86	.563/.045/.86
Synth. 2PL	8	–/.068/.92	.538/.047/.90	.539/.039/.84	.539/.039/.84	.533/.037/.89	.534/.037/.89
Synth. 2PL	16	–/.070/.90	.516/.040/.92	.514/.034/.85	.514/.034/.85	.511/.034/.89	.511/.033/.89
Synth. 2PL	32	–/.069/.91	.502/.041/.90	.496/.029/.88	.496/.029/.88	.495/.028/.89	.495/.028/.89
Synth. 2PL	64	–/.067/.91	.494/.039/.92	.487/.027/.88	.487/.027/.88	.487/.027/.89	.487/.026/.89
Synth. guessing	4	–/.070/.90	.603/.054/.88	.612/.050/.77	.613/.051/.77	.599/.044/.88	.599/.045/.86
Synth. guessing	8	–/.073/.88	.579/.052/.89	.580/.042/.81	.580/.042/.81	.573/.039/.88	.574/.039/.88
Synth. guessing	16	–/.069/.90	.561/.046/.90	.559/.035/.86	.559/.035/.86	.556/.033/.91	.556/.033/.90
Synth. guessing	32	–/.072/.89	.549/.044/.92	.545/.031/.88	.545/.031/.88	.543/.030/.90	.543/.030/.91
Synth. guessing	64	–/.068/.91	.542/.044/.91	.534/.028/.89	.534/.028/.89	.533/.026/.91	.533/.026/.91
Synth. 2-D	4	–/.069/.90	.582/.053/.89	.596/.054/.74	.596/.053/.74	.581/.049/.81	.582/.051/.79
Synth. 2-D	8	–/.069/.90	.556/.048/.89	.562/.048/.76	.562/.048/.76	.555/.045/.81	.555/.045/.81
Synth. 2-D	16	–/.072/.89	.539/.047/.88	.539/.043/.79	.538/.042/.79	.535/.040/.84	.536/.041/.84
Synth. 2-D	32	–/.068/.92	.526/.042/.91	.525/.038/.83	.526/.038/.83	.524/.038/.85	.524/.038/.85
Synth. 2-D	64	–/.069/.90	.521/.046/.87	.516/.037/.82	.516/.037/.83	.514/.036/.83	.515/.036/.83

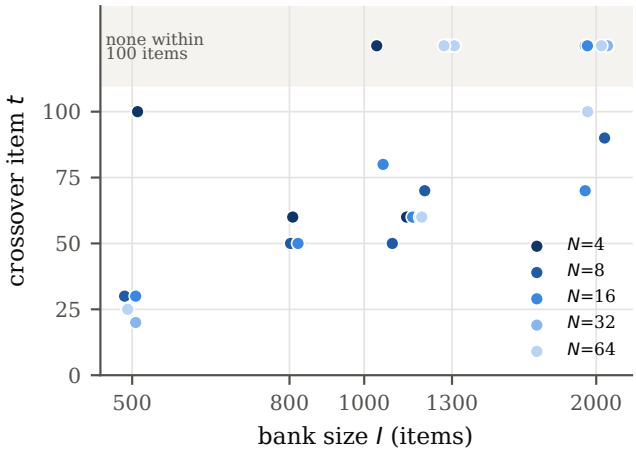


Figure 7: Exploratory test of the regime change of Proposition 7: the item after which entropy selection under the marginal model has a lower score error than EIG:mg at every later checkpoint, against the size of the bank. One point per benchmark and panel size (jittered horizontally); shaded band: no crossover within the 100-item budget.

Table 13: Exploratory (not in the frozen plan): Kendall's τ between the full-bank score estimated after t items and the true full-bank accuracy, across targets, at $N = 8$ (one adaptive test per target; mean over 5 replicates).

	$t = 10$				$t = 25$				$t = 50$			
Benchmark	Naive	MFI:pi	EIG:mg	EVR:mg	Naive	MFI:pi	EIG:mg	EVR:mg	Naive	MFI:pi	EIG:mg	EVR:mg
GSM8K	0.71	0.78	0.76	0.78	0.82	0.84	0.85	0.85	0.86	0.87	0.88	0.87
ARC-C	0.28	0.61	0.60	0.61	0.45	0.69	0.69	0.72	0.55	0.75	0.74	0.76
HellaSwag	0.25	0.64	0.65	0.61	0.37	0.75	0.75	0.76	0.47	0.79	0.80	0.80
WinoGrande	0.27	0.48	0.51	0.54	0.33	0.60	0.61	0.64	0.42	0.64	0.67	0.67
MMLU	0.42	0.67	0.70	0.72	0.56	0.74	0.77	0.79	0.62	0.77	0.80	0.81
SWE-bench V.	0.62	0.75	0.73	0.76	0.72	0.84	0.83	0.84	0.80	0.88	0.88	0.89
LiveCodeBench	0.57	0.72	0.76	0.70	0.73	0.79	0.82	0.80	0.76	0.86	0.87	0.86
CRUXEval-O	0.43	0.63	0.64	0.65	0.53	0.74	0.77	0.75	0.61	0.79	0.81	0.81

6.5 Uncertainty

Plug-in posteriors are overconfident (Figure 8): at $N = 8$ the 90% interval of plug-in Fisher CAT covered the true score 0.60 of the time on average, against 0.75 for EOST-M and 0.90 for the model-free Wilson interval of naive subsampling. Marginalization therefore removes a large part of the overconfidence, most at small N, but no model-based adaptive policy reaches nominal coverage on real data. On the synthetic banks (Figure 8b), EOST-M's score coverage at $N = 8$ is 0.89 when the model is well specified, 0.88 with guessing and 0.81 with a second ability dimension (plug-in Fisher CAT: 0.84, 0.81 and 0.76). Coverage thus degrades with misspecification, and the residual gap on real data is plausibly due to model misspecification rather than to the marginalization itself. The two panels of Figure 8 also

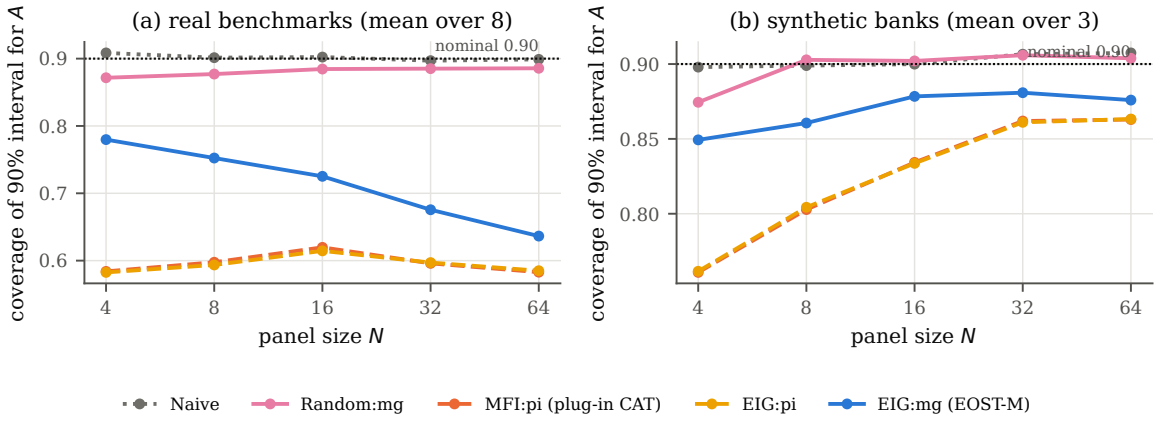


Figure 8: Coverage of the 90% interval for the full-bank score, averaged over steps, targets and benchmarks, as a function of the panel size. Dotted line: nominal level.

move in opposite directions as the panel grows: on the synthetic banks the coverage of the model-based intervals rises towards the nominal level, on real data it falls. More calibrators sharpen the item posteriors, which helps when the model is adequate and exposes its misfit when it is not.

6.6 Synthetic banks

On the synthetic banks the true abilities are known (Figure 9). At the primary panel size $N = 8$, EOST-M had the smallest mean absolute error of the ability estimate among the four policies of the figure on all three banks (0.244, 0.391 and 0.469 on the well-specified, guessing and two-dimensional banks, against 0.251, 0.415 and 0.478 for plug-in Fisher CAT), and its 90% credible interval covered the true ability more often (0.88, 0.67 and 0.59, against 0.75, 0.52 and 0.50). With four calibrators the ranking depends on the bank; from 16 calibrators on, EIG under the marginal model stays ahead of both plug-in policies on every bank. Coverage of the true ability falls for every adaptive policy as the panel grows: the posterior sharpens while the calibrated ability scale, which is fixed by the panel and the prior, only approximates the generating one. This identification issue is why our endpoints are defined on observable quantities (Section 3).

6.7 Sensitivity analyses

Table 14 repeats the four contrasts under the sensitivity variants specified in the frozen plan. Removing near-duplicate calibrators (Cohen's $\kappa \geq 0.8$ with the target, on the 5 leaderboard benchmarks) leaves the H2 gain at $N = 8$ essentially unchanged (+6.1%, against +4.9% for the main analysis on the same benchmarks). The empirical-Bayes hyperprior, by contrast, is an essential part of the method: with the fixed weakly-informative item prior, the H2 gain at $N = 8$ falls to a median of +2.8% (main: +5.2%) and EOST-M improves on plug-in Fisher CAT on only 5 of 8 benchmarks. For items that the panel cannot locate, the expected

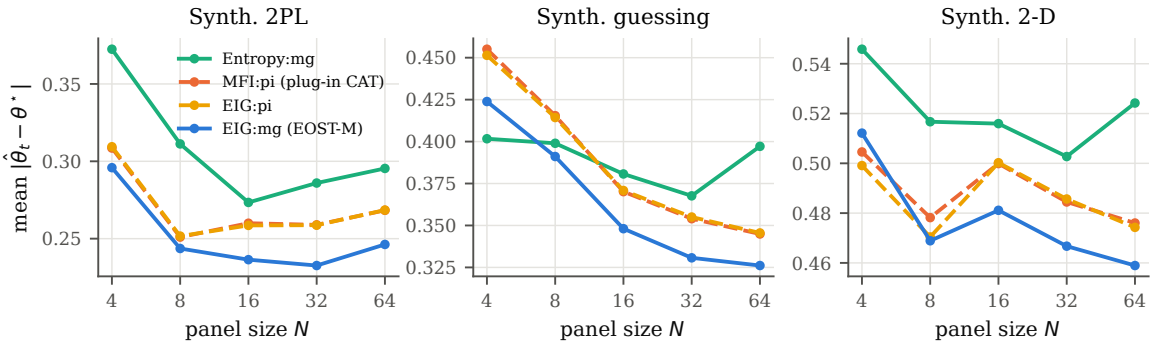


Figure 9: Error of the ability estimate with respect to the true generating ability on the three synthetic banks (well-specified 2PL; 60% of items with a 0.25 guessing floor; two correlated ability dimensions), averaged over the 100-item test.

response function is essentially the prior predictive, so integrating over the item posterior is only as good as the prior it integrates; this is the failure that development had revealed (Section 5.2).

The one-parameter variant specified in the frozen plan turned out to be uninformative. Its likelihood is invariant to a joint rescaling of the abilities, the difficulties and the common discrimination; with Gaussian priors on the former and a single log-normal prior term on the latter, the joint posterior mode shrinks the ability scale and inflates the common discrimination (median 1.5×10^4, up to 2.3×10^7, across calibrations). The resulting nearly deterministic response curves make every model-based policy grossly overconfident (EOST-M coverage 0.15 at $N = 8$), so we report this variant but do not interpret its contrasts. The two-parameter model is protected from this degeneracy because every item carries its own prior term on $\log a$. A post hoc variant with a strongly informative prior on the common discrimination, $\log a \sim \mathcal{N}(0, 0.05^2)$, is well behaved (median common discrimination 2.5; EOST-M coverage 0.84 at $N = 8$). It provides the check of Corollary 4 that the planned variant could not: when all items share one discrimination, the median log-loss difference between EIG and entropy at $N = 8$ is $+0.4\%$ under the marginal model and -0.4% under plug-in parameters, against $+1.9\%$ and $+1.6\%$ with item-specific discriminations. The information advantage of EIG over entropy therefore comes from the heterogeneity of discriminations, as the theory says. Under this constrained model the gain of marginalization over plug-in parameters is also small ($+0.7\%$), an indication that much of the H2 gain concerns discriminations that a small panel cannot estimate well; Section 6.8 tests this directly.

6.8 Post hoc probes that motivated study 2

After all frozen runs, three post hoc analyses with decision rules recorded before running probed the main result: strong baselines, a regularization path of the discrimination prior and a decomposition of interval coverage. The regularized one-parameter variant suggests

Table 14: Sensitivity analyses (30 targets, 3 replicates; main analysis for reference). Median relative improvement of EIG:mg across confirmatory benchmarks, in %, [benchmarks improved / tested], and mean coverage of EOST-M's 90% score interval. †Specified in the frozen plan, but its joint MAP is degenerate (the common discrimination diverges; Section 6.7), so its contrasts are not interpreted. ‡Post hoc: common discrimination with prior $\log a \sim \mathcal{N}(0, 0.05^2)$.

Variant	N	H1 log loss	H2 log loss	H3 IAE	H4 cov. gap	cov.
main (EB prior, 2PL)	8	+1.9 [8/8]	+5.2 [8/8]	+54.4 [8/8]	+44.2 [8/8]	0.75
main (EB prior, 2PL)	64	+1.8 [6/6]	+1.1 [6/6]	+62.1 [6/6]	+10.3 [6/6]	0.64
fixed item prior	8	+2.3 [8/8]	+2.8 [5/8]	+46.4 [8/8]	+35.8 [5/8]	0.70
fixed item prior	64	+2.2 [6/6]	+0.7 [4/6]	+63.2 [6/6]	+9.1 [5/6]	0.63
near-duplicates removed	8	+2.0 [5/5]	+6.1 [5/5]	+49.1 [5/5]	+46.2 [4/5]	0.73
near-duplicates removed	64	+2.2 [5/5]	+1.0 [5/5]	+66.4 [5/5]	+6.9 [3/5]	0.64
one-parameter†	8	−0.5 [4/8]	+38.9 [8/8]	−95.6 [1/8]	−3.8 [3/8]	0.15
one-parameter†	64	−0.4 [3/6]	+45.0 [4/6]	−140.1 [1/6]	+0.9 [4/6]	0.20
one-parameter, regularized‡	8	+0.4 [7/8]	+0.7 [7/8]	+51.8 [8/8]	+14.3 [8/8]	0.84
one-parameter, regularized‡	64	−0.2 [2/6]	0.0 [3/6]	+44.8 [6/6]	−3.3 [3/6]	0.81

a competing explanation for H2: EOST-M may beat plug-in Fisher CAT mainly because item-specific discriminations estimated from a small panel are unreliable, a failure that a constrained plug-in model could avoid without any marginalization. We tested this post hoc against strong baselines on the confirmatory design (same targets, replicates, holdouts and panels; $N \in \{4, 8, 16\}$), under decision rules recorded before the runs (Table 15). The recomputed EOST-M and MFI:pi rows reproduced the confirmatory artifacts exactly (12,570 rows, recomputed days later after several machine restarts).

Better point estimates do not fix plug-in testing: with posterior-mean instead of posterior-mode item parameters, held-out log loss is 15.5% worse than EOST-M's at $N = 8$ (7 of 8 benchmarks significant), because the posterior mean of a poorly located item's discrimination inherits the long right tail of its posterior. A regularized Rasch-type plug-in test—a common discrimination with prior $\log a \sim \mathcal{N}(0, 0.05^2)$—is a much stronger baseline. EOST-M's held-out log loss is lower on all 8 benchmarks, but only by a median of 2.1% at $N = 8$ (significant on 4), its score error is lower on 5 of 8 (by a median of 2.9%, and of 15.3% at $N = 16$, where a single discrimination becomes a visible misspecification), and the constrained model's intervals are better calibrated (coverage 0.81 against 0.75). By the rule fixed in advance, the advantage of EOST-M over plug-in CAT is therefore largely an advantage over *unregularized item-specific discriminations*: integrating their uncertainty and constraining them are two remedies for the same small-panel failure, with integration ahead on prediction and score estimation, and the constrained model ahead on interval calibration. Finally, selecting by the posterior-expected Fisher information under the marginal model gives the same tests as EIG in practice (median difference 0.0% at $N = 8$, no benchmark significant): the optimism that Corollary 6 attributes to averaged information did not change which items were worth asking in our banks.

Table 15: Post hoc comparison with strong baselines (not in the frozen plan; same targets, replicates, holdouts and calibration panels as the confirmatory study; decision rules recorded before the runs). Median relative improvement of EOST-M (EIG:mg) over each baseline across the eight benchmarks, in %, with [benchmarks with a Holm-significant difference favouring EOST-M / favouring the baseline], Holm over this post hoc family, so that the counts for MFI:pi differ from Table 5; last column: mean coverage of the baseline's 90% score interval at $N = 8$.

	holdout log loss			score IAE	$\lvert\text{cov} - 0.9\rvert$	cov.
Policy	$N = 4$	$N = 8$	$N = 16$	$N = 8$	$N = 8$	$N = 8$
EOST-M (EIG:mg), reference	–	–	–	–	–	0.75
plug-in 2PL (MFI:pi)	+7.1 [8/0]	+5.2 [5/0]	+3.6 [2/0]	+7.5 [1/0]	+44.2 [5/0]	0.60
plug-in posterior-mean 2PL, MFI	+20.4 [8/0]	+15.5 [7/0]	+10.0 [7/0]	+7.3 [2/0]	+52.5 [6/0]	0.54
plug-in posterior-mean 2PL, EIG	+21.2 [8/0]	+14.7 [7/0]	+9.8 [7/0]	+7.7 [2/0]	+53.7 [6/0]	0.53
plug-in regularized 1PL, MFI	+2.9 [4/0]	+2.1 [4/0]	+3.3 [5/0]	+2.9 [2/0]	−25.9 [0/0]	0.81
marginal regularized 1PL, EIG	+1.0 [3/0]	+1.8 [5/0]	+2.8 [4/0]	+6.7 [2/0]	−54.0 [0/1]	0.83
expected Fisher information (EFI:mg)	0.0 [0/0]	0.0 [0/0]	0.0 [0/0]	+0.9 [0/0]	−3.0 [0/0]	0.76

Can the two remedies be combined? In a further exploratory experiment, also planned before running, we kept EOST-M's calibration but fixed the prior standard deviation of $\log a$ to values between 0.05 and 0.5 instead of its empirical-Bayes estimate (median 0.68), shrinking the discriminations towards a common value (Figure 10). No setting met the criterion fixed in advance (coverage of at least 0.80 with a median held-out log loss at most 0.5% above EOST-M's): coverage stayed between 0.74 and 0.76 (EOST-M: 0.75) while held-out log loss rose by 0.2% to 2.4%. The calibration advantage of the Rasch-type test therefore does not come from equalizing the discriminations as such.

Where does it come from? A decomposition of the post hoc runs, again planned before computing (Table 16), answers this by the rule fixed in advance: the Rasch-type intervals are wider than EOST-M's on all 8 benchmarks (median 29% wider, Holm-significant on 8), while its score error is not lower on 4 of the 7 benchmarks where its coverage is higher. Across all benchmark–model cells, coverage is almost a monotone function of the ratio of mean interval width to score IAE (Spearman $\rho = 0.98$ over 32 cells), and that ratio stays below the 4.12 of a calibrated 90% interval for unbiased Gaussian errors in every cell (at most 3.97; on average 2.19 for plug-in 2PL, 3.02 for EOST-M, 3.48 for the Rasch-type test and 3.59 for the marginal one-parameter model). The constrained model is better calibrated because it is less confident, not because it is more accurate, and under adaptive selection every model we tried states intervals too narrow for its errors. The same marginal model with uniformly sampled items covers 0.88 of the time on the same banks, and EOST-M covers 0.89 on the well-specified synthetic bank (Figure 8): the shortfall arises when adaptive selection meets a misspecified model, not from either alone.

Because the regularized Rasch-type test is a strong competitor found after the fact, study 2 tests it, together with a Bayesian CAT that also integrates the calibrators' abilities, under a new frozen plan.

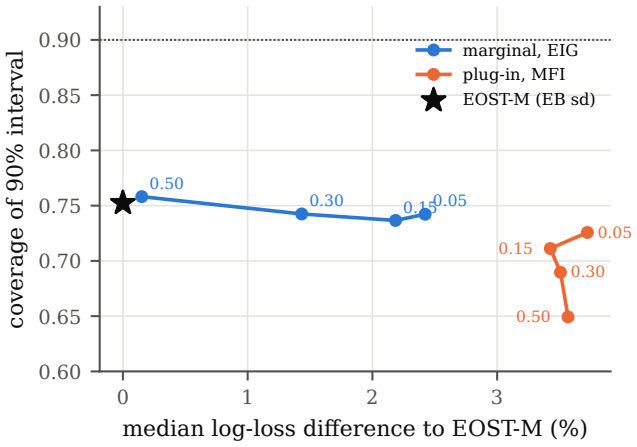


Figure 10: Exploratory regularization path ($N = 8$; planned before running): EOST-M's calibration with the prior standard deviation of $\log a$ fixed to the value next to each point instead of its empirical-Bayes estimate, for the marginal model with EIG and the plug-in model with MFI. Horizontal axis: median over benchmarks of the relative held-out log-loss difference to EOST-M; vertical axis: mean coverage of the 90% score interval (dotted: nominal).

Table 16: Where the coverage of the 90% score interval comes from (exploratory analysis of the post hoc runs, planned before computing; $N = 8$). Coverage and score IAE (in percentage points) of EOST-M (EIG:mg) and of the regularized Rasch-type plug-in test (MFI:1pr), median over targets of the ratio of their mean interval widths (Rasch-type over EOST-M; Holm-significant on 8 of 8 benchmarks), and ratio of mean interval width to score IAE for plug-in 2PL (MFI:pi), EOST-M, the Rasch-type test and the marginal regularized one-parameter model (EIG:1prmg, column 1PL:mg). A calibrated 90% interval for unbiased Gaussian errors has ratio $2 \cdot 1.645/\sqrt{2/\pi} = 4.12$.

	coverage		width	score IAE		width / IAE			
Benchmark	EOST-M	Rasch	ratio	EOST-M	Rasch	MFI:pi	EOST-M	Rasch	1PL:mg
GSM8K	0.70	0.78	1.12	4.4	4.1	2.62	2.89	3.34	3.33
ARC-C	0.74	0.87	1.44	2.9	3.0	2.20	2.97	3.97	3.83
HellaSwag	0.76	0.79	1.14	1.8	1.7	2.00	2.88	3.21	3.46
WinoGrande	0.75	0.75	1.52	2.9	4.2	1.84	2.97	3.06	2.99
MMLU	0.77	0.81	1.76	3.2	5.3	1.90	3.22	3.40	3.68
SWE-bench V.	0.76	0.81	1.20	3.5	3.5	2.39	2.98	3.51	3.70
LiveCodeBench	0.76	0.81	1.08	2.9	2.7	2.26	3.11	3.53	3.83
CRUXEval-O	0.77	0.86	1.39	3.1	3.4	2.29	3.17	3.85	3.89

7 Study 2: independent targets and strong competitors

Study 2 evaluated 267 targets (between 12 and 40 per benchmark, one per strict cluster) under the frozen plan of Section 5.3. Table 17 lists every decision, Table 19 the main contrasts per benchmark, and Table 18 the policy means. Of the 13 frozen decisions, 6 were supported. Apart from the recalibration contrast C5b, every superiority contrast favoured EOST-M on at least seven of the eight benchmarks and no competitor was significantly better on any benchmark; four of the five superiority hypotheses that were not supported (C2, C4, C7, C8) missed the frozen threshold by one benchmark, with four of eight significant and none against. CRUXEval-O, with only 12 independent clusters, did not reach significance in any contrast.

7.1 Replication with independent targets

With one target per cluster, EOST-M's held-out log loss was lower than plug-in Fisher CAT's on 8 of 8 benchmarks (median 5.9%), Holm-significantly on 5 and in favour of the baseline on 0: replication R2 is supported. Its score error was 53.9% below that of naive subsampling (R3, supported; 6 benchmarks significant), and its intervals were less miscalibrated than plug-in Fisher CAT's by a median of 41.6% (R4, supported). With independent targets the study-1 result therefore holds: the gain over plug-in Fisher CAT was positive on every benchmark, larger with four calibrators (median 8.3%, 5 benchmarks significant) and smaller

Table 17: Study 2: decisions under the frozen plan (primary panel size $N = 8$; one target per strict family cluster). Holm sig.: benchmarks with a Holm-significant difference favouring EOST-M / favouring the competitor (Holm over the 88 superiority tests); positive: benchmarks on which EOST-M is better; median: median over benchmarks of the relative improvement of EOST-M (%; for C3 the relative difference of the posterior-draw marginal (PDM) Bayesian CAT, for C5a the range of coverage). Supported: at least five benchmarks for EOST-M and none against (C3: equivalence on at least six; C5a: inside on at least six). *C5a, C5b and C6 concern EOST-M's recalibrated interval.

	EOST-M versus	Endpoint	Holm sig.	positive	median	Supported
R2	plug-in 2PL (MFI:pi)	log loss	5/0	8/8	+5.9	yes
R3	Naive	score IAE	6/0	8/8	+53.9	yes
R4	plug-in 2PL (MFI:pi)	$\lvert\text{cov} - 0.9\rvert$	5/0	8/8	+41.6	yes
C1	Rasch-type plug-in	log loss	5/0	7/8	+2.4	yes
C2	Rasch-type marginal	log loss	4/0	8/8	+1.8	no
C4	Random:mg	score IAE	4/0	8/8	+21.1	no
C5b	raw EOST-M interval*	interval score	0/2	4/8	+1.0	no
C6	Naive (Wilson)*	interval score	6/0	7/8	+35.0	yes
C7	ATLAS-style 3PL CAT	score IAE	4/0	8/8	+26.0	no
C8	tinyBenchmarks	error @50	4/0	8/8	+29.6	no
C9	MINCE	error @50	5/0	8/8	+44.8	yes
C3	PDM Bayesian CAT (equivalence)	log loss	4/8 within ±1%		+0.32	no
C5a	(coverage in $[0.85, 0.95]$)*	coverage	5/8 banks inside		0.81–0.91	no

Table 18: Study 2: policy means at $N = 8$ averaged over the eight benchmarks: integrated held-out log loss, integrated absolute score error, mean coverage and width of the 90% score interval, and integrated interval score ($\alpha = 0.1$; lower is better). Recalibrated intervals share the estimates of the raw policy.

Policy	log loss	score IAE	coverage	width	interval score
EOST-M (EIG:mg)	0.3582	0.0324	0.74	0.095	0.201
EOST-M, recalibrated	(same)	0.0324	0.86	0.134	0.206
Rasch-type plug-in (MFI:1pr)	0.3680	0.0344	0.81	0.121	0.220
Rasch-type, recalibrated	(same)	0.0344	0.91	0.271	0.321
Rasch-type marginal (EIG:1prmg)	0.3660	0.0366	0.82	0.128	0.214
PDM Bayesian CAT, BMI (frozen burn-in)	0.3599	0.0339	0.76	0.101	0.221
PDM Bayesian CAT, EIG (frozen burn-in)	0.3589	0.0326	0.77	0.101	0.210
plug-in 2PL (MFI:pi)	0.3825	0.0352	0.59	0.076	0.304
ATLAS-style 3PL CAT	0.3825	0.0448	0.48	0.085	0.417
Random:mg	0.3651	0.0422	0.88	0.162	0.211
Naive (Wilson)	–	0.0640	0.90	0.235	0.290
EOST-M, cross-fitted EB	0.3583	0.0319	0.74	0.094	0.197
EOST-M, Student-t EB	0.3601	0.0319	0.70	0.088	0.212
mixed (every 4th random)	0.3583	0.0326	0.77	0.102	0.189

Table 19: Study 2 per benchmark ($N = 8$): number of targets (independent clusters) and relative improvement (%) of EOST-M over each competitor; * Holm-significant. †Recorded aggregate verdicts.

		held-out log loss			score IAE	
Benchmark	targets	MFI:pi	Rasch plug-in	Rasch marg.	Random:mg	ATLAS
GSM8K	40	+5.3*	+1.7	+0.3	+22.0*	+3.4
ARC-C	40	+4.5	+2.0*	+1.7*	+12.8	+51.2*
HellaSwag	40	+12.2*	+6.4*	+1.3	+43.1*	+38.0*
WinoGrande	40	+5.3*	+2.8*	+3.2*	+30.0*	+48.7*
MMLU	40	+7.2*	+4.7*	+5.0*	+5.0	+24.1
SWE-bench V.	40	+6.9*	+1.8	+2.1*	+20.2	+16.3*
LiveCodeBench†	15	+6.4	+3.4*	+1.9	+46.7*	+27.9
CRUXEval-O†	12	+5.4	−0.4	+0.2	+2.0	+1.4

with sixteen (4.6%, none significant), and of the same size for deterministic (6.1%) and recorded aggregate verdicts (5.9%). Plug-in Fisher CAT's 90% intervals covered the full-bank score only 0.59 of the time, against 0.74 for EOST-M.

7.2 Integration, constraint, and Bayesian CATs

Against the regularized Rasch-type plug-in test, EOST-M's held-out log loss was lower on 7 of 8 benchmarks (median 2.4%; 5 Holm-significant for EOST-M, 0 for the competitor): C1 is supported. Against the same one-parameter model integrated over its item posterior, the median improvement was 1.8% (4/0; C2 not supported). Integration therefore beat constraint on held-out prediction, as the post hoc probe of study 1 suggested, but by a small

margin, of similar size for deterministic and aggregate verdicts (2.4% and 1.5%); the margin was larger with sixteen calibrators than with four (median 4.0% against 2.9%), where a single discrimination becomes a visible misspecification. On the score, the two tests were close (median 1.6% in favour of EOST-M, no benchmark significant), and the Rasch-type intervals were again better calibrated before recalibration (coverage 0.81 against 0.74; Section 7.4).

The posterior-draw marginal (PDM) Bayesian CAT, which also integrates the calibrators' abilities and the hyperprior through posterior draws, differed from EOST-M by a median of 0.32% in held-out log loss and was practically equivalent within ±1% on 4 of 8 benchmarks (C3 not supported; Table 21). A review of the code during the run, before any result was inspected, found that the frozen burn-in of 10 sweeps was too short on some benchmarks: the translation move is restricted to shifts that keep every sampled difficulty on the grid, so on HellaSwag, where many difficulties sit near the edge of the grid, it rarely acted and the mean calibrator ability drifted by a median of 0.51 during the retained sweeps (median over benchmarks 0.17). Both Bayesian-CAT arms were re-run post hoc with a burn-in of 60 sweeps on the same targets, panels, holdouts and chains, which brought the drift on HellaSwag down to 0.24, close to the other benchmarks (median 0.16). The correction did not change the conclusion (Table 20); Section 8 examines the sampler's convergence with long chains and better moves. With either burn-in, the PDM Bayesian CAT was not significantly different from EOST-M on any benchmark, in held-out log loss or in score error; its log loss was higher by a median of 0.39% with Bayesian maximum information and 0.50% with EIG (frozen: 0.32% and 0.18%), within ±1% on only 3 (BMI) and 3 (EIG) benchmarks, and its intervals covered 0.76 of the time against 0.74 for EOST-M. The PDM Bayesian CAT follows van der Linden and Ren (2020) and averages each item's response function over the posterior draws separately, which treats the shared uncertainty about the location of the calibration as independent item-level noise (Lemma 1). The exploratory joint fully Bayesian CAT, which keeps this uncertainty shared by mixing over draws of the whole calibration, was the closest to EOST-M with EIG selection (median 0.05%, within ±1% on 5 benchmarks; with Bayesian maximum information 0.33%), as we had expected before running it. With eight calibrators, integrating their abilities and the hyperprior on top of the item parameters therefore made no significant difference; equivalence within ±1% was not shown, and on a few benchmarks the bootstrap intervals still allow differences of about one per cent.

The post hoc re-analysis of the frozen arm reproduces the equivalence test of Table 21 exactly, and with the frozen burn-in the correction runs reproduce the frozen rows exactly (maximum difference 0 on the first target of every benchmark), so the corrected arms differ from the frozen ones only by the burn-in. At $N = 4$ (Table 22) the corrected PDM Bayesian CAT was within ±1% of EOST-M on 4 (BMI) and 2 (EIG) benchmarks, with median differences of 0.21% and 0.45%, 0 and 0 benchmarks significantly in its favour and 0 and 1 in EOST-M's, and coverage 0.81 (EOST-M: 0.78); the location of the calibration is more uncertain with four calibrators, and the chains' drift was correspondingly larger. The sampler is not expensive: with eight calibrators the frozen chain (42 sweeps) took a median of 4.0 s per panel, against 0.7 s for EOST-M's whole calibration and 1.5 s for the leave-one-calibrator-out recalibration.

Table 20: Study 2, Bayesian CATs against EOST-M at $N = 8$ (post hoc correction of the burn-in and exploratory joint model; same targets, panels, holdouts and chains). Equivalence: 90% bootstrap interval of the relative difference in held-out log loss within ±1% (rule of C3: at least six benchmarks); median relative difference of log loss (positive: Bayesian CAT worse); benchmarks with a Holm-significant difference favouring EOST-M / the Bayesian CAT; the same for the score error (median relative improvement of EOST-M); mean coverage of the 90% score interval (EOST-M: 0.74).

	held-out log loss			score IAE		
Variant	equiv.	median %	sig.	median %	sig.	coverage
PDM Bayesian, BMI (frozen, burn-in 10)	4/8	+0.32	0/0	+5.1	0/0	0.76
PDM Bayesian, EIG (frozen, burn-in 10)	5/8	+0.18	0/0	−0.8	0/0	0.77
PDM Bayesian, BMI (burn-in 60)	3/8	+0.39	0/0	+5.0	0/0	0.76
PDM Bayesian, EIG (burn-in 60)	3/8	+0.50	0/0	+4.2	0/0	0.76
joint fully Bayesian, BMI (exploratory)	5/8	+0.33	0/0	+2.8	0/0	0.74
joint fully Bayesian, EIG (exploratory)	5/8	+0.05	0/0	−1.5	0/0	0.75

Table 21: Study 2, hypothesis C3: relative difference (%) of held-out log loss of the posterior-draw marginal (PDM) Bayesian CAT (Bayesian maximum information) with respect to EOST-M at $N = 8$, its 90% bootstrap interval over targets, and equivalence within ±1%.

Benchmark	difference	90% interval	equivalent
GSM8K	+1.31	[-0.71, +3.27]	no
ARC-C	+0.29	[-0.12, +0.73]	yes
HellaSwag	+0.09	[-0.32, +0.49]	yes
WinoGrande	-0.06	[-0.28, +0.16]	yes
MMLU	+0.26	[-0.20, +0.69]	yes
SWE-bench V.	+0.56	[-0.02, +1.10]	no
LiveCodeBench	+0.99	[-0.30, +2.27]	no
CRUXEval-O	+0.35	[-0.81, +1.61]	no

7.3 Adaptivity and benchmark-reduction methods

All methods below use the same eight calibrators per target. ATLAS and tinyBenchmarks were designed for panels of hundreds to thousands of systems, so these comparisons concern the small-panel regime, not the methods in their native settings; MINCE targets few calibration models. Under the same calibrated model, adaptive selection reduced the score error of random items by a median of 21.1% (C4 not supported; 4 benchmarks significant). Against the ATLAS-style 3PL CAT the median improvement in score error was 26.0% (C7 not supported). After 50 items, EOST-M's score error was 29.6% below tinyBenchmarks' gp-IRT (C8 not supported) and 44.8% below MINCE's subset mean (C9 supported); Table 23 gives the errors at 25, 50 and 100 items. The adaptive gain over Random:mg grew with the panel (median 17.2% with four calibrators, 29.3% with sixteen) and was similar for

Table 22: Study 2, Bayesian CATs against EOST-M at $N = 4$ (post hoc correction of the burn-in and exploratory joint model; same targets, panels, holdouts and chains). Equivalence: 90% bootstrap interval of the relative difference in held-out log loss within ±1% (rule of C3: at least six benchmarks); median relative difference of log loss (positive: Bayesian CAT worse); benchmarks with a Holm-significant difference favouring EOST-M / the Bayesian CAT; the same for the score error (median relative improvement of EOST-M); mean coverage of the 90% score interval (EOST-M: 0.78).

	held-out log loss			score IAE		
Variant	equiv.	median %	sig.	median %	sig.	coverage
PDM Bayesian, BMI (frozen, burn-in 10)	4/8	+0.33	0/0	+2.6	0/0	0.82
PDM Bayesian, EIG (frozen, burn-in 10)	4/8	+0.45	0/0	+6.5	0/0	0.79
PDM Bayesian, BMI (burn-in 60)	4/8	+0.21	0/0	+1.6	0/0	0.81
PDM Bayesian, EIG (burn-in 60)	2/8	+0.45	1/0	+5.8	1/0	0.79
joint fully Bayesian, BMI (exploratory)	4/8	−0.03	0/0	+1.0	0/0	0.79
joint fully Bayesian, EIG (exploratory)	4/8	+0.17	0/0	−0.8	0/0	0.79

deterministic and aggregate verdicts (21.1% and 24.3%). The ATLAS-style test had the largest score error of all model-based policies and covered the score only 0.48 of the time: with eight calibrators its item parameters are plug-in estimates, as in plug-in Fisher CAT, and its randomesque choice among the five most informative items trades efficiency for exposure control (ATLAS itself calibrates on thousands of models; our implementation follows its model and selection rule, not its code). tinyBenchmarks and MINCE choose their items without looking at the target. With eight calibrators, tinyBenchmarks' K-Means cannot always find as many anchors as its budget: its item embeddings come from the panel's plug-in fit and can only take as many distinct values as there are distinct response patterns among the items, which is far fewer than the 2^8 possible ones when many items are solved by all or by none of the calibrators. At a budget of 100 items, fewer anchors were available in some cells of 6 of the 8 benchmarks and in 100% of the cells of the most affected one (Table 24). A review of the code during the run found this, and a post hoc audit decided before the results were seen shows that it does not change C8: at the 50-item budget it affected at most 7% of the cells of a benchmark, and restricting C8 to full-budget cells, or comparing with EOST-M after the number of items tinyBenchmarks actually used, again gave 4 significant benchmarks under the frozen Holm family, as in the frozen analysis (median 29.6%). Because the frozen code did not record how many anchors were used, they were counted post hoc with the same seeds; the recomputed errors equal the frozen ones exactly (maximum difference 0). MINCE's own stopping rule asked for a median of 158 items and then had a score error of 2.50 points, more than EOST-M after 100 items (2.13). Table 23 also lists tinyBenchmarks' p-IRT and anchor-only estimators; C8 concerns gp-IRT, the estimator its authors recommend.

Table 23: Study 2: absolute error of the full-bank score (percentage points) after 25, 50 and 100 items at $N = 8$, averaged over targets and benchmarks. tinyBenchmarks and MINCE select their items without adapting to the target.

Method	25 items	50 items	100 items
EOST-M (EIG:mg)	3.39	2.81	2.13
ATLAS-style 3PL CAT	4.46	4.20	3.79
Random:mg	4.78	3.34	2.42
tinyBenchmarks gp-IRT	4.96	4.39	4.49
tinyBenchmarks p-IRT	4.69	3.92	3.31
tinyBenchmarks anchors	6.23	5.62	5.37
MINCE subset mean	6.71	4.69	3.19
Naive (Wilson)	7.05	4.88	3.26

Table 24: Study 2, post hoc audit of tinyBenchmarks at $N = 8$: percentage of cells (target $\times$ replicate) in which fewer anchor items than the budget were available: K-Means returns at most one anchor per distinct item embedding, and with eight calibrators the embeddings take only as many distinct values as there are distinct response patterns among the items, often fewer than 100; last column: median number of anchors actually used at a budget of 100.

	cells with fewer anchors than budget (%)			anchors used
Benchmark	25	50	100	at budget 100
GSM8K	1	7	76	82
ARC-C	0	0	0	100
HellaSwag	0	0	29	100
WinoGrande	0	0	1	100
MMLU	0	0	0	100
SWE-bench V.	0	2	100	71
LiveCodeBench	0	0	98	67
CRUXEval-O	0	0	8	100

7.4 Calibrated uncertainty

Model-based intervals remained too narrow with eight calibrators (Figure 11): EOST-M covered the full-bank score 0.74 of the time, the PDM Bayesian CAT 0.76 (after the burn-in correction), the Rasch-type test 0.81 and plug-in Fisher CAT 0.59, whereas Random:mg reached 0.88 and Naive's Wilson interval 0.90. Leave-one-calibrator-out recalibration raised EOST-M's coverage to 0.86 (from 0.81 on CRUXEval-O to 0.91), inside $[0.85, 0.95]$ on 5 benchmarks, one short of the frozen criterion (C5a not supported). It did not improve the interval score, which penalizes width as well as misses (C5b not supported; significantly worse on HellaSwag and WinoGrande): the wider intervals cost about as much as the misses they removed. Recalibrated EOST-M intervals had a much better interval score than Naive's Wilson intervals (C6 supported, median 35.0%), because at a coverage of 0.86 they were much narrower (mean width 0.134 against 0.235). The same recalibration brought the Rasch-type test inside $[0.85, 0.95]$ on all eight benchmarks (0.91 on average, between 0.86

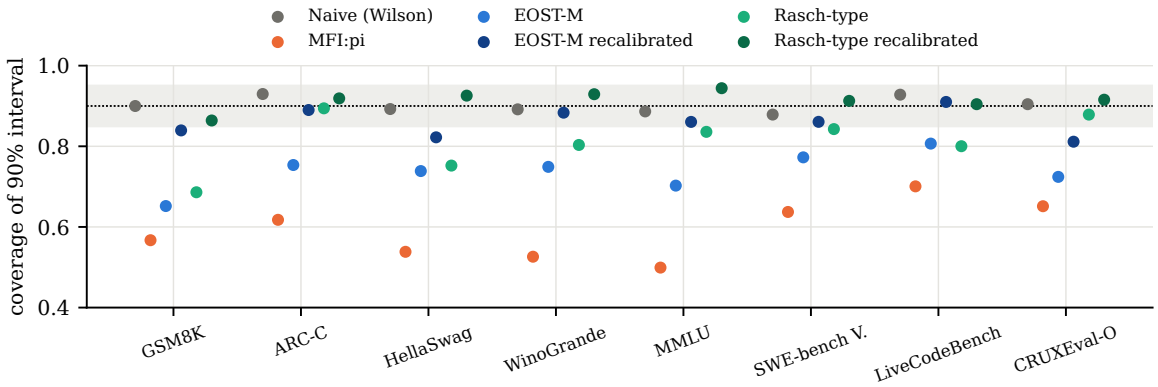


Figure 11: Study 2: coverage of the 90% score interval per benchmark at $N = 8$ (mean over steps, replicates and targets). Shaded band: the $[0.85, 0.95]$ criterion of hypothesis C5a; dotted: nominal.

and 0.94), but with intervals about twice as wide as recalibrated EOST-M's (0.271) and a worse interval score on 5 benchmarks (none better). Recalibration needs calibrators to serve as pseudo-targets and helped more as the panel grew: EOST-M's recalibrated coverage was 0.82, 0.86 and 0.88 at $N = 4$, 8 and 16, and its interval score was worse than the raw interval's at $N = 4$ (0.235 against 0.209) but better at $N = 16$ (0.179 against 0.184). Mixing one uniformly drawn item in four into the adaptive test raised coverage only to 0.77.

7.5 Temporal validation and robustness

Temporal validation. On the 31 SWE-bench Verified submissions of the latest quarter, calibrated only on earlier submissions, 10 of the 11 frozen contrasts favoured EOST-M at a Holm-adjusted $p < 0.05$ within the component and none favoured a competitor; only the recalibration contrast C5b was not significant ($p = 0.116$). EOST-M's held-out log loss was 6.7% lower than plug-in Fisher CAT's (27 of 31 targets) and 3.1% lower than the Rasch-type test's, and its score error 26.2% lower than Random:mg's. Coverage was 0.77 raw and 0.86 recalibrated (Rasch-type: 0.77 and 0.87; Naive: 0.90).

Synthetic banks. Where the fitted 2PL is correct, recalibration moved coverage away from the nominal level (0.86 raw, 0.82 recalibrated at $N = 8$), as it did on the bank with guessing (0.89 and 0.86); it moved coverage towards it only on the bank with a second ability dimension (0.81 and 0.85). Recalibration from a handful of pseudo-targets is noisy, and it pays only when the model's own intervals are clearly too narrow.

Statistical conventions. As in study 1, the direction of a significant contrast is read from the mean difference while the test is rank-based; across both studies 0 of the 815 Holm-significant contrasts had a mean difference and a rank-biserial correlation of opposite signs.

Table 25: Stability of the empirical-Bayes hyperparameters (post hoc; three panels per benchmark and panel size). Medians over benchmarks of: the standard deviation across panels (between), the bootstrap standard deviation over items with abilities fixed (items), the absolute difference between fits on two random halves of the items (halves), and at $N = 8$ the bootstrap standard deviation over calibrators (calibrators).

	$N = 4$			$N = 8$			$N = 16$			$N = 8$
	between	items	halves	between	items	halves	between	items	halves	calibrators
μ_b	0.17	0.09	0.16	0.07	0.07	0.11	0.08	0.07	0.07	0.10
σ_b	0.11	0.09	0.10	0.16	0.07	0.10	0.18	0.07	0.05	0.15
μ_a	0.06	0.05	0.06	0.08	0.04	0.07	0.07	0.04	0.04	0.14
σ_a	0.05	0.03	0.05	0.05	0.03	0.03	0.04	0.03	0.04	0.06

7.6 Robustness of the empirical-Bayes hyperprior

The empirical-Bayes hyperprior is load-bearing: with a fixed weakly-informative prior, the study-1 gain of EOST-M over plug-in Fisher CAT halves (Section 6.7). Because it is estimated from the same panel, it could act as an opportunistic tuning parameter. Stability and predictive checks (post hoc) and three alternatives run in study 2 under its frozen plan address this.

Stability. Hyperparameters are re-estimated on bootstrap resamples of the items (abilities held fixed), on two random halves of the items, and at $N = 8$ on bootstrap resamples of the calibrators (Table 25). Resampling items moves them by at most 0.07 (standard deviation, $N = 8$): they pool hundreds of items and are well determined given the panel. What varies is the panel: resampling calibrators moves the spread of the log-discriminations by 0.06 and that of the difficulties by 0.15, comparable to the variation between panels (0.05 for σ_a). The hyperprior therefore reflects which systems are in the panel, not noise in the items.

Prior predictive checks. For each of the 72 panels (three per benchmark and panel size), 200 response matrices were simulated from the fitted hyperprior given the calibrators' abilities and compared with the observed matrix on three statistics; under a correct model about 5% of the two-sided predictive p-values would fall below 0.05. The share of items that every calibrator or no calibrator answers correctly matters most for Proposition 5, because such items are located by the prior alone. It is reproduced on average (median observed 0.42, simulated 0.42) but not panel by panel: $p < 0.05$ for 19% of the panels (observed share higher than simulated in 8%, lower in 11%), most of them on HellaSwag and MMLU. The spread of the items' proportions correct is reproduced (observed 0.32, simulated 0.31; $p < 0.05$ for 3%). The mean item–rest correlation is mostly lower than simulated (observed 0.39, simulated 0.41; $p < 0.05$ for 32% of the panels, 25% below and 7% above), again mainly on the multiple-choice banks of the Open LLM Leaderboard: real systems agree with each other less, item by item, than a unidimensional 2PL with independent items implies. The hyperprior therefore captures the distribution of item locations on which EOST-M

relies, while the response model is misspecified in its dependence structure, one reason why model-based score intervals are too narrow (Section 7.4).

Leave-item-block-out, other families, hierarchical Bayes. Study 2 ran EOST-M with a cross-fitted hyperprior (estimated, for every item, on the other half of the items, so that no item informs its own prior) and with Student-t hyperprior families ($\nu = 4$) for both item parameters; the Bayesian CATs integrate the hyperparameters instead of plugging them in, which is the hierarchical Bayesian benchmark. Relative to EOST-M, the cross-fitted variant left held-out log loss unchanged (median relative difference 0.0%) and the Student-t variant made it slightly worse on every benchmark (median 0.4%), with no benchmark significant, and the Bayesian CATs were not significantly different on any benchmark (Section 7.2). The gain of EOST-M therefore does not come from fitting the prior to the very items it is applied to, nor from the Gaussian family or the plug-in hyperparameters.

8 Robustness follow-ups after review

A second review raised two methodological questions about study 2—whether the response-derived clusters favour EOST-M, and whether the Bayesian CATs were estimated with enough Markov chain Monte Carlo—and asked whether the effects generalize across benchmarks under a less conservative criterion than the frozen one. Three follow-ups answer them (RF1–RF3); their plans and decision rules were written in the research log before they were run, and none of them changes a frozen decision. Further, exploratory follow-ups of the MCMC question (RF2b–RF2d) were designed after the diagnostics of RF2 had been seen, and a sensitivity analysis of RF3 (RF3b) after a third review; their plans, too, were written before they were run.

8.1 Clusters that never see the target's responses (RF1)

The strict clusters join systems whose 2PL residuals correlate, and those residuals are computed from every system's full response vector, the target's included. Excluding the calibrators that resemble a target could make its panel harder to calibrate and so favour a method built for calibration uncertainty. We kept the frozen targets, holdouts and panel seeds and changed only the systems excluded from each panel: heuristic families alone (no response data), leave-bank-out clusters (residual correlations computed on the other benchmarks that contain the same systems), and the frozen construction at thresholds 0.4, 0.6 and 0.7. Recomputing the frozen construction reproduced all 4,005 frozen rows exactly. Under every alternative rule, R2 and C1 kept their frozen median improvements (5.9% and 2.4%), R2 remained positive on all eight benchmarks with 4–6 of them Holm-significant, and C1 remained positive on seven with 5 significant, none against (Table 26; per benchmark in Table 27): the robustness criterion fixed in advance was met. The response-derived links matter only on the Open LLM Leaderboard banks, where hundreds of fine-tunes share a few base models; on the other benchmarks the clusters are essentially the heuristic families (Table 28). Adaptivity (C4) had 3 significant benchmarks under the alternative rules (frozen: 4), with medians between 18.3% and 20.5%.

Table 26: Robustness of study 2 to the clustering (follow-up RF1, planned before running). Frozen targets, holdouts and panel seeds; only the systems excluded from each target's panel change. For each exclusion rule: targets used, and for R2 (EOST-M vs plug-in Fisher CAT, log loss), C1 (vs the Rasch-type plug-in test, log loss) and C4 (vs Random:mg, score IAE): median relative improvement of EOST-M over benchmarks (%), benchmarks with a positive difference, and benchmarks Holm-significant for / against EOST-M (frozen 88-test family with the rule's p-values substituted).

		R2			C1			C4	
Panel exclusion rule	targets	median	pos.	sig.	median	pos.	sig.	median	sig.
frozen (joint residuals, $r \geq 0.5$)	267	+5.9	8/8	5/0	+2.4	7/8	5/0	+21.1	4/0
metadata only (no responses)	267	+5.9	8/8	4/0	+2.4	7/8	5/0	+20.5	3/0
leave-bank-out residuals	267	+5.9	8/8	5/0	+2.4	7/8	5/0	+19.6	3/0
joint residuals, $r \geq 0.4$	267	+5.9	8/8	6/0	+2.4	7/8	5/0	+19.5	3/0
joint residuals, $r \geq 0.6$	267	+5.9	8/8	4/0	+2.4	7/8	5/0	+20.1	3/0
joint residuals, $r \geq 0.7$	267	+5.9	8/8	4/0	+2.4	7/8	5/0	+18.3	3/0

Table 27: Follow-up RF1 per benchmark: relative improvement (%) of EOST-M in held-out log loss over plug-in Fisher CAT (R2) and over the Rasch-type plug-in test (C1) under each panel-exclusion rule; * Holm-significant (frozen 88-test family with the rule's p-values substituted). HS: HellaSwag; WG: WinoGrande; LCB: LiveCodeBench.

Rule		GSM8K	ARC	HS	WG	MMLU	SWE-V	LCB	CRUX-O
metadata only	R2	+4.8*	+4.5	+12.4*	+5.3	+7.1*	+6.9*	+6.4	+5.4
metadata only	C1	+1.9	+1.9*	+6.7*	+2.9*	+4.9*	+1.8	+3.4*	−0.4
leave-bank-out	R2	+4.0	+4.6*	+12.2*	+5.2*	+6.7*	+6.9*	+6.4	+5.4
leave-bank-out	C1	+1.8	+1.9*	+6.4*	+2.9*	+5.1*	+1.8	+3.4*	−0.4
joint, $r \geq 0.4$	R2	+5.1*	+4.5*	+12.5*	+5.1*	+7.6*	+6.9*	+6.4	+5.4
joint, $r \geq 0.4$	C1	+1.8	+1.9*	+6.7*	+2.8*	+3.9*	+1.8	+3.4*	−0.4
joint, $r \geq 0.6$	R2	+4.8	+4.6*	+12.2*	+5.2	+7.1*	+6.9*	+6.4	+5.4
joint, $r \geq 0.6$	C1	+1.9	+1.9*	+6.4*	+3.0*	+5.0*	+1.8	+3.4*	−0.4
joint, $r \geq 0.7$	R2	+4.8*	+4.5	+12.2*	+5.3	+7.0*	+6.9*	+6.4	+5.4
joint, $r \geq 0.7$	C1	+1.7	+1.9*	+6.6*	+2.9*	+5.1*	+1.8	+3.4*	−0.4

8.2 MCMC quality of the Bayesian CATs (RF2–RF2d)

On six study-2 targets per benchmark, drawn with a fixed seed (replicate 0, $N = 8$; 48 panels), we ran the post hoc copy of the frozen sampler, which reproduces it exactly, with four chains. Chain 1 started at EOST-M's estimates, as the frozen arm did; chains 2–4 started from dispersed points (abilities perturbed by $\mathcal{N}(0, 0.5^2)$; $\mu_b \pm 0.5$; $\sigma_b \times 0.7$ or 1.4; $\mu_a \pm 0.3$; $\sigma_a \times 0.7$ or 1.4). Each chain ran 250 burn-in sweeps and 300 further sweeps, every third of which was retained (400 retained draws in all). Diagnostics use the 300 post-burn-in sweeps of each chain: rank-normalized split $\hat{R}$ (the larger of the bulk and folded versions), bulk effective sample sizes (ESS) with Geyer's initial monotone sequence, and Monte Carlo standard errors from the ESS of the raw draws (Vehtari et al., 2021; Geyer, 1992). The

Table 28: Clusters under each panel-exclusion rule of follow-up RF1: number of clusters / size of the largest cluster. The five Open LLM Leaderboard banks share their systems and their joint clusters; leave-bank-out clusters differ per leaderboard bank (GSM8K shown).

Rule	Open LLM LB (GSM8K)	SWE-bench V.	LCB	CRUX-O
frozen, $r \geq 0.5$	176/39	92/64	15/4	12/9
metadata only	227/13	94/64	15/4	12/9
leave-bank-out	147/72	94/64	15/4	12/9
joint, $r \geq 0.4$	134/113	92/64	15/4	12/9
joint, $r \geq 0.6$	196/34	92/64	15/4	12/9
joint, $r \geq 0.7$	209/27	92/64	15/4	12/9

posterior-draw marginal arms average the item posteriors over the 400 pooled draws; the joint arms mix over every tenth draw (40). Table 29 gives the convergence per benchmark, Table 31 the arms on the same cells and Figure 12 the traces of the first panel of each benchmark.

The overall location converged ($\hat{R} < 1.05$ on 92% of panels, median bulk ESS 778), and so did μ_b on most panels (90%), but σ_b, μ_a and σ_a did not: $\hat{R} < 1.05$ on at most 8% of panels, median $\hat{R}$ 1.27–1.50, median bulk ESS 8–11, and Monte Carlo standard errors of 30–38% of their posterior standard deviations. Neither did the ability scale, the standard deviation of the calibrators' abilities ($\hat{R} < 1.05$ on 0% of panels): the ratio of the largest to the smallest chain mean had a median of 1.21 (at most 1.51). Multiplying all abilities and difficulties by a constant and dividing the discriminations by it leaves the likelihood unchanged, so with eight calibrators only their $\mathcal{N}(0, 1)$ prior fixes the scale of the calibration; the frozen sampler has a translation move but no move along this ridge, on which the spread of the calibrators' abilities moved together with σ_b (median correlation 0.88) and μ_a (-0.86). The four chains' held-out log losses differed by more than 5% on 10% of panels (at most 43%). Under the rule fixed in advance the study-2 statement about the Bayesian CATs was not confirmed: on individual cells the long chains changed the held-out log loss by a median of 0.61–1.02% relative to a burn-in of 60, more than the 0.5% allowed, although the signed median changes were small (-0.25 to 0.13%). As the better estimate, the long-chain arms differed from EOST-M by a median of -0.45 to 0.20% in held-out log loss, with the benchmarks split (Table 31); on these cells they covered the score 0.86 of the time, against 0.78 for EOST-M.

Exploratory follow-ups, designed after these diagnostics and each planned before it was run, added the missing moves. RF2b added to every sweep a Metropolis–Hastings move along the scale ridge, applied to the posterior of the abilities and hyperparameters with the item parameters summed out on the grid: the abilities, μ_b and σ_b are multiplied by s and μ_a is decreased by $\log s$, with $\log s = \pm 0.2$ or ± 0.4 (multiples of the step of the log-discrimination grid, so that the move maps that grid onto itself); the acceptance ratio contains the Jacobian s^{N+2}, and the item parameters are then drawn from their exact conditional, which makes the scheme a partially collapsed Gibbs sampler (van Dyk and Park, 2008) with a discrete form of parameter expansion (Liu and Wu, 1999). With this move (median acceptance 21%;

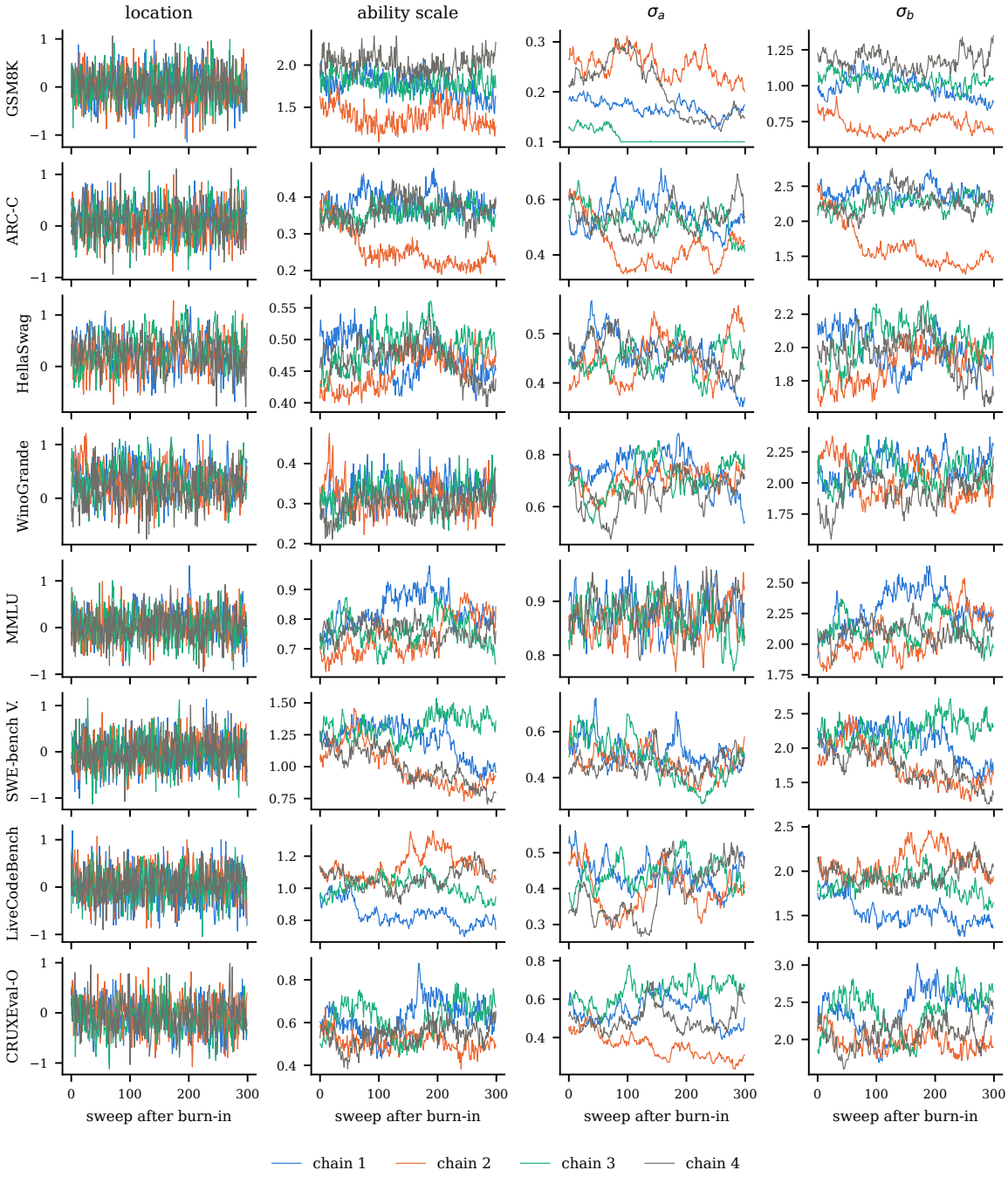


Figure 12: Follow-up RF2: traces of the overall location (mean calibrator ability), the ability scale (standard deviation of the calibrators' abilities), σ_a and σ_b for the first selected panel of each benchmark; four chains (chain 1 starts at EOST-M's estimates), 300 sweeps after a burn-in of 250.

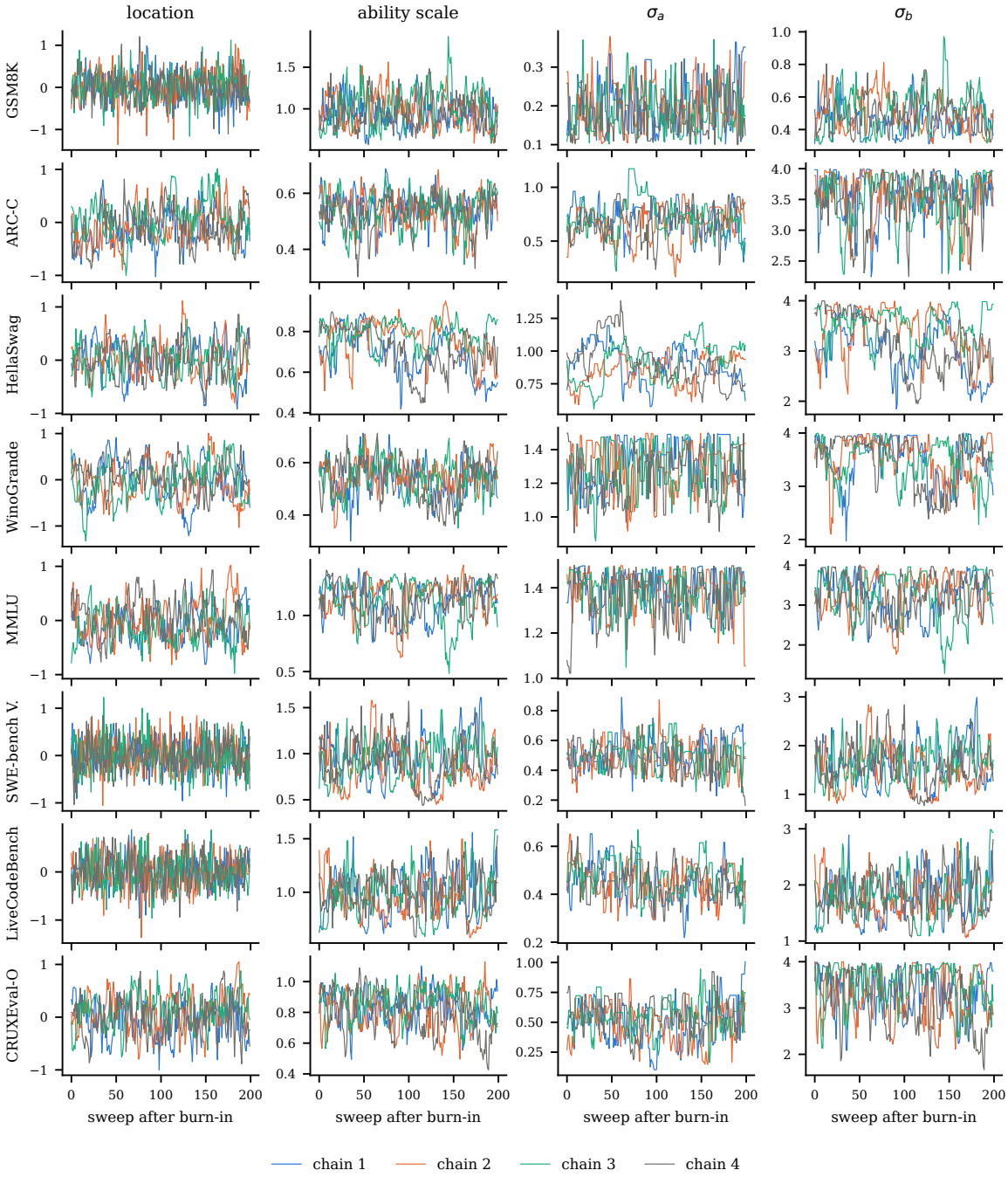


Figure 13: Exploratory follow-up RF2d (exact sampler of the model of Lemma 1, with the hyperparameters, the scale and the location moved with the item parameters summed out): traces of the same quantities and panels as in Figure 12; four chains, 200 sweeps after a burn-in of 150.

Table 29: MCMC convergence of the Bayesian CATs' sampler (follow-up RF2: the frozen sampler with long chains; 6 targets per benchmark, replicate 0, $N = 8$; four chains, one from EOST-M's estimates and three from dispersed starts; burn-in 250, then 300 sweeps per chain with every third retained, 400 retained draws). Per benchmark and parameter: number of the six panels with rank-normalized split $\hat{R} < 1.01$ / $\hat{R} < 1.05$ / bulk ESS ≥ 100. Location: mean calibrator ability; scale: standard deviation of the calibrators' abilities (explanatory, not in the plan of RF2); μ_a, σ_a: mean and standard deviation of the log discriminations, μ_b, σ_b: of the difficulties. Last two columns: median and largest relative spread (%) of the held-out log loss of the PDM Bayesian CAT (BMI) across the four chains. Summary rows over all panels: percentages; median (maximum) $\hat{R}$; median (minimum) bulk ESS; median Monte Carlo standard error as a percentage of the posterior standard deviation.

	panels with $\hat{R} < 1.01$ / $\hat{R} < 1.05$ / ESS ≥ 100						chain spread (%)	
Benchmark	location	μ_b	σ_b	μ_a	σ_a	scale	median	max
GSM8K	6/6/6	0/3/2	0/0/0	0/0/0	0/0/0	0/0/0	8.9	43.3
ARC-C	1/6/6	6/6/6	0/0/0	0/0/0	0/0/0	0/0/0	1.0	1.7
HellaSwag	0/3/3	2/6/5	0/0/0	0/0/0	0/0/0	0/0/0	1.7	4.2
WinoGrande	0/5/4	4/5/5	0/0/0	0/1/0	0/0/0	0/0/0	0.9	1.3
MMLU	5/6/6	5/6/6	0/0/0	0/0/0	0/3/2	0/0/0	0.4	3.2
SWE-bench V.	6/6/6	6/6/6	0/0/0	0/0/0	0/0/0	0/0/0	1.1	2.6
LiveCodeBench	2/6/5	1/5/5	0/0/0	0/0/0	0/0/0	0/0/0	1.7	5.6
CRUXEval-O	3/6/6	6/6/6	0/0/0	0/0/0	0/1/0	0/0/0	0.8	5.5
all panels (%)	48/92/88	62/90/85	0/0/0	0/2/0	0/8/4	0/0/0	1.0	43.3
$\hat{R}$, median (max)	1.01 (1.10)	1.01 (1.21)	1.47 (2.46)	1.50 (2.77)	1.27 (3.04)	1.40 (2.44)		
bulk ESS, median (min)	778 (35)	612 (13)	8 (5)	8 (5)	11 (5)	9 (5)		
MCSE / sd (%)	4	4	37	38	30	35		

Table 30), $\hat{R} < 1.05$ was reached on 75% of panels for the ability scale, 73% for σ_b and 79% for μ_a (RF2: 0, 0 and 2%), but only on 10% for σ_a (median bulk ESS 11), so the convergence criterion fixed for RF2b was not met; the four chains' held-out log losses now differed by at most 6.0% (RF2: 43%). Its arms differed from EOST-M by a median of 0.01 to 0.29% in held-out log loss (descriptive, since the chains had not converged).

The frozen sampler's conjugate hyperparameter draws, which RF2b kept, mix slowly when each item carries little information, and they treat the grid prior as a continuous density: they condition on the sampled grid points, whereas EOST-M normalizes the prior over the grid, and the two differ when a standard deviation approaches the grid step (0.2 for $\log a$), where the unnormalized version accumulates at the lower bound of σ_a (2 of RF2's 192 chains and 3 of RF2b's 192 spent most of their sweeps there; Figure 12, GSM8K). RF2c and RF2d therefore sample the model of Lemma 1 exactly. RF2c replaces the conjugate draws by random-walk Metropolis moves on μ_a, $\log \sigma_a$, μ_b and $\log \sigma_b$ with the item parameters summed out (two rounds per sweep; cheap because the grid is the product of a discrimination axis and a difficulty axis; step sizes adapted during burn-in only), keeps the scale move and the frozen ability update, and multiplies the weights of the translation move by the factor $(Z(\eta)/Z(\eta'))^I$ that the normalized prior requires. RF2c was stopped after 14 panels: on GSM8K every

Table 30: MCMC convergence of the exploratory samplers (follow-ups RF2b and RF2d; same panels, starting points and columns as Table 29; 400 retained draws). Upper block (RF2b): a Metropolis–Hastings move along the scale ridge added to every sweep of the frozen sampler. Lower block (RF2d): the model of Lemma 1 (item prior normalized over the grid) sampled with the hyperparameters, the scale and the location moved by Metropolis–Hastings steps with the item parameters summed out.

	panels with $\hat{R} < 1.01$ / $\hat{R} < 1.05$ / ESS ≥ 100						chain spread (%)	
Benchmark	location	μ_b	σ_b	μ_a	σ_a	scale	median	max
RF2b: with the scale move, 4 chains, burn-in 200, 200 further sweeps per chain								
GSM8K	6/6/6	5/6/6	0/5/4	1/5/5	0/0/0	1/5/4	1.2	6.0
ARC-C	6/6/6	6/6/6	0/5/5	0/4/5	0/1/0	0/5/3	1.1	3.1
HellaSwag	1/2/2	2/6/5	0/1/1	0/0/1	0/0/0	0/1/1	2.0	5.6
WinoGrande	4/6/6	6/6/6	0/3/3	0/6/5	0/0/0	0/3/2	0.6	1.6
MMLU	6/6/6	5/6/6	0/5/5	0/5/6	0/4/0	1/5/5	0.5	1.3
SWE-bench V.	6/6/6	5/6/6	0/6/6	1/6/6	0/0/0	0/6/6	0.7	4.7
LiveCodeBench	5/6/6	5/6/6	0/4/4	0/6/6	0/0/0	0/5/5	1.9	3.0
CRUXEval-O	5/6/6	6/6/6	1/6/6	1/6/6	0/0/0	2/6/6	0.7	2.3
all panels (%)	81/92/92	83/100/98	2/73/71	6/79/83	0/10/0	8/75/67	1.0	6.0
$\hat{R}$, median (max)	1.00 (1.09)	1.00 (1.04)	1.04 (1.25)	1.03 (1.32)	1.31 (2.03)	1.03 (1.26)		
bulk ESS, median (min)	552 (35)	544 (97)	153 (13)	171 (10)	11 (5)	132 (12)		
MCSE / sd (%)	4	4	8	8	32	9		
RF2d: exact sampler of Lemma 1's model, 4 chains, burn-in 150, 200 further sweeps per chain								
GSM8K	6/6/6	3/6/6	0/2/3	0/3/3	1/6/5	0/4/3	1.3	26.0
ARC-C	1/5/5	0/4/5	0/6/5	0/2/1	0/1/1	0/6/5	0.3	0.8
HellaSwag	1/4/2	0/2/0	0/1/0	0/2/2	0/2/0	0/1/0	0.5	3.6
WinoGrande	0/4/3	0/0/0	0/0/0	0/0/0	0/5/5	0/2/1	0.7	1.5
MMLU	0/5/4	0/4/3	0/4/1	0/4/1	2/6/6	0/4/1	0.7	1.0
SWE-bench V.	6/6/6	5/6/6	0/2/0	0/1/0	1/6/5	0/1/0	1.2	3.1
LiveCodeBench	5/6/5	2/4/4	0/3/3	0/3/3	0/4/3	0/4/3	1.1	5.8
CRUXEval-O	2/6/6	1/6/4	0/5/4	0/5/3	0/5/5	0/6/4	0.6	5.7
all panels (%)	44/88/77	23/67/58	0/48/33	0/42/27	8/73/62	0/58/35	0.7	26.0
$\hat{R}$, median (max)	1.02 (1.09)	1.03 (1.19)	1.05 (1.25)	1.06 (1.16)	1.03 (1.20)	1.04 (1.21)		
bulk ESS, median (min)	162 (42)	129 (16)	77 (13)	72 (20)	127 (15)	85 (15)		
MCSE / sd (%)	8	9	12	12	9	11		

recorded quantity converged (largest $\hat{R}$ 1.05), but on ARC-C and HellaSwag the overall location did not ($\hat{R}$ up to 2.8), because when the difficulty prior extends beyond the edge of the grid (HellaSwag) the corrected translation weights, raised to the power of the number of items, block almost every shift. RF2d therefore adds a collapsed translation—the abilities and μ_b shifted by 1, 2 or 4 grid steps with the item parameters summed out, which items at the edge of the grid cannot block—and finer scale steps ($\log s \in \{\pm 0.05, \pm 0.1, \pm 0.2, \pm 0.4\}$): it moves the hyperparameters, the scale and the location by Metropolis–Hastings steps with the item parameters summed out. All samplers were checked before they were run: with the new move switched off, RF2b's sampler reproduces RF2's draws exactly, and the stationary laws of the scale moves, the hyperparameter moves and both translation moves match exact computations on small problems. RF2b used a burn-in of 200 sweeps and RF2d one of 150, each followed by 200 sweeps per chain with every second retained (400 draws).

Table 31: Bayesian CATs with frozen, corrected and long chains against EOST-M on the same 48 cells (follow-up RF2; 6 targets per benchmark, replicate 0, $N = 8$). Median over cells of the relative difference in held-out log loss to EOST-M (%; positive: worse than EOST-M); benchmarks whose median favours the arm / EOST-M; benchmarks with a median within ±1%; for the long chains, median signed and median absolute relative difference (%) to the same arm with burn-in 60; mean coverage of the 90% intervals. Exploratory follow-ups RF2b (sampler with the scale move) and RF2d (exact sampler of the model of Lemma 1) are compared with the same arm with burn-in 60. PDM: posterior-draw marginal; joint: joint fully Bayesian (mixture over 40 of the retained draws); BMI: Bayesian maximum information; EIG: expected information gain.

	log loss vs EOST-M			vs burn-in 60 (%)		
Arm	median (%)	banks	±1%	signed	abs.	coverage
EOST-M (empirical Bayes, no chain)	–	–	–	–	–	0.78
frozen arm of study 2: 1 chain, burn-in 10, 8 draws						
PDM, BMI	+0.08	4/4	6	–	–	0.82
PDM, EIG	+0.06	2/6	8	–	–	0.85
post hoc correction: 1 chain, burn-in 60, 8 draws						
PDM, BMI	+0.37	3/5	4	–	–	0.81
PDM, EIG	+0.10	3/5	6	–	–	0.84
joint, EIG	+0.10	4/4	5	–	–	0.81
joint, BMI	+0.11	5/3	7	–	–	0.80
RF2, long chains: 4 chains, burn-in 250, 400 draws (joint: 40)						
PDM, BMI	+0.17	3/5	8	+0.13	1.02	0.86
PDM, EIG	+0.04	4/4	7	+0.11	0.77	0.86
joint, EIG	+0.20	4/4	8	+0.05	0.61	0.86
joint, BMI	−0.45	6/2	7	−0.25	0.75	0.86
RF2b, scale move: 4 chains, burn-in 200, 400 draws (joint: 40)						
PDM, BMI	+0.10	4/4	5	−0.19	0.77	0.85
PDM, EIG	+0.01	4/4	6	−0.05	0.61	0.86
joint, EIG	+0.07	1/7	5	+0.15	0.81	0.76
joint, BMI	+0.29	4/4	7	−0.02	0.76	0.82
RF2d, exact sampler: 4 chains, burn-in 150, 400 draws (joint: 40)						
PDM, BMI	−0.31	5/3	7	−0.69	1.62	0.78
PDM, EIG	+0.15	3/5	6	−0.29	1.62	0.76
joint, EIG	+0.20	2/6	6	−0.16	1.27	0.77
joint, BMI	−0.13	4/4	7	−0.14	1.13	0.79

RF2d came much closer to convergence (median acceptance: translation 68%, scale 47%, hyperparameter moves about 40%). $\hat{R} < 1.05$ held on 88% of panels for the location, 58% for the ability scale, and 67%, 48%, 42% and 73% for μ_b, σ_b, μ_a and σ_a, with medians of $\hat{R}$ between 1.020 and 1.058 across these quantities, a largest value of 1.25 (against 3.04 in RF2), and median bulk ESS between 72 and 162 (Table 30, Figure 13). No chain stayed at the lower bound of σ_a, but on 11 panels σ_b and on 6 panels σ_a spent more than a quarter of the sweeps within 5% of the upper bound of the prior box inherited from the frozen sampler (4 and 1.5), which therefore constrains the posterior there. With $\hat{R} < 1.05$ on only 42% of panels for the slowest quantity, short of the 90% fixed in advance, the convergence criterion was not met, and a fully converged comparator was not obtained. The four chains' held-out

log losses differed by a median of 0.7% (at most 26.0%). Under the model of Lemma 1, the posterior mean of the ability scale exceeded EOST-M's point estimate on 65% of panels (median ratio 1.15): the calibration that EOST-M conditions on has a smaller spread than the posterior.

Across all 18 combinations of Bayesian arm and sampler on these cells, from the frozen chain to RF2d, the median difference from EOST-M in held-out log loss stayed within ±0.45%, with the benchmarks split (RF2d: −0.31 to 0.20%; Table 31 gives every arm). The study-2 comparison did not move as the sampler improved, which makes an artefact of the frozen chain unlikely; our statements about the Bayesian CATs nevertheless concern the implementations studied, not an exactly computed fully Bayesian test. The RF2d arms covered the score as often as EOST-M did on the same cells (0.76–0.79 against 0.78), so integrating the calibrators' abilities and the hyperprior did not repair the under-coverage; the higher coverage of RF2's long chains (0.86) did not survive the exact model. In score error the RF2d arms were lower than EOST-M by a median of 5–11% over cells but on only 4–5 of 8 benchmarks, a possible advantage that 48 cells cannot establish.

8.3 Effects across benchmarks (RF3, RF3b)

The frozen rule asks for significance on at least five of eight benchmarks after a Holm correction over 88 tests, so an effect can favour EOST-M on all eight benchmarks and still be declared not supported. A random-effects meta-analysis asks a different, secondary question—does an effect point the same way across benchmarks?—without changing any decision (Table 32). Treating the benchmarks as independent, the pooled improvement excluded zero for 10 of the 11 superiority hypotheses, including the 4 that missed the frozen

Table 32: Cross-benchmark effect of each study-2 superiority hypothesis (follow-up RF3, secondary; no frozen decision changes): random-effects pooled relative improvement of EOST-M (%) over the eight benchmarks treated as independent, Hartung–Knapp–Sidik–Jonkman 95% interval, 95% prediction interval for a new benchmark, I^2 (%), benchmarks with a positive effect and exact two-sided sign-test p; "sources": 95% interval when the five Open LLM Leaderboard benchmarks, which share their 395 systems, count as one source (RF3b; four sources, 3 degrees of freedom); frozen decision. *Recalibrated EOST-M interval.

EOST-M versus	pooled	95% CI	95% PI	I^2	pos.	sign p	sources	frozen
R2 plug-in 2PL, log loss	+6.0	[4.7, 7.2]	[4.3, 7.6]	7	8/8	0.008	[5.7, 7.6]	yes
R3 Naive, score IAE	+50.5	[37.0, 64.0]	[24.2, 76.8]	87	8/8	0.008	[43.5, 57.5]	yes
R4 plug-in 2PL, |cov − 0.9|	+41.9	[35.3, 48.6]	[33.6, 50.3]	6	8/8	0.008	[34.0, 45.5]	yes
C1 Rasch-type plug-in, log loss	+2.6	[1.2, 4.1]	[−0.2, 5.5]	71	7/8	0.070	[−0.4, 5.0]	yes
C2 Rasch-type marginal, log loss	+2.0	[0.8, 3.2]	[−0.6, 4.7]	75	8/8	0.008	[1.4, 2.8]	no
C4 Random:mg, score IAE	+23.4	[9.8, 37.1]	[−16.5, 63.3]	88	8/8	0.008	[−5.0, 52.6]	no
C5b raw interval*, interval score	−5.8	[−19.9, 8.3]	[−42.7, 31.0]	82	4/8	1.000	[−16.1, 10.0]	no
C6 Naive (Wilson)*, interval score	+29.8	[9.9, 49.7]	[−17.9, 77.5]	91	7/8	0.070	[18.7, 51.1]	yes
C7 ATLAS-style, score IAE	+27.1	[11.2, 43.0]	[−20.0, 74.2]	93	8/8	0.008	[−1.1, 42.4]	no
C8 tinyBenchmarks, error @50	+34.5	[20.0, 49.0]	[−3.9, 72.9]	80	8/8	0.008	[−1.8, 76.7]	no
C9 MINCE, error @50	+41.9	[26.0, 57.8]	[1.2, 82.6]	85	8/8	0.008	[34.1, 49.3]	yes

threshold by one benchmark (C2, C4, C7, C8; I^2 75–93%), and the 95% prediction interval for a new benchmark excluded zero only for R2, R3, R4, C9. But five benchmarks share the same 395 Open LLM Leaderboard systems, a dependence the model ignores. Counting them as one source (RF3b: 4 sources, standard error from a bootstrap over the shared systems), 6 of the 11 intervals excluded zero (R2, R3, R4, C2, C6, C9); C1, C4, C7, C8 stayed positive in 3–4 of the 4 sources but with intervals including zero, so of the four near misses only C2 kept an interval above zero. The dependence through shared systems was small (median correlation 0.02); the loss comes from having four sources and from their heterogeneity. Most effects thus point the same way across benchmarks; the analysis does not show that they generalize to a new one.

9 What adaptive evaluation costs

Adaptive testing saves items only after somebody has answered the whole bank: the calibration panel. Counting one administered item as the unit of cost, evaluating M new systems costs MI with the full bank of I items, and $NI + MT$ with a fresh panel of N calibrators and adaptive tests of T items, so adaptive testing pays off once $M > NI/(I - T)$. With eight calibrators and $T = 100$ this break-even lies between 8.4 and 10 new systems on every benchmark (Table 33), and for 100 new systems the total cost is about 16% of full evaluation.

The fairer comparison is with uniform subsampling, which needs no panel at all. At matched precision, EOST-M needed a median of 22.5 items to reach the score error that 100 uniformly sampled items reach with the sample mean, so it saves about 78 items per system but must first pay NI; the break-even is then a median of 122 new systems (from 67 to 352 across benchmarks). Against Random:mg, which pays for the same panel, adaptivity alone saves items from the first system on 7 of 8 benchmarks: EOST-M matched 100 random items scored by the same model after a median of 65 items. Two situations change the arithmetic. When the calibrators' full results already exist—public leaderboards, previous internal evaluations—the panel is free and adaptive testing is cheaper from the first system. When items are expensive to administer and a benchmark is used for many systems, as for agentic benchmarks, the per-system saving dominates quickly. The computation itself is negligible: calibrating a panel and running a 100-item test takes seconds on one CPU core (Section 5).

Table 33: Cost of adaptive evaluation with a fresh calibration panel of $N = 8$ systems that must first answer all I items (study 1). Break-even: number of new systems from which EOST-M with 100 items costs less than full evaluation; items for which EOST-M matches the score error of 100 uniformly sampled items scored by the sample mean (Naive) or by the same calibrated model (Random:mg), and the resulting break-even against Naive. When the calibrators' full results already exist (public leaderboards), the calibration cost vanishes and EOST-M is cheaper from the first system.

Benchmark	I	break-even vs full evaluation	items to match Naive@100	break-even vs Naive	items to match Random:mg@100
GSM8K	1319	8.7	70	352	80
ARC-C	1172	8.7	20	117	90
HellaSwag	2000	8.4	8	174	25
WinoGrande	1267	8.7	20	127	60
MMLU	2000	8.4	15	188	100
SWE-bench V.	500	10.0	40	67	70
LiveCodeBench	1055	8.8	25	113	60
CRUXEval-O	800	9.1	25	85	60

10 Discussion

Three messages. The evidence supports the three messages of the introduction, each with a qualification. (1) Small panels break plug-in adaptive testing: item parameters estimated from four to eight systems and then treated as known—mainly poorly located difficulties, and inflated discriminations (Section 6)—make Fisher-information CAT predict worse and state intervals that cover the score only 0.59 of the time; the effect replicated with independent targets and on later systems, and it fades as the panel grows. (2) Accounting for calibration uncertainty fixes much of this, in more than one way: integrating the item posterior (EOST-M) and constraining the discriminations (a Rasch-type test) are two remedies, integration predicting slightly better and constraint giving wider, better calibrated intervals; integrating the panel-level uncertainty as well, in the Bayesian tests we ran, made no significant difference (equivalence was not shown either). (3) Once uncertainty is handled, the choice among information criteria matters little (descriptively), and the original hypothesis that information beats entropy on held-out prediction is consistent in sign but small. What remains unsolved is calibrated uncertainty: no remedy we tried gave nominal coverage with intervals as narrow as EOST-M's.

Small-panel calibration uncertainty is the problem. With four or eight calibrators, a large fraction of items is answered identically by every calibrator, so that their difficulty is located only by the prior (Figure 3). Plug-in testing treats these poorly located parameters as known: Fisher selection credits such items with information they do not carry, the plug-in posterior becomes overconfident, and the predicted outcomes of the unobserved items inherit the parameter errors, which is the bias floor of Figure 5. The expected response function caps the information of an item whose difficulty is uncertain—at $2/(\pi s_i^2)$ for a Gaussian difficulty posterior with standard deviation s_i (Proposition 5)—and predicts its outcome with

the uncertainty it actually has; the empirical-Bayes hyperprior keeps the predicted success probabilities of unlocated items sensible. Both effects fade as the panel grows (Figure 2 and Figure 4). For the very large panels used by recent work (Li et al., 2026), plug-in Fisher information is an adequate approximation; for new, private or expensive benchmarks, self-generated item pools, or the four-to-eight-system panels that a single laboratory can afford to run, it is not.

Integrating and constraining are two remedies. The post hoc probes of study 1 suggested that constraining the discriminations to a common value is almost as good as integrating their uncertainty. Study 2 tested this under a frozen plan with independent targets. Integration was better on held-out prediction (C1 supported, median 2.4%), more so with sixteen calibrators, where one common discrimination becomes a visible misspecification; the score errors of the two tests were close; and the constrained model's raw intervals were better calibrated because they are wider, not because its estimates are more accurate (Section 6.8). The one-parameter model integrated over its item posterior sat in between (C2, median 1.8%). The practical reading is that a small panel must not be trusted with item-specific discriminations: integrating their uncertainty is preferable when prediction and score accuracy matter most, and constraining them gives a simpler and more conservative test.

Why not always use a fully Bayesian adaptive test? A fully Bayesian adaptive test is the principled answer to calibration uncertainty, and Fink et al. (2025) found it the most accurate correction in simulations with hundreds of calibration responses per item. With eight calibrators it did not help measurably: integrating the calibrators' abilities and the hyperprior changed held-out prediction, score error and coverage by no significant amount, although equivalence within $\pm 1\%$ was not shown either. The form that averages each item over the posterior draws separately (the PDM Bayesian CAT) was, if anything, slightly worse than EOST-M in study 2, which would fit its treating the shared uncertainty about the location of the calibration as independent item noise; but with the better samplers of Section 8 neither form was consistently worse, so we do not read this difference as a property of the posterior-draw approximation. By Lemma 1, EOST-M is the fully Bayesian model with the calibrators' abilities and the hyperprior held fixed; in this regime what matters is the item-level uncertainty that both integrate rather than the panel-level uncertainty that only the fully Bayesian model adds, as far as our data can tell. A fully Bayesian test also needs a Markov chain per panel, whose tuning is a source of error in its own right: our frozen burn-in was too short on some benchmarks, and with eight calibrators the posterior has a ridge in the scale of the calibration along which a standard Gibbs sampler barely moves, so that even long chains did not converge for the scale hyperparameters; an exact sampler with collapsed moves came close but did not meet our convergence criterion either (Section 8). EOST-M is the simpler default. Two leads remain open. The better samplers left held-out log loss essentially unchanged but suggest a possible advantage of Bayesian integration for estimating the score: the near-converged exact sampler was ahead of EOST-M in score error on most of the 48 cells it was run on, but on only about half of the benchmarks, too few to establish it. And panel-level uncertainty may matter more when the calibrators' abilities are poorly determined, for instance with short banks; we did not test this.

The criterion matters less than first hypothesized. The pilot program and study 1 agree on the direction of the original thesis—EIG learns more about competence than predictive entropy—but also bound its size: once both criteria are computed under the same calibration-marginal model, the gain on held-out prediction is small, and EIG, maximum and posterior-weighted Fisher information and the score-targeted criterion are practically interchangeable (Table 10; a descriptive comparison—the failure of H1 is not a test of equivalence). Corollary 4 explains why: entropy and information disagree only through the spread of discriminations among items with similar success probability, and the expected response function shrinks discriminations towards what the panel can resolve. The post hoc one-parameter variant confirms the mechanism: with a common discrimination the EIG–entropy difference vanishes (Section 6.7). For the bank score rather than competence, Proposition 7 shows that uncertainty sampling becomes the right one-step criterion once competence is pinned down relative to the size of the unobserved bank; the score-error curves show both regimes (Section 6.3). This is a negative result about the original hypothesis, and a useful one: what an adaptive test for LLMs must get right is the treatment of calibration uncertainty, not the choice among information criteria.

Honest uncertainty. The review of study 1 asked whether the under-coverage of model-based intervals could be fixed. Study 2 tried four remedies under a frozen plan: integrating the calibrators' abilities and the hyperprior (the Bayesian CATs), leave-one-calibrator-out recalibration, mixing uniformly drawn items into the adaptive test, and recalibrating a constrained model. None gave nominal coverage with intervals as narrow as EOST-M's. The Bayesian CATs' intervals and the mixed test stayed below 0.80; recalibration brought EOST-M to 0.86 on average, inside the frozen band on 5 of 8 benchmarks, with an interval score no better than before; the recalibrated Rasch-type test reached 0.91 with intervals twice as wide. Random items scored by the same model reached 0.88 without any recalibration. The exploratory coverage decomposition of study 1 suggests that the shortfall arises when adaptive selection meets a misspecified model, and the prior predictive checks locate a misspecification in the dependence structure of real systems' verdicts (Section 7.6). Recalibration is therefore not yet a validated remedy: it raised coverage, but neither frozen criterion for it was met, it made the interval score worse on two benchmarks, and on the well-specified synthetic bank it moved coverage away from the nominal level. Practitioners should report model-based intervals together with the panel size and their expected under-coverage, may add recalibrated intervals as a sensitivity analysis, and should use a Wilson interval from uniformly sampled items when nominal coverage matters more than width.

Adaptive testing or benchmark reduction? If the aim is to evaluate the next system with a fraction of a benchmark and only a handful of calibration systems exist, which method should be used? In study 2, EOST-M's score error after 50 items was 29.6% lower than tinyBenchmarks' and 44.8% lower than MINCE's, and over the whole test 26.0% lower than that of an ATLAS-style 3PL CAT, in the same direction on every benchmark and significantly on four or five of eight (only the comparison with MINCE met the frozen criterion). These methods were designed for other regimes: tinyBenchmarks for hundreds of calibration models, whose anchors then cover the item space; ATLAS for thousands; MINCE for sizing a random subset. With a handful of calibrators the plug-in parameters they rely on are unreliable, and tinyBenchmarks cannot always find as many anchors as its budget. In

this regime our evidence favours adaptive selection under calibration uncertainty, although it is conclusive by the frozen criterion only against MINCE; with large panels the differences should shrink, as the gain over plug-in Fisher CAT does.

A protocol for live evaluation. Every result of this paper replays recorded verdicts. A live deployment differs in two ways: the evaluated model has never been seen, and its answers are produced and graded at the moment the test asks for them, with their sampling variability. We release the protocol and code for such an experiment (`scripts/live_eost.py`); it could not be run in our execution environment, which has no access to model weights. The protocol has three phases. (1) *Freeze*: a calibration panel of N leaderboard systems is drawn with a fixed seed, excluding the target's family by a name pattern; the panel is calibrated as in Section 3; the leave-one-calibrator-out recalibration factors are computed; the panel, seed, rule and a hash of the bank are written to `frozen.json` before the first query. (2) *Adaptive phase*: the model, served through any OpenAI-compatible endpoint with temperature 0, answers the item chosen by EIG; an independent deterministic oracle grades the answer (GSM8K: exact match of the final number after the five-shot chain-of-thought prompt; CRUXEval-O: execution of `assert f(input) == prediction` in an isolated subprocess); the estimate and both intervals are recorded at every step, with checkpoints at 20, 40 and 60 items. (3) *Truth phase*: the remaining items are administered (answers already obtained are reused) and the full-bank score is computed only after the adaptive phase is complete. Every query, answer and verdict is appended to a log, so an interrupted run resumes.

The live engine performs exactly the computations of the replay engine; a unit test feeds it recorded verdicts and checks that the selected items, estimates and intervals coincide with the replay. A mock end-to-end run, in which a recorded CRUXEval-O system answers through the live code path, reproduces the replay. The OpenAI-compatible client is tested against a local server. Two differences from the replay studies must be kept in mind when interpreting a live run: the target's prompt and decoding may differ from those that produced the calibrators' verdicts (the Open LLM Leaderboard used its own five-shot harness; CRUXEval-O verdicts aggregate samples at temperature 0.2), and such a protocol shift is itself part of what a live run measures.

Limitations. (i) All evidence replays recorded verdicts. Replay is exact for the recorded data, but it does not include the sampling variability of stochastic decoding, and the live protocol above could not be run in our execution environment; the temporal validation on later SWE-bench submissions approximates, but does not replace, a prospective deployment. (ii) Two confirmatory benchmarks (LiveCodeBench, CRUXEval-O) carry recorded aggregate verdicts, binarized pass rates over several samples; graded responses would need a different measurement model. (iii) The measurement model is a unidimensional 2PL; prior predictive checks show that real systems agree with each other less, item by item, than the model implies, mainly on multiple-choice banks (Section 7.6). (iv) Strict clusters are themselves a heuristic: a residual correlation of 0.5 can miss distant relatives (distillation, shared training data) and merge unrelated systems; one target per cluster removes the dependence between targets, but calibrators drawn from other clusters may still share ancestry with a target. The study-2 conclusions on R2 and C1 did not depend on how the clusters were built—from metadata alone, from residuals on other benchmarks, or with thresholds from 0.4 to 0.7 (Section 8). (v) The leave-one-calibrator-out recalibration is a cross-validated correction

without a finite-sample guarantee; with eight calibrators it rests on very few pseudo-targets. (vi) The frozen Bayesian CATs used a grid-based Gibbs sampler with eight retained draws, whose scale hyperparameters did not converge even in long chains; exploratory samplers with collapsed moves came close to convergence and left the comparison with EOST-M unchanged, but a fully converged comparator—for instance by Hamiltonian Monte Carlo on a continuous parametrization—remains to be run (Section 8). (vii) Both analysis plans were frozen by hashing and committing them, and the second was also sent to the first author before execution, but no public registry or external time stamp was reachable; a reader has to trust the git history. (viii) Items were treated as equally costly and item exposure was not controlled; an adaptive benchmark that serves the same few informative items to every model would accelerate their contamination (Wainer, 2000). (ix) Panels are calibrated by joint maximum a posteriori estimation, which made the planned one-parameter sensitivity analysis of study 1 degenerate (Section 6.7); marginal maximum likelihood over the calibrators' abilities avoids this class of problems. (x) The self-testing side of the framework rests only on a historical pilot (Section 5.4); a prospective study in which models generate items, an independent oracle verifies them and a fresh panel calibrates them is the natural next step. (xi) The cross-benchmark meta-analysis is secondary and treats the benchmarks as independent, although five of them share the same systems; counted as one source, they leave fewer pooled effects clear of zero (Section 8).

Recommendations. For adaptive evaluation of LLMs with a small calibration panel: (1) do not plug in item-specific discriminations calibrated on fewer than a few dozen systems; integrate their uncertainty with an empirical-Bayes item prior (the expected response function is cheap), or constrain them to a common value when a simple, conservative test matters more than accuracy; (2) select by expected information gain or any information criterion computed from the marginal model—the choice matters little—and, when the estimand is the benchmark score, let the score-targeted criterion take over late in the test; (3) the Bayesian treatments of the calibrators that we ran made no significant difference with panels of this size, so the simpler calibration-marginal model, exact given the calibrators, is a reasonable default; (4) report model-based intervals with the panel size and their expected under-coverage, and report coverage alongside accuracy; treat interval recalibration as promising but not yet validated; (5) exclude the target's whole cluster of related systems, not only its nominal family, from its panel, and analyse with clusters as units; (6) count the calibration panel in the cost: with eight fresh calibrators who answer the whole bank, adaptive testing costs less than full evaluation after 8.4–10 new systems, but less than uniform subsampling of equal precision only after a median of 122; when the calibrators' results already exist, it is cheaper from the first system.

11 Conclusion

Adaptive testing can make LLM evaluation cheaper, but with the small calibration panels of new, private or expensive benchmarks it works only if the uncertainty of the item calibration is taken seriously. Integrating that uncertainty through expected response functions with an empirical-Bayes prior is simple and exact given the calibrators. In two studies with frozen analysis plans, it predicted held-out verdicts better than plug-in Fisher-information testing, and in the second study also better than a regularized Rasch-type test, whose score errors

were similar; it estimated benchmark scores more accurately than uniform subsampling and MINCE; and, with eight calibrators, it was ahead of random items scored by the same model, ATLAS-style adaptive testing and tinyBenchmarks on every benchmark without meeting the frozen criterion for these three comparisons. Bayesian CATs that also integrate the calibrators' abilities were not significantly different from it, without being shown equivalent, and better-mixing samplers did not change that comparison. The original thesis of this project, that information rather than entropy is what an adaptive test should maximize, holds in sign but is modest on prediction once uncertainty is integrated, and the choice among information criteria hardly matters. The open problem is honest uncertainty: model-based intervals remain too narrow, and recalibration from the calibration panel improved coverage only at the price of width. Every number in this paper can be regenerated from pinned public data with a Docker-only pipeline and frozen analysis plans, and its qualitative statements about results are checked automatically against the artifacts.

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